

Bulk trace laws and algebraic boundary interaction for the sine Loewner operator

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Abstract

The sine Loewner operator is a spatial compression of a whole-line Fourier exchange whose spectrum is contained in $\{-1, 0, 1\}$. We determine two effects of truncation: a logarithmic transition in the outer spectral bands and algebraically interacting states in the central gaps. An exact reduction of the infinite Bloch fiber gives a trace law with a remainder bounded uniformly by the first two derivative norms. We evaluate its Schatten and determinant coefficients and transfer it to finite matrices with an error independent of dimension. Every fixed nonzero central-gap level occurs at exactly one half-line boundary phase modulo one and is simple. An explicit Mellin equation determines the physical tail exponent and the full leading interaction matrix, including the complementary-resolvent correction. Exact interval certificates isolate a phase-zero mode in $(0.405, 0.428)$ and prove its normalized tail amplitude has magnitude greater than $1/100$, giving a nonzero even-odd splitting. Further consequences include phase-dependent third terms, parity determinant asymptotics and odd logarithmic moments generated by additive frequency triples.

1 Introduction

A whole-line operator can have only three spectral values and still acquire rich spectral structure when restricted to a long interval. The sine Loewner operator provides an explicit example. Its whole-line Fourier representation exchanges two adjacent frequency windows, with spectrum contained in $\{-1, 0, 1\}$. Spatial truncation produces a logarithmic transition near the two outer spectral bands and isolated states in the central gaps. The outer transition admits a uniform trace law; the gap states follow a one-period boundary flow and interact across the interval at an explicitly determined algebraic rate.

The finite matrices are

$$A_N(\varepsilon)_{mn} = \begin{cases} \frac{\sin(2\pi m\varepsilon) - \sin(2\pi n\varepsilon)}{\pi(m - n)}, & m \neq n, \\ 2\varepsilon \cos(2\pi m\varepsilon), & m = n, \end{cases} \quad -N \leq m, n \leq N, \quad (1.1)$$

where $N \geq 1$, $d = 2N + 1$ and $0 < \varepsilon < 1/2$. Their critical integral operator on $L^2[-1, 1]$ has kernel

$$K_\tau(x, y) = \frac{\sin(2\pi\tau x) - \sin(2\pi\tau y)}{\pi(x - y)}, \quad K_\tau(x, x) = 2\tau \cos(2\pi\tau x). \quad (1.2)$$

Both are self-adjoint contractions and commute with reflection. The kernel is the commutator of multiplication by $\sin(2\pi\tau x)$ and the finite Hilbert transform with kernel $1/[\pi(x - y)]$. The critical scale is $N\varepsilon$; an exact covariance comparison uses the slightly different parameter $d\varepsilon/2$.

Three main conclusions

A uniform trace law and a finite-matrix bridge. For $r_t = \sqrt{t^2 + (1 - t)^2}$, set

$$\varkappa(f) = \frac{1}{2\pi^2} \int_0^1 \frac{f(r_t) + f(-r_t) - f(1) - f(-1)}{t(1 - t)} dt. \quad (1.3)$$

The integral converges for Hölder f : the endpoint numerator is $O([t(1-t)]^\alpha)$ for its Hölder exponent $\alpha > 0$.

Theorem 1.1 (Uniform trace law). *Every real $f \in C^2([-1, 1])$ with $f(0) = 0$ satisfies, for $\tau \geq 2$,*

$$\mathrm{tr} f(\mathcal{K}_\tau) = 2\tau[f(1) + f(-1)] + \varkappa(f) \log \tau + \mathbf{R}_\tau(f), \quad (1.4)$$

where, with the absolute constant $C_{\mathrm{tr}} = 143$,

$$|\mathbf{R}_\tau(f)| \leq C_{\mathrm{tr}}(\|f'\|_\infty + \|f''\|_\infty). \quad (1.5)$$

The bound is uniform in f and τ .

The two bulk masses lie at ± 1 . The logarithmic correction is supported on $g \leq |\lambda| \leq 1$, where $g = 1/\sqrt{2}$, with the endpoint subtraction in (1.3) essential. The theorem extends to globally Lipschitz tests that are $C^{1,1}$ only on these outer bands. It gives all Schatten powers $|x|^p$, $p \geq 1$, closed determinant coefficients and signed transition counts. In particular, for $w \geq 0$ the coefficient in $\log \det(I + w\mathcal{K}_\tau^2)$ is

$$-\frac{4}{\pi^2} \arcsin^2 \left(\frac{1}{2} \sqrt{\frac{2w}{1+w}} \right),$$

which equals $-1/9$ at $w = 1$.

Theorem 4.2 compares $A_N(\varepsilon)$ with $\mathcal{K}_{d\varepsilon/2}$ independently of dimension. After the exact linear correction, its C^2 trace error is at most $L_\varepsilon \|f''\|_\infty$, with $L_\varepsilon = 2\varepsilon \csc(2\pi\varepsilon) - 1/\pi = O(\varepsilon^2)$. The Lipschitz error is $O(\varepsilon)$. Thus the trace laws also apply at fixed sampling scale, uniformly when ε stays below $1/2 - \delta$, $\delta > 0$.

Physical gap states and one-period flow. Dilation identifies $\mathcal{K}_\tau$ with the compression to $(-\tau, \tau)$ of the fixed whole-line exchange $\mathcal{G}$ with kernel $[\sin(2\pi X) - \sin(2\pi Y)]/[\pi(X - Y)]$. Let $B_b = P_{(b, \infty)} \mathcal{G} P_{(b, \infty)}$. For each fixed $\alpha \in (-g, 0) \cup (0, g)$ there is exactly one b modulo one for which α is an eigenvalue of B_b , and that eigenvalue is simple. Long-interval clusters converge to the two boundary spectra with exact multiplicity (Theorems 7.2 and 7.5). This information is finer than the bulk and logarithmic trace terms.

The full algebraic interaction and a certified nonzero example. For an isolated level α shared by the two boundaries, let η_L, η_R be the normalized physical tail amplitudes of Section 8. For an interval of length L with fixed endpoint phases, the two nearby eigenvalues are

$$\alpha + L^{-\gamma(\alpha)} \lambda_j(M_\alpha) + o(L^{-\gamma(\alpha)}), \quad j = 1, 2,$$

where

$$\gamma(\alpha) = \frac{2}{\pi} \arctan \sqrt{1 - 2\alpha^2}, \quad M_\alpha = -\frac{2\pi(1 - \alpha^2)\sqrt{1 - 2\alpha^2}}{\alpha} \begin{pmatrix} 0 & \overline{\eta_L} \eta_R \\ \overline{\eta_R} \eta_L & 0 \end{pmatrix}.$$

This matrix includes the complementary-resolvent correction, which has the same order as the direct overlap (Theorem 8.1). Centered intervals have opposite even and odd coefficients; the odd state lies farther from zero when the amplitude is nonzero. Exact interval certificates isolate a phase-zero mode in $(0.405, 0.428)$, prove its uniqueness in $[0.3565, 0.4765]$, and establish $|\eta| > 1/100$ for every normalized physical eigenfunction there. They therefore give an unconditional instance of the nonzero splitting law. General nonvanishing is not assumed or proved.

Other consequences and scope

At small continuum scale, the four leading signed/parity branches begin at orders $\tau, \tau^3, \tau^5, \tau^7$. For a simple isolated parity eigenvalue in the finite critical limit, physical eigenvector reconstructions converge weakly to zero while magnification reveals nontrivial endpoint profiles. These are distinct from the long-window boundary modes. Physical boundary localization also gives continuous phase-dependent third terms. For finite matrices these hold whenever $\varepsilon_N \rightarrow 0$ and $\tau_N = (2N + 1)\varepsilon_N/2 \rightarrow \infty$, without a further condition on $(2N + 1)\varepsilon_N^3$ or convergence of the fractional phases. Reflection resolves information invisible in the leading signed trace: for $w > 0$,

$$\frac{\det(I + w\mathcal{K}_{\tau,+}^2)}{\det(I + w\mathcal{K}_{\tau,-}^2)} \longrightarrow 1 + \frac{w}{2},$$

where $+$ and $-$ denote spatial parity. For a fixed finite commensurate frequency family, a bipartite bulk graph is sign-symmetric, whereas additive frequency triples produce an odd logarithmic term.

The estimates have different scopes, summarized below. In particular, relative gap indices do not determine unspecified absolute ranks, and no uniform interaction law at zero or at a band edge is asserted.

Question	Parameter and regularity range
Critical finite limit	$N\varepsilon_N \rightarrow \tau \in (0, \infty)$; all fixed signed/parity eigenvalue branches.
Uniform trace remainder	$\tau \geq 2$; global C^2 , or global Lipschitz and band-local $C^{1,1}$.
Finite covariance transfer	$0 < \varepsilon < 1/2$; constants independent of N , with stated dependence on ε .
Phase-dependent constant	Global Lipschitz and band-local C^2 ; continuous periodic phase profile.
Boundary interaction	Fixed isolated $\alpha \in (-g, 0) \cup (0, g)$; fixed endpoint phases and integer increases of interval length.
Certified nonvanishing	The isolated phase-zero mode enclosed in $(0.405, 0.428)$.

Relation to earlier work

Loewner matrices are classical [34]; their inertia is studied in [10]. De Andrade Silva recorded the arithmetic divided-difference block [41, 40]; its finite arithmetic dictionary and cutoff-derivative geometry occur in [22, 23]. Compressed translations appear in [24, Section 5.1], and the commutator framework is classical [14]. Positive time–band concentration and discrete prolate sequences are classical [19, 30, 31, 43, 44, 45]; quantitative bounds include [13, 26]. No priority is claimed for these objects or for the variable $N\varepsilon$. The same matrix also appears in Shi [39, eqs. (3.4)–(3.7)]: in that paper’s notation, $2L \mathbf{G}_{N,L}(x) = A_N(1 - x/L)$ for $L/2 < x < L$, including the diagonal entries. Shi studies a finite arithmetic model and its rank-two displacement structure; the critical scaling and uniform trace asymptotics considered here are different conclusions.

The two-term structure of Theorem 1.1 is that of the Widom–Sobolev formula [12, 46, 49]. In the notation of [12, eq. (2.27)], $\varkappa(f) = 2\mathfrak{U}(f; X_{1,0}, X_{0,-1})$, where $X_{1,0}$ and $X_{0,-1}$ are the two one-sided limits of the Bloch symbol of Proposition 2.4, the factor 2 counts the two ends of the truncation, and $\pm\sqrt{t^2 + (1 - t)^2}$ is the nonzero spectrum of their linear interpolation (Section 3.1). Matrix-valued trace formulas, including the jump integral displayed in [38, eq. (1.3)], provide direct precedents for the reduced three-dimensional symbol. Block determinant asymptotics and operator-valued Szegő–Widom theory supply further settings [8, 11]; Bloch-function trace asymptotics for a lattice-periodic operator appear in [37]. The infinite physical fiber must be reduced with control of the compression error before the finite-matrix reference can be used. The positive logarithmic counting and logistic rank laws are already in Landau–Widom [31, pp. 469–470]. What is proved here is the reduction of the exchange symbol to a fixed three-dimensional jump, the resulting trace law with a remainder bounded by the first two derivative

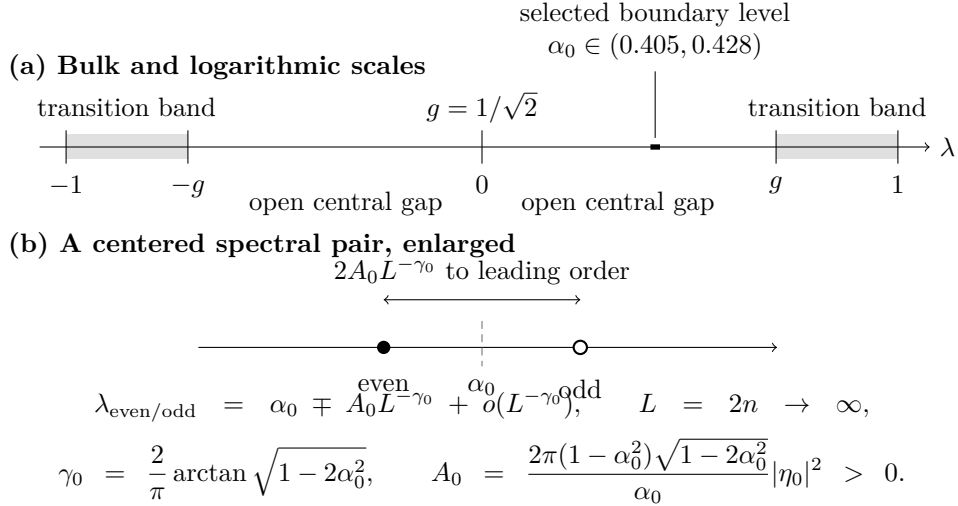


Figure 1: Spectral scales for the single-frequency operator. The shaded bands support the logarithmic trace density, whereas the bulk values are ± 1 . The marked interval encloses the certified phase-zero half-line level; it is not a census of the central gaps. The enlarged panel shows the leading locations of the two eigenvalues of $\mathcal{K}_n$ converging to that level. Their placement is schematic, not numerical data. Here η_0 is the normalized physical tail amplitude. The coefficient includes the complementary-resolvent contribution, and $|\eta_0| > 1/100$ makes it nonzero.

norms uniformly in f and τ , its band-local extension, the folded signed profile, and the quantified dimension-independent transfer back to finite matrices. The Hilbert–Carleman comparison has a direct antecedent in Otte [36], and the smooth compression estimate is a version of Laptev–Safarov [32]. We prove all operator comparisons needed here. Recent work also treats moving thresholds and low-regularity traces for positive localization operators [6, 29]. Our signed transition estimate is uniform on compact subsets of the outer bands; it does not identify individual eigenvalues with positive prolate eigenvalues or assert their sharper counting errors for this operator.

One-period edge flow has precedents for periodic differential operators [21, Proposition 3.21]; the proof here uses an analytic bandlimited covariance and a relative projection index. The endpoint model is related to classical finite-Hilbert-transform analysis [28, 9, 2]. Our frequency pencil retains the same-half Hilbert blocks together with a phase-dependent exchange; the scalar reductions for adjacent or overlapping intervals do not directly diagonalize that pencil. The full complementary correction has the structure of a Feshbach–Schur reduction [16], but its coefficient is evaluated here for the physical boundary modes. Block Toeplitz determinant constants [8] and scalar Toeplitz-plus-Hankel parity formulas [7] provide the reference results; we prove the localization and reflected limits needed to transfer them to the present operator-valued symbol.

Organization

Section 2 gives the exchange representation, strict signed monotonicity and critical limit. Section 3 develops the logarithmic density, coefficients and complete uniform-remainder proof. Section 4 gives finite-scale estimates and endpoint geometry; Section 5 treats commensurate frequencies. The boundary analysis proceeds from phase constants (Section 6) to half-line flow and the endpoint equation (Section 7), the full interaction (Section 8) and its certificates (Section 9). Section 10 resolves parity traces and counts. The endpoint spectral measure and differential obstruction in Section 7, together with the final open questions, delineate what these methods do and do not determine. All mathematical proofs are included in the paper; the ancillary files contain reproducible programs and exact rational witnesses.

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2 Exchange structure and critical scaling

2.1 The two operators and their exchange representations

On $L^2(\mathbb{T})$, $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, put $e_m(t) = e^{2\pi i m t}$ and let P_N project onto $V_N = \text{span}\{e_m : |m| \leq N\}$. For $\delta = 1 - \varepsilon$, define

$$(S_\delta f)(t) = \mathbf{1}_{[\delta, 1]}(t)f(t - \delta), \quad T_\delta = S_\delta + S_\delta^*.$$

S_δ carries the source arc $[0, \varepsilon]$ onto the target arc $[1 - \varepsilon, 1]$; these are disjoint because $\varepsilon < 1/2$. Thus T_δ exchanges them, vanishes off their union, and has norm one.

Lemma 2.1. *The matrix of $P_N T_\delta P_N$ is (1.1). At the critical scale,*

$$A_N(\tau/N)_{mn} = N^{-1} K_\tau(m/N, n/N). \quad (2.1)$$

In particular $\|A_N(\varepsilon)\| \leq 1$.

Proof. The matrix entry of $S_{1-\varepsilon}$ is $e^{2\pi i m \varepsilon} \int_0^\varepsilon e^{2\pi i (n-m)t} dt$; the adjoint entry has the same integral multiplied by $e^{-2\pi i n \varepsilon}$. Their sum gives (1.1), including the diagonal. Substitution gives (2.1). $\square$

Write $\mathcal{R}f(x) = f(-x)$, $(\mathcal{R}_N a)_m = a_{-m}$ and $\mathcal{H}_p = \ker(\mathcal{R} - pI)$ for $p = \pm 1$. The largest and smallest spectral values in these sectors are denoted by $\lambda_p^\pm(\tau)$ for $\mathcal{K}_\tau$ and by $\lambda_{N,p}^\pm(\varepsilon)$ for $A_N(\varepsilon)$; $F_p(\tau) = \|\mathcal{K}_\tau|_{\mathcal{H}_p}\|$. We use the unitary Fourier transform $(\mathcal{F}f)(\xi) = \hat{f}(\xi) = \int_{\mathbb{R}} f(x)e^{2\pi i x \xi} dx$ throughout. Let E extend functions on $[-1, 1]$ by zero and put $U = \mathcal{F}E$.

Proposition 2.2 (Cross-window structure). *Let R_τ denote multiplication by $\mathbf{1}_{(-\tau, \tau)}$ on $L^2(\mathbb{R})$ and*

$$(J_\tau g)(\xi) = \begin{cases} g(\xi - \tau), & 0 < \xi < \tau, \\ g(\xi + \tau), & -\tau < \xi < 0, \\ 0, & |\xi| \geq \tau. \end{cases} \quad (2.2)$$

Then $J_\tau = J_\tau^*$, $J_\tau^2 = R_\tau$, and

$$\mathcal{K}_\tau = U^* J_\tau U, \quad \langle f, \mathcal{K}_\tau f \rangle = 2 \operatorname{Re} \int_0^\tau \overline{\widehat{f}(\xi - \tau)} \widehat{f}(\xi) d\xi. \quad (2.3)$$

Define C_θ to have kernel $c_\theta(x - y) = \sin(\pi\theta(x - y))/[\pi(x - y)]$ on $[-1, 1]$. With $\Pi_\pm = (R_\tau \pm J_\tau)/2$ and $C_\pm = U^* \Pi_\pm U$,

$$C_\pm \geq 0, \quad C_+ + C_- = C_{2\tau}, \quad C_+ - C_- = \mathcal{K}_\tau, \quad \mathcal{K}_\tau^2 \leq C_{2\tau}. \quad (2.4)$$

$\mathcal{K}_\tau$ is trace class with $\|\mathcal{K}_\tau\|_1 \leq 4\tau$, and

$$\operatorname{tr} \mathcal{K}_\tau = \frac{2 \sin(2\pi\tau)}{\pi}, \quad \operatorname{tr}(\mathcal{K}_\tau|_{\mathcal{H}_p}) = \frac{\sin(2\pi\tau) + p \operatorname{Si}(2\pi\tau)}{\pi}, \quad (2.5)$$

where $\operatorname{Si}(x) = \int_0^x \sin(t) t^{-1} dt$. Moreover $\|\mathcal{K}_\tau\| < 1$ for every finite $\tau > 0$.

Proof. Direct Fourier integration yields (2.3); equivalently, commuting the multiplier $i \operatorname{sgn} \xi$ of the Hilbert transform on $\mathbb{R}$ with the two frequency shifts in the sine, then compressing to $[-1, 1]$, gives J_τ . The operators $\Pi_\pm$ are orthogonal projections. The restriction $A = R_\tau U : L^2[-1, 1] \rightarrow L^2(-\tau, \tau)$ is Hilbert–Schmidt with $\|A\|_2^2 = 4\tau$. Thus $C_\pm = A^* \Pi_\pm A$ are positive trace class, their sum is $A^* A = C_{2\tau}$, and their total trace is 4τ . Also $U^* J_\tau U U^* J_\tau U \leq U^* J_\tau^2 U = C_{2\tau}$. Since $C_\pm$ and $\frac{1}{2}(I + p\mathcal{R})C_\sigma$, $p, \sigma = \pm 1$, are positive trace-class operators with continuous kernels, their traces are the integrals of their diagonals (Mercer). Hence $\operatorname{tr} \mathcal{K}_\tau = \int_{-1}^1 2\tau \cos(2\pi\tau x) dx$, $\operatorname{tr}(\mathcal{R}\mathcal{K}_\tau) = \int_{-1}^1 \sin(2\pi\tau x)/(\pi x) dx = 2 \operatorname{Si}(2\pi\tau)/\pi$, and $\operatorname{tr}(\mathcal{K}_\tau|_{\mathcal{H}_p}) = \frac{1}{2}[\operatorname{tr} \mathcal{K}_\tau + p \operatorname{tr}(\mathcal{R}\mathcal{K}_\tau)]$ proves (2.5). If $\mathcal{K}_\tau$ had an eigenvalue of modulus one, equality in the contraction $U^* J_\tau U$ would force Uf to vanish outside $(-\tau, \tau)$. The Fourier transform of a compactly supported function is entire, so this forces $f = 0$, a contradiction. $\square$

The same exchange identifies exactly where compression loses mass. Let $\mathcal{G}_\tau = \mathcal{F}^* J_\tau \mathcal{F}$, $P = EE^*$, and $\tilde{\mathcal{K}}_\tau = \mathcal{G}_\tau E$. Since $J_\tau^2 = R_\tau$, Plancherel gives $\tilde{\mathcal{K}}_\tau^* \tilde{\mathcal{K}}_\tau = C_{2\tau}$ and

$$C_{2\tau} - \mathcal{K}_\tau^2 = E^* \mathcal{G}_\tau (I - P) \mathcal{G}_\tau E \geq 0, \quad 1 - \|\mathcal{K}_\tau f\|^2 = \|(I - P) \tilde{\mathcal{K}}_\tau f\|^2 + \int_{|\xi| > \tau} |\widehat{Ef}(\xi)|^2 d\xi \quad (\|f\| = 1). \quad (2.6)$$

The two terms are output escaping the physical interval and input outside the frequency band. For an eigenvector the left side is $1 - \lambda^2$. These identities follow by inserting $P + (I - P) = I$; the band has width 2τ , so its concentration kernel is $\sin(2\pi\tau(x - y))/[\pi(x - y)]$.

The elementary identity

$$K_\tau(x, y) = 2 \operatorname{Re} \int_0^\tau e^{2\pi i[x(\tau-s)+ys]} ds = 2 \cos(\pi\tau(x + y)) c_\tau(x - y) \quad (2.7)$$

will be used repeatedly.

Theorem 2.3 (Continuum inertia and strict monotonicity). *For every $\tau > 0$, $\mathcal{K}_\tau$ is injective and has infinitely many positive and infinitely many negative eigenvalues in each parity sector. Write $\xi_{k,p}^+(\tau) > 0$ for the positive eigenvalues in nonincreasing order and $\xi_{k,p}^-(\tau) < 0$ for the negative eigenvalues in nondecreasing order, counting multiplicity; thus $\xi_{1,p}^\pm = \lambda_p^\pm$. For every $k \geq 1$, $p = \pm 1$ and $0 < \tau_1 < \tau_2$,*

$$0 < \xi_{k,p}^+(\tau_1) < \xi_{k,p}^+(\tau_2) < 1, \quad -1 < \xi_{k,p}^-(\tau_2) < \xi_{k,p}^-(\tau_1) < 0. \quad (2.8)$$

Each ordered eigenvalue is continuous in τ , tends to zero as $\tau \downarrow 0$, and tends to its respective sign as $\tau \rightarrow \infty$.

These statements do not assert simplicity at every parameter.

Proof. Both $A = R_\tau U$ and A^* are injective: the Fourier transform of a function supported on either bounded interval is entire, and cannot vanish on the other interval unless it vanishes identically. In the polar decomposition $A = BV$, $B = (AA^*)^{1/2}$ is therefore positive and injective, V is unitary, and both respect reflection. The last assertion follows from $A\mathcal{R} = \mathcal{R}A$ between the two spaces and uniqueness of the polar decomposition. Thus $\mathcal{K}_\tau$ is unitarily equivalent, sector by sector, to $BJ_\tau B$. This proves injectivity.

In each parity sector the involution J_τ on $L^2(-\tau, \tau)$ has infinite-dimensional eigenspaces of both signs: for any $p, \sigma = \pm 1$, choose $h \in L^2(0, \tau)$ with $h(\tau - \xi) = \sigma p h(\xi)$, and set $g = h$ on $(0, \tau)$ and $g(\xi) = \sigma h(\xi + \tau)$ on $(-\tau, 0)$. Then $J_\tau g = \sigma g$ and $\mathcal{R}g = pg$. The range of B is dense in either parity sector. Approximate any n orthonormal vectors in this joint eigenspace by $Bf_1, \dots, Bf_n$ with $f_j \in \mathcal{H}_p$. The matrix $(\sigma \langle Bf_i, J_\tau Bf_j \rangle)_{i,j}$ can be made arbitrarily close to the identity, hence positive definite. The minimax principle gives at least n eigenvalues of sign σ for $BJ_\tau B$ on the sector p .

For monotonicity use the parity-preserving isometry $\mathcal{D}_a f(x) = a^{-1/2} f(x/a) \mathbf{1}_{[-a,a]}(x)$, $0 < a < 1$. The kernel gives $\mathcal{D}_a^* \mathcal{K}_\tau \mathcal{D}_a = \mathcal{K}_{a\tau}$. Fix k, p, σ , with $\sigma = \pm 1$, and let $c > 0$ be the k th largest eigenvalue of $\sigma \mathcal{K}_{a\tau}|_{\mathcal{H}_p}$. The dilated span M of its first k eigenvectors has dimension k , and every unit vector in M has Rayleigh quotient at least c for $\sigma \mathcal{K}_\tau$. Minimax gives weak monotonicity. If the k th largest eigenvalue at τ were also c , the spectral subspace E of $\sigma \mathcal{K}_\tau|_{\mathcal{H}_p}$ for eigenvalues strictly larger than c would have dimension at most $k - 1$. There would be a unit vector $v \in M \cap E^\perp$. Its Rayleigh quotient is at least c because $v \in M$ and at most c because $v \in E^\perp$. Equality and the spectral theorem force $\sigma \mathcal{K}_\tau v = cv$. Every nonzero eigenfunction is analytic, because the kernel is entire in its first variable, locally uniformly in its second. But v vanishes on $(a, 1)$, a contradiction. This proves strict monotonicity for both signs; strict contractivity gives the outer inequalities in (2.8). Norm continuity of the kernel and the minimax principle give continuity of each ordered eigenvalue, and $\|\mathcal{K}_\tau\| \leq 4\tau$ gives the limit at zero. For the limit at infinity use the full-line operator $\mathcal{G} = \mathcal{F}^* J \mathcal{F}$ of Section 2.1.1. The construction above with $\tau = 1$ gives infinitely many orthonormal eigenvectors of J of either sign and either reflection parity. Their inverse Fourier transforms give the same eigenspaces for $\mathcal{G}$. Fix k orthonormal vectors $g_1, \dots, g_k$ of sign σ and parity p . For $P_\tau = P_{(-\tau, \tau)}$, strong convergence $P_\tau \rightarrow I$ gives

$$(\langle P_\tau g_i, P_\tau g_j \rangle)_{i,j} \rightarrow I_k, \quad (\sigma \langle P_\tau g_i, \mathcal{G} P_\tau g_j \rangle)_{i,j} \rightarrow I_k.$$

The truncated vectors retain parity p . Thus every unit vector in their span has Rayleigh quotient for $\sigma \mathcal{G}$ tending uniformly to one. The dilation identity in Section 2.1.1 and minimax imply $\liminf_{\tau \rightarrow \infty} |\xi_{k,p}^\sigma(\tau)| \geq 1$. Contractivity supplies the reverse bound, proving the asserted limits. $\square$

2.1.1 The global exchange and its Bloch symbol

On $L^2(\mathbb{R})$ define

$$(Jg)(\xi) = \mathbf{1}_{(0,1)}(\xi)g(\xi - 1) + \mathbf{1}_{(-1,0)}(\xi)g(\xi + 1), \quad \mathcal{G} = \mathcal{F}^* J \mathcal{F}.$$

Then $J = J^*$, $J^2 = \mathbf{1}_{(-1,1)}$ and $\|J\| = 1$. For compactly supported inputs, the two Fourier integrals give the kernel

$$G(X, Y) = 2 \cos(\pi(X + Y)) \frac{\sin(\pi(X - Y))}{\pi(X - Y)} = \frac{\sin(2\pi X) - \sin(2\pi Y)}{\pi(X - Y)}.$$

In particular $\mathcal{G}^2 = \mathcal{F}^* \mathbf{1}_{(-1,1)} \mathcal{F}$ and $G(X + 1, Y + 1) = G(X, Y)$.

Dilation gives the identities

$$(\mathcal{D}_\tau f)(X) = \tau^{-1/2} f(X/\tau), \quad \mathcal{D}_\tau \mathcal{K}_\tau \mathcal{D}_\tau^* = P_{(-\tau, \tau)} \mathcal{G} P_{(-\tau, \tau)}.$$

Here P_I multiplies by $\mathbf{1}_I$; the dilated kernel is $\tau^{-1}K_\tau(X/\tau, Y/\tau) = G(X, Y)$. The trace-class property, the norm bound and the trace formulas are in Proposition 2.2.

Proposition 2.4 (Exact Bloch symbol). *Identify $L^2(\mathbb{R})$ with $\ell^2(\mathbb{Z}; H)$, $H = L^2(0, 1)$, by $f_n(t) = f(n + t)$. Then $\mathcal{G}$ is the Laurent operator with blocks*

$$g_k(t, s) = G(k + t, s) = \int_0^1 \varsigma(\theta)(t, s) e^{-2\pi i k \theta} d\theta, \quad \varsigma(\theta)(t, s) = 2 \cos(\pi(t + s)) e^{i\pi(t-s)(1-2\theta)}. \quad (2.9)$$

For $e_j(t) = e^{2\pi i j t}$ and $W_\theta = M_{e^{2\pi i \theta t}}$,

$$\varsigma(\theta) = W_\theta^* X_{1,0} W_\theta, \quad X_{j,k} = |e_j\rangle\langle e_k| + |e_k\rangle\langle e_j|.$$

It has eigenvalues $1, -1, 0$ and one jump on the circle: $\varsigma(0) = X_{1,0}$, $\varsigma(1) = X_{0,-1}$.

Proof. Periodicity gives the cell blocks $G(k + t, s)$. With $z = t - s$,

$$\int_0^1 e^{i\pi z(1-2\theta)} e^{-2\pi i k \theta} d\theta = \frac{\sin(\pi z)}{\pi(k + z)}.$$

The removable case $k = z = 0$ follows by continuity. Multiplication by $2 \cos(\pi(t + s))$ proves (2.9); Fubini is valid because the kernels are bounded on the unit square. Expanding the cosine expresses the symbol as the exchange of the orthonormal pair $e^{2\pi i(1-\theta)t}, e^{-2\pi i \theta t}$, proving the rest. $\square$

The jump is not removable by a continuous periodic gauge to a constant symbol: its two endpoint limits would then be conjugated by the same unitary and would have to coincide. The proof below instead uses a nonconstant symbol on a fixed three-dimensional subspace.

2.2 Critical-window convergence

Theorem 2.5 (Explicit operator-norm limit). *For every $0 < T < \infty$ and every integer $N > 2T$, which ensures that $\tau/N < 1/2$ for $0 \leq \tau \leq T$, there are reflection-intertwining isometries U_N from $\mathbb{C}^{2N+1}$ into $L^2[-1, 1]$, independent of τ , under which the embedded matrix $\tilde{A}_{N,\tau} = U_N A_N(\tau/N) U_N^*$ acts as zero off its step-function range and*

$$\sup_{0 \leq \tau \leq T} \|\tilde{A}_{N,\tau} - \mathcal{K}_\tau\| \leq \frac{2T + 8\pi T^2}{N}. \quad (2.10)$$

Pointwise in τ , the numerator may be replaced by $2\tau + 8\pi\tau^2$. The same estimate holds in each parity sector.

Proof. Put $h = 2/d$, and partition $[-1, 1]$ into equal cells B_m of width h centered at $z_m = hm$, $-N \leq m \leq N$. Define $(U_N a)(x) = h^{-1/2} a_m$ on B_m . Since $N^{-1} = h(2N + 1)/(2N)$, the embedded matrix has step kernel

$$\tilde{K}_{N,\tau}(x, y) = \frac{2N + 1}{2N} K_\tau(m/N, n/N), \quad (x, y) \in B_m \times B_n. \quad (2.11)$$

Formula (2.7) gives

$$|K_\tau| \leq 2\tau, \quad |\partial_x K_\tau|, |\partial_y K_\tau| \leq 2\pi\tau^2. \quad (2.12)$$

Since $z_m = hm$ and $|z_m - m/N| = |m|/(N(2N + 1)) \leq 1/(2N + 1) = h/2$, while $|x - z_m| \leq h/2$ on B_m , we have $|x - m/N| \leq h$. The scalar mismatch in (2.11) and the two coordinate errors therefore give

$$|\tilde{K}_{N,\tau}(x, y) - K_\tau(x, y)| \leq \frac{\tau + 4\pi\tau^2}{N}.$$

The square $[-1, 1]^2$ has area 4, so the operator norm is at most the Hilbert–Schmidt norm, which is at most twice this supremum. Reflection intertwining follows from the symmetric cells. $\square$

Differentiating (2.7) also gives

$$\|\mathcal{K}_\tau - \mathcal{K}_{\tau'}\| \leq (4 + 8\pi T)|\tau - \tau'|, \quad 0 \leq \tau', \tau \leq T. \quad (2.13)$$

Hence if $\tau_N = N\varepsilon_N \rightarrow \tau < \infty$, then with $T = \sup_N \tau_N < \infty$ and every $N > 2T$,

$$\|\tilde{A}_{N,\tau_N} - \mathcal{K}_\tau\| \leq \frac{2\tau_N + 8\pi\tau_N^2}{N} + (4 + 8\pi T)|\tau_N - \tau| \rightarrow 0. \quad (2.14)$$

The parity norms and every isolated nonzero parity spectral cluster therefore converge; the same holds for an individual signed edge whenever it is separated from zero.

3 Uniform trace laws and their coefficients

3.1 The logarithmic density and spectral counts

Proposition 3.1 (Density and even moments). *For Hölder f ,*

$$\varkappa(f) = \int_{1/\sqrt{2}}^1 [f(x) + f(-x) - f(1) - f(-1)] \varpi(x) dx, \quad \varpi(x) = \frac{2x}{\pi^2(1-x^2)\sqrt{2x^2-1}}. \quad (3.1)$$

For integers $m \geq 1$,

$$\pi^2 \varkappa(x^{2m}) = \sum_{j=1}^m \binom{m}{j} (-2)^j \frac{(j-1)!^2}{(2j-1)!}, \quad \varkappa(x^{2m+1}) = 0. \quad (3.2)$$

Proof. The two halves of the t integral in (1.3) coincide. On either half substitute $x = \sqrt{1-2t(1-t)}$ to obtain the density. Its singularity at $1/\sqrt{2}$ is integrable; at 1 the endpoint subtraction supplies the Hölder factor. For the moments expand $(1-2t(1-t))^m - 1$ and integrate each term using $B(j, j) = (j-1)!^2/(2j-1)!$. Odd tests cancel identically. $\square$

Thus the coefficients for x^2, x^4, x^6 are respectively $-2, -10/3, -64/15$ in units of π^{-2} . In particular Theorem 1.1, applied to the C^2 test $f(x) = x^2 - x^4$, for which $f(0) = 0$ and $f(1) + f(-1) = 0$, gives

$$\mathrm{tr}(\mathcal{K}_\tau^2 - \mathcal{K}_\tau^4) = \frac{4}{3\pi^2} \log \tau + O(1). \quad (3.3)$$

The gap comes from the jump, not from the physical spectrum. The two one-sided exchanges interpolate as

$$M_t = (1-t)X_{0,-1} + tX_{1,0} = \begin{pmatrix} 0 & 1-t & 0 \\ 1-t & 0 & t \\ 0 & t & 0 \end{pmatrix}, \quad \mathrm{spec} M_t = \{0, \pm r_t\}.$$

Thus the jump trace is $f(r_t) + f(-r_t) - f(1) - f(-1)$, giving (1.3) in the Widom normalization [49]. In contrast, the gauge symbol $a(\theta)$ in Section 3.4 has values $0, \pm 1$ at every θ . The classical prolate interpolation traverses $[0, 1]$; here the moving values stay in the two outer bands. The subtraction at ± 1 is essential because ϖ has infinite mass there.

3.1.1 Band-local regularity and the trace norm

Put $g = 2^{-1/2}$ and retain C_V, C_*, B_1, B_2 and the strip constant $b(\cdot)$ of (3.27) from Section 3.4. Define

$$A_{\mathrm{band}} = 2C_V + 4b(1/2) + 2 + \frac{2\log(5/2)}{\pi^2} + \sqrt{2}(C_* + B_1 + B_2).$$

Here $C^{1,1}$ means continuously differentiable with Lipschitz derivative, using one-sided derivatives at an interval's endpoints. Trace asymptotics for test functions with finitely many Hölder singularities are treated in [33, 47] and [12, Theorem 4.22], with remainders of lower order than the logarithmic term. The hypothesis below is of a different kind: beyond a global Lipschitz bound, no regularity is imposed inside the central gap, which the exact gap of the model makes possible.

Theorem 3.2 (Regularity only on the model bands). *Let $f : [-1, 1] \rightarrow \mathbb{R}$ be L -Lipschitz, with $f(0) = 0$. Suppose its restrictions to $[-1, -g]$ and $[g, 1]$ are $C^{1,1}$, and let M be the larger Lipschitz constant of their derivatives. Then, for $\tau \geq 2$,*

$$|\mathrm{tr} f(\mathcal{K}_\tau) - 2\tau[f(1) + f(-1)] - \varkappa(f) \log \tau| \leq A_{\mathrm{band}}L + B_2M.$$

For the phase-shifted operator $\mathcal{K}_{\tau,\phi}$ of Proposition 3.30, the same estimate holds uniformly in ϕ with A_{band} replaced by $A_{\mathrm{band}} + 4b(1/2)$.

Proof. In the scalar model write $C = T_n(c)$ and $S = T_n(s)$. Since $c + s \geq 1$, $C + S \geq I$ and $H_n = C^2 + S^2 = [(C + S)^2 + (C - S)^2]/2 \geq I/2$. Thus $D_n(u) + D_n(v) = I - H_n$ and $D_n(v_0)$ have spectra in $[0, 1/2]$, the latter by (3.40). The star consequently has exactly n eigenvalues of each sign, all of magnitude at least g . For h of (3.41), restricted to $0 \leq x \leq 1/2$, put $r = \sqrt{1 - x}$. Then

$$h' = \frac{f'(-r) - f'(r)}{2r}, \quad h'' = \frac{f''(r) + f''(-r)}{4r^2} - \frac{f'(r) - f'(-r)}{4r^3} \quad \text{a.e.}$$

Hence $h \in C^{1,1}$, $h(0) = 0$, $\mathrm{Lip}(h) \leq \sqrt{2}L$ and $\mathrm{Lip}(h') \leq M + \sqrt{2}L$; integrate the a.e. bound on h'' to obtain the last assertion. Scalar Taylor remainder is therefore bounded by $(M + \sqrt{2}L)|x - y|^2/2$, so the spectral-measure proof of Lemma 3.26 applies to h . Extend it affinely from $1/2$ to 1 , preserving its Lipschitz bound; all second-derivative estimates and spectral evaluations remain in $[0, 1/2]$. Signed singular-value pairing gives exactly

$$\mathrm{tr} f(T_n(a)) = n[f(1) + f(-1)] + \mathrm{tr} h(D_n(u) + D_n(v)).$$

Equations (3.36), (3.39) and (3.56) now bound the error at $n = 2q$ by $2C_V L + (C_* + B_1)\sqrt{2}L + B_2(M + \sqrt{2}L)$. Integer transfer, bulk adjustment and (3.58) add $[4b(1/2) + 2 + 2\pi^{-2} \log(5/2)]L$, giving A_{band} . The phase comparison adds $4b(1/2)L$. $\square$

Corollary 3.3 (Schatten norms and a sharp coefficient bound). *For $p \geq 1$ put*

$$\varkappa_p = \varkappa(|x|^p) = \frac{1}{\pi^2} \int_0^1 \frac{r_t^p - 1}{t(1 - t)} dt < 0.$$

Then $\|\mathcal{K}_\tau\|_p^p = 4\tau + \varkappa_p \log \tau + O_p(1)$ as $\tau \rightarrow \infty$, and $p \mapsto \varkappa_p$ is strictly decreasing and convex. For $p = 1$ the remainder has modulus at most A_{band} for $\tau \geq 2$. The coefficients at $p = 1$ and $p = 3$ are

$$\begin{aligned} \varkappa_1 &= -\frac{2}{\pi^2} [\sqrt{2} \log(1 + \sqrt{2}) - \log 2] = -0.112122\dots, \\ \varkappa_3 &= -\frac{2}{\pi^2} \left[\frac{1}{2} - \log 2 + \frac{5}{2\sqrt{2}} \log(1 + \sqrt{2}) \right] = -0.276589\dots, \end{aligned}$$

and $\varkappa_{2m}$ is given by (3.2). Moreover, for every Lipschitz f ,

$$|\varkappa(f)| \leq |\varkappa_1| \mathrm{Lip}(f), \tag{3.4}$$

with equality for $f(x) = |x|$. This sharpens the crude bound (3.58) by the factor $2/(\pi^2 |\varkappa_1|) = 1.8073$.

Proof. For $p \geq 1$ the function $|x|^p$ is Lipschitz on $[-1, 1]$ and smooth away from the origin, so Theorem 3.2 applies and gives the expansion with $f(1) + f(-1) = 2$. The displayed integral is (1.3) for an even f . Since $r_t < 1$ off the endpoints, $p \mapsto r_t^p$ is strictly decreasing and convex, and these properties pass through the positive weight $[t(1-t)]^{-1}$. For $p = 1$, $L = 1$ and $M = 0$ in the band-local theorem. Rationalizing $r_t - 1$ in (1.3) and setting $u = 2t - 1$, the integrand being even in u , gives

$$\kappa(|x|) = -\frac{2}{\pi^2} \int_0^1 \frac{du}{1 + \sqrt{(1+u^2)/2}}.$$

For $u = \sinh v$ and $V = \log(1 + \sqrt{2})$ the integral equals $\sqrt{2}V - 2 \int_0^V (\sqrt{2} + \cosh v)^{-1} dv$. Substitution $z = \tanh(v/2)$ evaluates the last integral as $2 \operatorname{artanh}((\sqrt{2} - 1)^2) = \frac{1}{2} \log 2$.

For $p = 3$ the same substitution $u = 2t - 1$, with $r = \sqrt{(1+u^2)/2}$ and $r^3 - 1 = (r-1)(1+r+r^2)$, gives

$$\kappa_3 = -\frac{2}{\pi^2} \int_0^1 \frac{1+r+r^2}{1+r} du,$$

which is the stated closed form, since $(1+r+r^2)/(1+r) = r + (1+r)^{-1}$ and $\int_0^1 r du = \frac{1}{2} + \log(1 + \sqrt{2})/(2\sqrt{2})$. For (3.4) bound each of $|f(\pm r_t) - f(\pm 1)|$ by $\operatorname{Lip}(f)(1 - r_t)$ in (1.3) and compare with the same integral for $|x|$, which realizes both bounds with equality. $\square$

The explicit logarithmic coefficients proved in this paper are collected in Table 1.

statistic	test f	bulk	$\kappa(f)$
trace norm	$ x $	4τ	-0.112122
squared Hilbert–Schmidt norm	x^2	4τ	$-2/\pi^2 = -0.202642$
Schatten $\ \cdot\ _3^3$	$ x ^3$	4τ	-0.276589
Schatten $\ \cdot\ _4^4$	x^4	4τ	$-10/(3\pi^2) = -0.337737$
quartic defect	$x^2 - x^4$	0	$4/(3\pi^2) = +0.135094$
$\log \det(I + \mathcal{K}_{\tau,\phi}^2)$	$\log(1 + x^2)$	$4\tau \log 2$	$-1/9$

Table 1: Bulk and logarithmic coefficients of Theorem 1.1. Every entry is exact; decimals truncate the closed forms of Corollary 3.3, Proposition 3.1 and Remark 3.11. The last row concerns the phase-shifted operator $\mathcal{K}_{\tau,\phi}$ of Proposition 3.30, for which the same law holds uniformly in ϕ .

3.1.2 Gap counts and transition quantiles

Let $\mu_j^\pm(\tau)$ be the positive eigenvalues and the magnitudes of negative eigenvalues, respectively, in nonincreasing order with multiplicity. Both lists are infinite. They include both reflection sectors. Set $N_\pm(a; \tau) = \#\{j : \mu_j^\pm(\tau) > a\}$ and $C_0 = 2C_V + 4b(1/2)$.

Proposition 3.4 (Gap mass and the count plateau). *For $w_g(x) = \min\{|x|, (g - |x|)_+\}$ and $\tau \geq 2$,*

$$\sum_j w_g(\lambda_j(\mathcal{K}_\tau)) \leq C_0, \quad \sum_{|\lambda_j| \leq a} |\lambda_j| \leq C_0 \max\{1, a/(g-a)\} \quad (0 < a < g).$$

For either sign and $0 < a < g$,

$$2\tau - 1 - \frac{C_0}{g-a} \leq N_\pm(a; \tau) \leq 2\tau + 1 + \frac{C_0}{a}. \quad (3.5)$$

Proof. For a nearest integer q to τ put $n = 2q$ and $M_n = T_n(a(\cdot))$. The star has n eigenvalues of each sign in the outer bands and all others zero. Equations (3.31) and (3.36) give $|\operatorname{tr} F(\mathcal{K}_\tau) - \operatorname{tr} F(M_n)| \leq C_0 \operatorname{Lip}(F)$ for every real Lipschitz F with $F(0) = 0$. Apply this first to the

nonnegative 1-Lipschitz function w_g , which vanishes on the star spectrum. The second mass bound follows from $w_g(x) \geq \min\{1, (g-a)/a\}|x|$ for $|x| \leq a$. For positive counts the ramps $F_a^+(x) = \min\{1, x_+/a\}$ and $F_a^-(x) = \min\{1, (x-a)_+/(g-a)\}$ bracket $\mathbf{1}_{(a,1]}$, have Lipschitz constants $1/a, 1/(g-a)$, and both have star trace n . Reflection gives the negative count; $|n - 2\tau| \leq 1$ finishes the proof. $\square$

Theorem 3.5 (Cumulative transition law). *For $g < a < 1$ put*

$$J(a) = \int_g^a \varpi(x) dx = \frac{2}{\pi^2} \operatorname{artanh} \sqrt{2a^2 - 1}. \quad (3.6)$$

As $\tau \rightarrow \infty$, each sign satisfies

$$N_{\pm}(a; \tau) = 2\tau - J(a) \log \tau + O((\log \tau)^{2/3}), \quad (3.7)$$

uniformly for a in every fixed compact subset of $(g, 1)$.

Proof. Fix $[a_0, a_1] \subset (g, 1)$ and $\delta_0 = \min\{(a_0 - g)/2, (1 - a_1)/2, 1\}$. The quintic step $S(t) = 10t^3 - 15t^4 + 6t^5$ on $[0, 1]$, continued by zero and one, is nondecreasing and C^2 , with derivative bounds $15/8, 15$. For $0 < \delta \leq \delta_0$ the functions $F_{a,\delta}^- = S((x-a)/\delta)$ and $F_{a,\delta}^+ = S((x-a+\delta)/\delta)$ bracket $\mathbf{1}_{(a,1]}$. Their endpoint values are 0, 1, they vanish at zero, and $\|(F^{\pm})'\|_{\infty} + \|(F^{\pm})''\|_{\infty} \leq 135/(8\delta^2)$. With $M = \max_{[a_0-\delta_0, a_1+\delta_0]} \varpi$, the density gives $|\varkappa(F_{a,\delta}^{\pm}) + J(a)| \leq M\delta$. The trace theorem therefore implies

$$|N_+(a; \tau) - 2\tau + J(a) \log \tau| \leq M\delta \log \tau + 135C_{\text{tr}}/(8\delta^2).$$

Choose $\delta = \min\{\delta_0, (\log \tau)^{-1/3}\}$. Reflection gives the other sign and integration of ϖ gives J . $\square$

Corollary 3.6 (Signed transition quantiles). *Fix $\nu \neq 0$ and put $j_{\tau} = \lfloor 2\tau - \nu \log \tau \rfloor$, which is positive for large τ . For $\nu > 0$,*

$$\mu_{j_{\tau}}^{\pm}(\tau) = \lambda(\nu) + O_{\nu}((\log \tau)^{-1/3}), \quad \lambda(\nu) = \sqrt{\frac{1 + \tanh^2(\pi^2 \nu / 2)}{2}}. \quad (3.8)$$

For $\nu < 0$, one has $\mu_{j_{\tau}}^{\pm}(\tau) = O_{\nu}((\log \tau)^{-1})$.

Proof. For $\nu > 0$, $a_{\nu} = \lambda(\nu)$ uniquely solves $J(a_{\nu}) = \nu$. On a compact neighborhood, $J' = \varpi$ is bounded below by a positive constant and the counting error is uniform. Writing $L = \log \tau$, a sufficiently large fixed D_{ν} therefore gives

$$N_{\pm}(a_{\nu} + D_{\nu}L^{-1/3}; \tau) < j_{\tau} \leq N_{\pm}(a_{\nu} - D_{\nu}L^{-1/3}; \tau).$$

Strict-count inversion gives the asserted error, including threshold multiplicities; the floor costs less than one. For $\nu < 0$ put $c = |\nu|$, $a = 2(C_0 + 1)/(cL)$. Eventually $a < g$, $cL \geq 4$, and the plateau gives $N_{\pm}(a; \tau) \leq 2\tau + 1 + cL/2 \leq 2\tau + cL - 1 < j_{\tau}$. Thus $\mu_{j_{\tau}}^{\pm} \leq a$. $\square$

No conclusion for the central index $\nu = 0$, or for separate reflection sectors, is asserted by these quantile estimates.

The profile is a fold of the classical Landau–Widom logistic law [31]: for $0 < t < 1/2$, $J(r_t) = \pi^{-2} \log((1-t)/t)$ and $\lambda(\nu) = r_{(1+e^{\pi^2 \nu})^{-1}}$. This follows from $\sqrt{2r_t^2 - 1} = |1 - 2t|$. The result is the stated counting and quantile error for both signed physical lists, which need not themselves be sign-symmetric.

Corollary 3.7 (Gap counts and signed transition counts). *Eigenvalues below are counted with multiplicity. For fixed $0 < a \leq b < 1/\sqrt{2}$,*

$$\#\{j : a \leq |\lambda_j(\mathcal{K}_\tau)| \leq b\} \leq \frac{C_0}{\min\{a, 1/\sqrt{2} - b\}} \quad (\tau \geq 2). \quad (3.9)$$

For fixed $1/\sqrt{2} < a < b < 1$, each sign separately satisfies

$$\#\{j : \lambda_j(\mathcal{K}_\tau) \in [a, b]\} = I_{a,b} \log \tau + O_{a,b}((\log \tau)^{2/3}), \quad (3.10)$$

and the same formula holds for $[-b, -a]$, where $I_{a,b} = \int_a^b \varpi(x) dx$ and ϖ is given in Proposition 3.1. With $v(x) = \sqrt{2x^2 - 1}$,

$$I_{a,b} = \kappa(\mathbf{1}_{[a,b]}) = \frac{1}{\pi^2} \log \frac{(1+v(b))(1-v(a))}{(1-v(b))(1+v(a))}, \quad I_{0.72, 0.995} = 0.4967058 \dots \quad (3.11)$$

Here the indicator notation denotes the convergent density integral; the trace theorem is applied only to smooth comparison functions.

Proof. On $a \leq |x| \leq b$ the weight of Proposition 3.4 is at least $\min\{a, 1/\sqrt{2} - b\}$, proving the gap count.

For the transition interval, put $L = \log \tau$ and $\delta = L^{-1/3}$. For sufficiently large τ , all thresholds below lie in a fixed compact subset of $(1/\sqrt{2}, 1)$. The strict counts obey

$$N_+(a; \tau) - N_+(b; \tau) \leq \#\{j : \lambda_j \in [a, b]\} \leq N_+(a - \delta; \tau) - N_+(b; \tau).$$

Theorem 3.5, uniform on that compact set, and $J' = \varpi$ bound both sides by $[J(b) - J(a)]L + O(L^{2/3})$, since $[J(a) - J(a - \delta)]L = O(\delta L)$. The same argument for N_- proves the negative interval formula. This bracketing also covers arbitrary multiplicity at both endpoints. Finally, $d[\log((1+v(x))/(1-v(x)))]/dx = \pi^2 \varpi(x)$ proves (3.11). $\square$

These results concern fixed compact windows. They do not assert an empty gap, bounded counts on the whole punctured gap, an $O(1)$ remainder for sharp transition counts, or uniformity when the endpoints approach 0, $1/\sqrt{2}$ or 1. The density integral diverges as $b \uparrow 1$. The exponent $2/3$ comes from this C^2 smoothing estimate; no optimality is claimed.

3.1.3 The regularity obstruction

Proposition 3.8 (Obstruction to an all-Hölder bounded remainder). *For every $0 < \alpha < 1$ and every closed interval $I = [a, b] \subset (1/\sqrt{2}, 1)$ with $a < b$, there is a real $f \in C^{0,\alpha}([-1, 1])$ supported in I for which*

$$\sup_{n \geq 2} |\operatorname{tr} f(\mathcal{K}_n) - \kappa(f) \log n| = \infty.$$

Consequently (1.4) cannot hold with an $O_f(1)$ remainder for every Hölder test function vanishing at zero.

Proof. Let X be the Banach space of real $C^{0,\alpha}$ functions supported in I , with norm $\|f\|_\infty + [f]_\alpha$. On X , $\kappa(f) = \int_I \varpi f$, where $\min_I \varpi > 0$. The functionals $L_n(f) = \sum_j f(\lambda_j(\mathcal{K}_n)) - (\log n) \int_I \varpi f$ are continuous: compactness leaves finitely many eigenvalues in I . Their number m_n is $O(\log n)$ by Corollary 3.7. Let Z_n contain those eigenvalues and the endpoints of I , and set $f_n(x) = \operatorname{dist}(x, Z_n)^\alpha$ on I , zero elsewhere. Since distance is 1-Lipschitz and $|r^\alpha - s^\alpha| \leq |r - s|^\alpha$, including across the endpoints, $\|f_n\|_X \leq 1 + (b - a)^\alpha$. Its spectral sum vanishes. The successive gap lengths ℓ_i have sum $b - a$ and number at most $m_n + 1$, so exact integration and convexity give

$$\int_I \varpi f_n \geq \frac{\min_I \varpi}{2^\alpha(1+\alpha)} \sum_i \ell_i^{1+\alpha} \geq \frac{\min_I \varpi (b-a)^{1+\alpha}}{2^\alpha(1+\alpha)(m_n+1)^\alpha}.$$

Thus $\|L_n\| \geq c(\log n)^{1-\alpha} \rightarrow \infty$. Uniform boundedness supplies one fixed $f \in X$ with unbounded remainder. $\square$

This obstruction concerns the combination of the entire Hölder class and an $O_f(1)$ remainder. It is compatible with the C^2 theorem, its band-local extension, and with the functional (1.3), which is defined on the larger class. The obstruction lies inside an outer transition band, where Theorem 3.2 still imposes derivative regularity. The counting results use admissible smooth or Lipschitz comparison functions.

The remainder cannot be replaced by $c(f) + o(1)$ as $\tau \rightarrow \infty$ through real values: already for $f(x) = x^2$ it contains the term $-2\pi^{-2}(\log 2) \cos(4\pi\tau)$ of Theorem 3.15, which does not converge. Along the integer ladder $\tau = n$ of Proposition 3.8 that term is the constant $-2\pi^{-2} \log 2$.

3.2 Resolvent and determinant coefficients

Proposition 3.9 (The resolvent). *For real z with $|z| > 1$,*

$$\kappa\left(\frac{\lambda}{\lambda - z}\right) = \frac{4z^2}{\pi^2(z^2 - 1)\sqrt{2z^2 - 1}} \operatorname{artanh} \frac{1}{\sqrt{2z^2 - 1}}. \quad (3.12)$$

Proof. Since κ annihilates constants, $\kappa(\lambda/(\lambda - z)) = z\kappa(1/(\lambda - z))$. Set $B = 2z^2 - 1 > 1$ and $v = 2\lambda^2 - 1$ in (3.1). Then $\pi^2 \varpi(\lambda) d\lambda = dv/[(1 - v)\sqrt{v}]$ and

$$\frac{1}{\lambda - z} + \frac{1}{-\lambda - z} - \frac{1}{1 - z} - \frac{1}{-1 - z} = \frac{4z(1 - v)}{(B - v)(B - 1)}.$$

The endpoint factor cancels, so

$$\pi^2 \kappa\left(\frac{1}{\lambda - z}\right) = \frac{4z}{B - 1} \int_0^1 \frac{dv}{(B - v)\sqrt{v}} = \frac{8z}{(B - 1)\sqrt{B}} \operatorname{artanh}(B^{-1/2}).$$

Multiply by z and use $B - 1 = 2(z^2 - 1)$. □

Since $\mathcal{K}_\tau(\mathcal{K}_\tau - z)^{-1}$ is trace class for $|z| > 1$ by Proposition 2.2, (3.12) specializes Theorem 1.1 to a resolvent trace formula: for each fixed real z with $|z| > 1$ the theorem gives $\operatorname{tr}[\mathcal{K}_\tau(\mathcal{K}_\tau - z)^{-1}] = 4\tau(1 - z^2)^{-1} + \kappa(\lambda/(\lambda - z)) \log \tau + O(1)$. Expanding (3.12) at $z = \infty$ returns $2\pi^{-2}z^{-2} + O(z^{-4})$, consistent with $\pi^2 \kappa(\lambda^2) = -2$; the two branch points at $|z| = 2^{-1/2}$ and $|z| = 1$ are the two edges of the support of ϖ .

Corollary 3.10 (A Fredholm determinant law). *For real $|z| < 1$ and the phase-shifted operator $\mathcal{K}_{\tau,\phi}$, with kernel $[\sin(2\pi\tau x + \phi) - \sin(2\pi\tau y + \phi)]/[\pi(x - y)]$ and treated in Proposition 3.30, as $\tau \rightarrow \infty$,*

$$\log \det(I - z\mathcal{K}_{\tau,\phi}) = 2\tau \log(1 - z^2) + \frac{2}{\pi^2} \operatorname{artanh}^2\left(\frac{z}{\sqrt{2 - z^2}}\right) \log \tau + O_z(1). \quad (3.13)$$

The remainder is uniform in ϕ and locally uniform in $z \in (-1, 1)$.

Proof. For $f_z(x) = \log(1 - zx)$, all factors $1 - z\lambda_j$ are positive and $\sum_j |f_z(\lambda_j)| < \infty$. Thus the real Fredholm logarithm is $\operatorname{tr} f_z(\mathcal{K}_{\tau,\phi})$. The test vanishes at zero and its first two derivatives are uniformly bounded when z ranges over a compact subinterval of $(-1, 1)$. Let $\beta(z) = \kappa(f_z)$. For $z \neq 0$, differentiation of the convergent coefficient integral and Proposition 3.9 give

$$\beta'(z) = z^{-1} \kappa\left(\frac{x}{x - 1/z}\right) = \frac{4 \operatorname{artanh}(z/\sqrt{2 - z^2})}{\pi^2(1 - z^2)\sqrt{2 - z^2}}.$$

This is the derivative of $2\pi^{-2} \operatorname{artanh}^2(z/\sqrt{2 - z^2})$. Both functions vanish at zero, so integration identifies the coefficient; the uniform phase trace law proves (3.13). □

Remark 3.11 (The square and a comparison). For $z \in \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$, define $\log \det(I - z\mathcal{K}_{\tau,\phi})$ as the holomorphic logarithm normalized to zero at $z = 0$, equivalently the absolutely convergent sum $\sum_j \text{Log}(1 - z\lambda_j)$, where λ_j are the eigenvalues of $\mathcal{K}_{\tau,\phi}$ and Log is the scalar principal logarithm. Applying the uniform phase trace law to the real and imaginary parts of this scalar test extends Corollary 3.10 to that domain. Likewise, for $w \in \mathbb{C} \setminus (-\infty, -1]$, put $\log \det(I + w\mathcal{K}_{\tau,\phi}^2) = \sum_j \text{Log}(1 + w\lambda_j^2)$. Then

$$\log \det(I + w\mathcal{K}_{\tau,\phi}^2) = 4\tau \text{Log}(1 + w) - \frac{4}{\pi^2} \arcsin^2\left(\frac{1}{2}\sqrt{\frac{2w}{1+w}}\right) \log \tau + O_w(1). \quad (3.14)$$

The squared inverse-sine expression denotes its analytic continuation from $w > 0$ throughout the stated slit plane, with value zero at $w = 0$. For $w > 0$, add the two logarithms at $z = \pm i\sqrt{w}$ and use $\text{artanh}(iy) = i \arctan y$ and $\arctan \sqrt{w/(2+w)} = \arcsin(\frac{1}{2}\sqrt{2w/(1+w)})$. Analytic continuation of the coefficient integral then gives the formula on the slit plane; the trace-law error bound applies directly at each w and is uniform in ϕ . At $w = 1$ the coefficient is $-\frac{1}{9}$, so $\log \det(I + \mathcal{K}_{\tau,\phi}^2) = 4\tau \log 2 - \frac{1}{9} \log \tau + O(1)$.

For comparison, Gebert and Küttler's generalized Hilbert matrix has entries $(H_N^\delta)_{jk} = \sin((1 + \delta)\pi)/[\pi(j + k + 1 + \delta)]$, $0 \leq j, k < N$. For $\delta < -1/2$ outside the negative integers and half-integers, their Corollary 1.3 gives $\log \det(I - (H_N^\delta)^2) = -\delta^2 \log N + O_\delta(1)$ [20]. This is an analogous determinant law for a different signed family; their critical half-integers have a weaker remainder, as specified in their Remark 1.4.

These formulas give bounded remainders. Theorem 6.1 proves their convergence along fixed phases by localizing the physical boundary corrections. For the unsquared determinant with real $|z| < 1$, the three-dimensional model symbol $a(\theta)$ of Section 3.4 has a coefficient of $\log \tau$ equal to the exponent $-\text{tr } \mathfrak{L}^2$, $\mathfrak{L} = (2\pi i)^{-1} \log[(I - zX_{1,0})^{-1}(I - zX_{0,-1})]$, of the block Fisher–Hartwig asymptotics [8, Corollary 2.4], which give a convergent constant term for the block Toeplitz truncation of $I - za$ itself.

3.3 An explicit quadratic formula

Let $C^{(a)}$ be the positive prolate operator on $L^2(-1, 1)$ with kernel $k_a(x - y) = \sin(a(x - y))/(\pi(x - y))$, and put $C_\tau = C^{(\pi\tau)}$, $C_{2\tau} = C^{(2\pi\tau)}$. Write $D_R = \text{tr}(C_{2\tau} - C_{2\tau}^2)$, $D_I = 4\tau - \text{tr } \mathcal{K}_\tau^2$, γ for Euler's constant, Si as in (2.5), and $\text{Ci}(x) = -\int_x^\infty \cos(t)/t dt$. The following independent calculation determines a persistent oscillation that the general bounded-remainder theorem does not specify.

Lemma 3.12 (Exact defect trace of the sinc kernel). *For every $a > 0$,*

$$\text{tr}(C^{(a)} - (C^{(a)})^2) = \frac{1}{\pi^2} [\log(4a) + 1 + \gamma - \text{Ci}(4a) - 2\mathfrak{J}(a)], \quad (3.15)$$

where $\mathfrak{J}(a) = \int_2^\infty \cos(2au) u^{-2} du = \frac{1}{2} \cos(4a) - 2a(\frac{\pi}{2} - \text{Si}(4a))$, with $|\text{Ci}(4a)| \leq 1/(2a)$ and $|\mathfrak{J}(a)| \leq 1/(4a)$. Consequently

$$\left| \text{tr}(C^{(a)} - (C^{(a)})^2) - \frac{\log(4a) + 1 + \gamma}{\pi^2} \right| \leq \frac{1}{\pi^2 a}, \quad D_R = \frac{\log(8\pi\tau) + 1 + \gamma}{\pi^2} + O(\tau^{-1}). \quad (3.16)$$

Proof. The displacement variable and Plancherel give $\text{tr}(C^{(a)})^2 = \int_{-2}^2 (2 - |u|) k_a(u)^2 du$ and $\text{tr } C^{(a)} = 4 \int_0^\infty k_a(u)^2 du = 2a/\pi$. Hence

$$\text{tr}(C^{(a)} - (C^{(a)})^2) = 4 \int_2^\infty k_a(u)^2 du + 2 \int_0^2 u k_a(u)^2 du = \frac{1 - 2\mathfrak{J}(a) + \gamma + \log(4a) - \text{Ci}(4a)}{\pi^2},$$

using $2\sin^2(au) = 1 - \cos(2au)$ and $\int_0^x (1 - \cos t)t^{-1} dt = \gamma + \log x - \text{Ci}(x)$. Integration by parts in the two directions gives

$$\begin{aligned}\text{Ci}(x) &= \frac{\sin x}{x} - \int_x^\infty \frac{\sin t}{t^2} dt, \\ \mathfrak{J}(a) &= -\frac{\sin(4a)}{8a} + \frac{1}{a} \int_2^\infty \frac{\sin(2au)}{u^3} du = \frac{\cos(4a)}{2} - 2a\left(\frac{\pi}{2} - \text{Si}(4a)\right).\end{aligned}$$

Thus $|\text{Ci}(4a)| \leq 1/(2a)$ and $|\mathfrak{J}(a)| \leq 1/(4a)$, proving all the stated bounds. $\square$

Remark 3.13. The leading $1/a$ contributions from $-\text{Ci}(4a)$ and $-2\mathfrak{J}(a)$ cancel. Further integration by parts gives the remainder $-\cos(4a)/(16\pi^2 a^2) + O(a^{-3})$ in (3.16). We use the elementary uniform bound $1/(\pi^2 a)$ below.

The exact formula is the classical sine-process number variance [17, 35]; see also [5, Theorem 4.3]. The lemma records its normalization and explicit remainder for use below. The logarithmic coefficient is consistent with Landau–Widom [31], and the lattice analogue with $O(L^{-2})$ error is in [1, eq. (D.6)].

Lemma 3.14. *One has*

$$\int_0^\infty \frac{\sin(2w)\sin^2 w}{w^2} dw = \log 2, \quad \int_0^\infty \frac{\cos(2w)\sin^2 w}{w^2} dw = 0,$$

and for $X > 0$ each tail $\int_X^\infty$ is at most $1/X$ in absolute value.

Proof. Since $\sin(2w)\sin^2 w = \frac{1}{2}\sin 2w - \frac{1}{4}\sin 4w = O(w^3)$ at 0, one integration by parts gives $\int_0^\infty (\frac{1}{2}\sin 2w - \frac{1}{4}\sin 4w)w^{-2} dw = \int_0^\infty (\cos 2w - \cos 4w)w^{-1} dw = \log 2$ by Frullani's formula. Likewise $\cos(2w)\sin^2 w = \frac{1}{4}(2\cos 2w - 1 - \cos 4w) = O(w^2)$ at 0, and $\int_0^\infty \frac{1}{4}(2\cos 2w - 1 - \cos 4w)w^{-2} dw = \int_0^\infty (-\sin 2w + \sin 4w)w^{-1} dw = -\frac{\pi}{2} + \frac{\pi}{2} = 0$. The tails are bounded by $\int_X^\infty w^{-2} dw$. $\square$

Theorem 3.15 (Hilbert–Schmidt norm of the critical kernel). *For every $\tau > 0$,*

$$\text{tr } \mathcal{K}_\tau^2 = 4\tau - \frac{2}{\pi^2} \left[\log(4\pi\tau) + 1 + \gamma + (\log 2) \cos(4\pi\tau) \right] + R_1(\tau), \quad |R_1(\tau)| \leq \frac{4}{\pi^3 \tau}. \quad (3.17)$$

Equivalently $D_J = 2 \text{tr}(C^{(\pi\tau)} - (C^{(\pi\tau)})^2) + 2\pi^{-2}(\log 2) \cos(4\pi\tau) + O(\tau^{-1})$, and

$$D_J - 2D_R = -\frac{2\log 2}{\pi^2}(1 - \cos 4\pi\tau) + O(\tau^{-1}). \quad (3.18)$$

Proof. By the product-to-sum formula, $\mathcal{K}_\tau(x, y) = 2\cos(\pi\tau(x+y))\sin(\pi\tau(x-y))/(\pi(x-y))$, so with $u = x - y$, $v = x + y$ and $s(u) = \sin^2(\pi\tau u)/(\pi^2 u^2) = k_{\pi\tau}(u)^2$,

$$\mathcal{K}_\tau(x, y)^2 = 4\cos^2(\pi\tau v)s(u).$$

The map $(x, y) \mapsto (u, v)$ satisfies $dx dy = \frac{1}{2} du dv$ and carries $[-1, 1]^2$ onto $\{|u| \leq 2, |v| \leq L\}$, $L = 2 - |u|$. Since $\int_{-L}^L 4\cos^2(\pi\tau v) dv = 4L + 2\sin(2\pi\tau L)/(\pi\tau)$,

$$\text{tr } \mathcal{K}_\tau^2 = 2 \int_{-2}^2 (2 - |u|) s(u) du + \frac{1}{\pi\tau} \int_{-2}^2 \sin(2\pi\tau(2 - |u|)) s(u) du. \quad (3.19)$$

The first term is $2 \text{tr}(C^{(\pi\tau)})^2 = 2(2\tau - \text{tr}(C^{(\pi\tau)} - (C^{(\pi\tau)})^2))$. In the second, expand $\sin(4\pi\tau - 2\pi\tau u) = \sin(4\pi\tau)\cos(2\pi\tau u) - \cos(4\pi\tau)\sin(2\pi\tau u)$ and substitute $w = \pi\tau u$:

$$\begin{aligned}\int_0^2 \sin(2\pi\tau u) s(u) du &= \frac{\tau}{\pi} \int_0^{2\pi\tau} \frac{\sin(2w)\sin^2 w}{w^2} dw = \frac{\tau}{\pi}(\log 2 - \rho_s), \\ \int_0^2 \cos(2\pi\tau u) s(u) du &= -\frac{\tau}{\pi} \rho_c,\end{aligned}$$

with $|\rho_s|, |\rho_c| \leq (2\pi\tau)^{-1}$ by Lemma 3.14. Hence the second term of (3.19) equals

$$-\frac{2\log 2}{\pi^2} \cos(4\pi\tau) + \frac{2}{\pi^2} (\rho_s \cos 4\pi\tau - \rho_c \sin 4\pi\tau),$$

whose remainder is at most $2/(\pi^3\tau)$. Lemma 3.12 with $a = \pi\tau$ contributes a further remainder of at most $2/(\pi^3\tau)$, and (3.17) follows. For (3.18) subtract twice (3.16) at $a = 2\pi\tau$ and use $\log(4\pi\tau) - \log(8\pi\tau) = -\log 2$. $\square$

3.3.1 The reflected second moment

For $0 < \ell < 1$, write $C_N(\ell)$ for the matrix with entries $c_\ell(m-n) = \sin(\pi\ell(m-n))/[\pi(m-n)]$ and diagonal ℓ .

Proposition 3.16 (Parity-resolved second moment). *For $\tau \geq 1$,*

$$\left| \text{tr}(\mathcal{R}\mathcal{K}_\tau^2) - \frac{1}{2} \right| \leq \frac{4\log(2\pi\tau) + 2 + \pi}{\pi^3\tau}. \quad (3.20)$$

Consequently, by Theorem 3.15, for $p = \pm 1$,

$$\begin{aligned} \text{tr}(\mathcal{K}_\tau^2|_{\mathcal{H}_p}) &= 2\tau - \frac{1}{\pi^2} [\log(4\pi\tau) + 1 + \gamma + (\log 2) \cos(4\pi\tau)] \\ &\quad + \frac{p}{4} + O\left(\frac{\log \tau}{\tau}\right). \end{aligned} \quad (3.21)$$

The two sectors carry the same bulk 2τ and the same logarithmic coefficient $-\pi^{-2}$, and separate in the limit only through the constant $p/4$. In the discrete frame the reflected second moment agrees with that of the prolate matrix $C_N(2\varepsilon)$:

$$\text{tr}(\mathcal{R}_N A_N(\varepsilon)^2) = \text{tr}(\mathcal{R}_N C_N(2\varepsilon)^2) = \sum_{|m|, |n| \leq N} c_{2\varepsilon}(m-n) c_{2\varepsilon}(m+n). \quad (3.22)$$

Proof. The product-to-sum identity $2\cos(\pi\tau u)k_{\pi\tau}(u) = k_{2\pi\tau}(u)$ gives the pointwise identity

$$K_\tau(-x, y)K_\tau(y, x) = k_{2\pi\tau}(x+y)k_{2\pi\tau}(x-y).$$

The trace of the product of the two Hilbert–Schmidt kernels is their diagonal integral. With $u = x - y$, $v = x + y$ and Jacobian $1/2$,

$$\text{tr}(\mathcal{R}\mathcal{K}_\tau^2) = \frac{1}{2} \iint_{|u|+|v| \leq 2} k_{2\pi\tau}(u)k_{2\pi\tau}(v) du dv.$$

Integrate in v and substitute $s = 2\pi\tau u$. For $T = 4\pi\tau$ this gives

$$\text{tr}(\mathcal{R}\mathcal{K}_\tau^2) = \frac{2}{\pi^2} G(T), \quad G(T) = \int_0^T \frac{\sin s}{s} \text{Si}(T-s) ds. \quad (3.23)$$

Write $G(T) - \frac{\pi^2}{4} = \int_0^T \frac{\sin s}{s} [\text{Si}(T-s) - \frac{\pi}{2}] ds + \frac{\pi}{2} [\text{Si}(T) - \frac{\pi}{2}]$ and use $|\text{Si}(x) - \pi/2| \leq \min\{\pi/2, 2/x\}$. The last term is at most π/T . In the integral, bound $T-s \geq T/2$ on $[0, T/2]$ and use $\int_0^X |\sin s| s^{-1} ds \leq 1 + \log X$ for $X \geq 1$; bound $s^{-1} \leq 2/T$ on $[T/2, T]$ and split that range at $T-1$, using $2/(T-s)$ on $[T/2, T-1]$ and $\pi/2$ on $[T-1, T]$. The three pieces contribute at most $4(1 + \log(T/2))/T$, $4\log(T/2)/T$ and π/T . Adding and multiplying by $2/\pi^2$ gives (3.20), since $T = 4\pi\tau \geq 2$. Then (3.21) follows from $\text{tr}(\mathcal{K}_\tau^2|_{\mathcal{H}_p}) = \frac{1}{2}[\text{tr} \mathcal{K}_\tau^2 + p \text{tr}(\mathcal{R}\mathcal{K}_\tau^2)]$ and (3.17).

The identical pointwise calculation on integer nodes, with τ replaced by ε , gives $A_{-m,n}A_{nm} = c_{2\varepsilon}(m+n)c_{2\varepsilon}(m-n) = C_{-m,n}C_{nm}$, including the removable diagonal cases. Summation proves (3.22). $\square$

Corollary 3.17 (Leading oscillation of the parity imbalance). *For $\tau \geq 1$,*

$$\begin{aligned} \operatorname{tr}(\mathcal{R}\mathcal{K}_\tau^2) &= \frac{1}{2} - \frac{[\log(8\pi\tau) + \gamma] \sin(4\pi\tau)}{2\pi^3\tau} - \frac{\cos(4\pi\tau)}{4\pi^2\tau} + E_{\text{par}}(\tau), \\ |E_{\text{par}}(\tau)| &\leq \frac{2\log(8\pi\tau) + 2\gamma + \pi - 5/2}{8\pi^4\tau^2}. \end{aligned} \quad (3.24)$$

In particular, the order $O(\log \tau / \tau)$ in (3.20) is sharp along suitable noninteger scales.

Proof. Put $T = 4\pi\tau$, $a(T) = \log(2T) + \gamma$, and $B = \pi/2$. Differentiating the exact convolution in (3.23) gives $G'(T) = \int_0^T \sin s \sin(T-s) / [s(T-s)] ds$. Symmetry, the identity $[s(T-s)]^{-1} = T^{-1}[s^{-1} + (T-s)^{-1}]$, and product-to-sum yield

$$G'(T) = \frac{\cos T [\operatorname{Ci}(2T) - \gamma - \log(2T)] + \sin T \operatorname{Si}(2T)}{T}, \quad \operatorname{Ci}(y) = - \int_y^\infty \frac{\cos t}{t} dt.$$

Here $\operatorname{Ci}(y) - \gamma - \log y = \int_0^y (\cos t - 1)t^{-1} dt$. Two integrations by parts give, for $y > 0$,

$$\begin{aligned} \operatorname{Ci}(y) &= \frac{\sin y}{y} - \frac{\cos y}{y^2} + 2 \int_y^\infty \frac{\cos t}{t^3} dt, \\ \operatorname{Si}(y) - B &= -\frac{\cos y}{y} - \frac{\sin y}{y^2} + 2 \int_y^\infty \frac{\sin t}{t^3} dt. \end{aligned}$$

Thus the errors after their first displayed terms are at most $2/y^2$. Substitution, with $\cos T \sin(2T) - \sin T \cos(2T) = \sin T$, gives

$$G'(T) = \frac{-a(T) \cos T + B \sin T}{T} + \frac{\sin T}{2T^2} + e(T), \quad |e(T)| \leq T^{-3}.$$

Set $F(T) = -[a(T) \sin T + B \cos T]/T$ and $H(T) = G(T) - \pi^2/4 - F(T)$. Direct differentiation gives

$$H'(T) = \frac{[3/2 - a(T)] \sin T - B \cos T}{T^2} + e(T).$$

For $T \geq 3$, $b(T) = [a(T) - 3/2]/T^2$ is positive and decreasing to zero: $b'(T) = [4 - 2a(T)]/T^3 < 0$. For any positive decreasing differentiable weight b tending to zero, integration by parts gives $|\int_T^\infty b(t)e^{it} dt| \leq 2b(T)$. Apply this to b and to B/T^2 ; also $\int_T^\infty |e(t)| dt \leq 1/(2T^2)$. The already proved estimate (3.20) implies $G(T) \rightarrow \pi^2/4$, while $F(T) \rightarrow 0$; hence $H(T) \rightarrow 0$. Integrating H' from T to infinity therefore proves

$$|H(T)| \leq \frac{2a(T) - 3 + \pi + 1/2}{T^2}.$$

Multiply by $2/\pi^2$ in (3.23) and substitute $T = 4\pi\tau$ to obtain (3.24). Along $\tau = n/2 + 1/8$, the sine equals one and the cosine vanishes, so the difference from $1/2$ is asymptotic to $-\log \tau / (2\pi^3\tau)$, proving sharpness of the order. $\square$

3.4 Proof of the uniform trace law

The proof combines a full-fiber unitary transport, scalar Hilbert-square corners and a Carleman–Mellin comparison. The classical comparison methods are in Otte [36, Lemmas 3.1–3.3], Fedele–Gebert [18, proof of Corollary 1.5] and Laptev–Safarov [32, Theorem 1.2]; all estimates used here are proved below.

We use Section 2.1.1 and Proposition 2.4: dilation identifies $\mathcal{K}_\tau$ with $P_{(-\tau, \tau)} \mathcal{G} P_{(-\tau, \tau)}$, where P_I denotes multiplication by $\mathbf{1}_I$ and

$$G(X, Y) = \frac{\sin(2\pi X) - \sin(2\pi Y)}{\pi(X - Y)}, \quad \mathcal{G} = \mathcal{G}^*, \quad \|\mathcal{G}\| = 1, \quad \mathcal{G}^2 = \mathcal{F}^* \mathbf{1}_{(-1, 1)} \mathcal{F}. \quad (3.25)$$

The global Fourier transform $\mathcal{F}$ retains the positive exponent of Section 2.1.1. Bloch coefficients below use a negative exponent, as in (2.9); the angular Fourier transform used after the Mellin change of variables will be defined separately. All Schatten norms are denoted by $\|\cdot\|_p$; in particular, $\mathfrak{S}_1$ is the trace class. Operators compressed to a subspace are extended by zero when they are compared on a common ambient space.

3.4.1 Trace comparisons and bounded endpoint changes

Lemma 3.18 (Scalar trace perturbation). *Let A, B be self-adjoint trace-class operators with spectra in a compact interval containing zero. If f is real and Lipschitz on that interval, with $f(0) = 0$, then*

$$|\operatorname{tr} f(A) - \operatorname{tr} f(B)| \leq \operatorname{Lip}(f) \|A - B\|_1. \quad (3.26)$$

Proof. For matrices, differentiate a polynomial along $A_t = A + t(B - A)$: $\frac{d}{dt} \operatorname{tr} p(A_t) = \operatorname{tr}(p'(A_t)(B - A))$. Integration gives the bound with $\|p'\|_\infty$. Uniform polynomial approximation of a continuous derivative, then convolution of a Lipschitz extension, gives every Lipschitz f without increasing its Lipschitz constant. For trace-class operators apply the matrix result on the sum of their finite spectral subspaces and pass to the limit. Both operator tails and scalar trace tails converge because $|f(\lambda)| \leq \operatorname{Lip}(f)|\lambda|$. This is a scalar trace inequality, not an operator-Lipschitz assertion. $\square$

Lemma 3.19 (Fixed strips). *For an interval E of length ℓ , put*

$$b(\ell) = \sqrt{2\ell} \int_0^1 e^{\pi \ell u^2} du. \quad (3.27)$$

Then $\|P_E \mathcal{G}\|_1 \leq b(\ell) \leq \sqrt{2\ell} e^{\pi \ell}$. If $I \triangle J$ consists of at most two intervals, each of length at most h , then

$$\|P_I \mathcal{G} P_I - P_J \mathcal{G} P_J\|_1 \leq 4b(h). \quad (3.28)$$

In particular, writing $A_I = P_I \mathcal{G} P_I$, every interval $I \subseteq \mathbb{R}$ and every $a_1, a_2 \in \mathbb{R}$ satisfy

$$\|A_{I+a_1} - A_{I+a_2}\|_1 \leq 4b(|a_1 - a_2|). \quad (3.29)$$

Proof. The restricted Fourier operator $R_E : L^2(-1, 1) \rightarrow L^2(E)$ with kernel $e^{2\pi i X \xi}$ satisfies $R_E R_E^* = P_E \mathcal{G}^2 P_E = (P_E \mathcal{G})(P_E \mathcal{G})^*$, by the symmetry of the frequency interval in (3.25). It has the same nonzero singular values as $P_E \mathcal{G}$. Translation of E gives a unitary factor in the frequency variable, so assume $E = [-\ell/2, \ell/2]$. Expand its kernel as $\sum_{j \geq 0} (2\pi i)^j X^j \xi^j / j!$. The j th rank-one term has trace norm

$$\frac{(2\pi)^j}{j!} \|X^j\|_{L^2(E)} \|\xi^j\|_{L^2(-1,1)} = \sqrt{2\ell} \frac{(\pi \ell)^j}{j!(2j+1)}.$$

Their sum is $b(\ell)$; uniform kernel convergence identifies the trace-norm sum with R_E . For $D = P_I - P_J$, the identity $P_I \mathcal{G} P_I - P_J \mathcal{G} P_J = D \mathcal{G} P_I + P_J \mathcal{G} D$ bounds the difference by $2\|D \mathcal{G}\|_1$. Decompose D into its at most two signed strip projections. Monotonicity of b gives (3.28); two translates have strips of length at most $|a_1 - a_2|$, including when they are disjoint. $\square$

Corollary 3.20 (Odd tests and integer parameters). *For every odd real Lipschitz function f and every $\tau > 0$,*

$$|\operatorname{tr} f(\mathcal{K}_\tau)| \leq 2b(1/2) \operatorname{Lip}(f) \leq 2e^{\pi/2} \operatorname{Lip}(f). \quad (3.30)$$

For $\tau \geq 2$ and a nearest integer q to τ ,

$$|\operatorname{tr} f(\mathcal{K}_\tau) - \operatorname{tr} f(\mathcal{K}_q)| \leq 4b(1/2) \operatorname{Lip}(f) \quad (3.31)$$

for every real Lipschitz f with $f(0) = 0$.

Proof. The kernel obeys $G(X + 1/2, Y + 1/2) = -G(X, Y)$. Thus the compression to $I + 1/2$ is unitarily equivalent to the negative of the compression to I . Their symmetric difference has at most two components, each of length at most $1/2$. Lemmas 3.18 and 3.19, followed by division by two, prove (3.30). For (3.31), the two endpoints of $(-\tau, \tau)$ each move by at most $1/2$ when replaced by those of $(-q, q)$. $\square$

3.4.2 A periodic gauge to a three-dimensional symbol

On the Bloch fiber $\mathcal{H} = L^2(0, 1)$ set $e_j(t) = e^{2\pi i j t}$ and $W_\theta = M_{e^{2\pi i \theta t}}$. Writing $X_{j,k} = |e_j\rangle\langle e_k| + |e_k\rangle\langle e_j|$, the symbol of Proposition 2.4 is

$$\sigma(\theta) := \varsigma(\theta) = W_\theta^* X_{1,0} W_\theta, \quad 0 \leq \theta \leq 1.$$

Define

$$c_\theta = \cos(\pi\theta/2), \quad s_\theta = \sin(\pi\theta/2), \quad w_\theta = c_\theta e_1 + s_\theta e_{-1}, \quad a(\theta) = |e_0\rangle\langle w_\theta| + |w_\theta\rangle\langle e_0|. \quad (3.32)$$

Both symbols have eigenvalues $1, -1, 0$, with simple nonzero eigenvalues, and the same endpoint operators: $a(0) = \sigma(0) = X_{1,0}$ and $a(1) = \sigma(1) = X_{0,-1}$. The range of a lies in the fixed space $E = \text{span}\{e_{-1}, e_0, e_1\}$.

Lemma 3.21 (Full-fiber gauge). *There is a unitary path V_θ on $\mathcal{H}$ such that*

$$V_0 = V_1 = I, \quad V - I \in C^\infty([0, 1]; \mathfrak{S}_1), \quad V_\theta^* \sigma(\theta) V_\theta = a(\theta). \quad (3.33)$$

Its periodic extension is continuous; matching derivatives at the two ends is not required.

Proof. For either $b = \sigma$ or $b = a$, form its three spectral projections

$$P_+ = (b^2 + b)/2, \quad P_- = (b^2 - b)/2, \quad P_0 = I - b^2.$$

The rank-one projections $P_\pm$ are smooth in trace norm, by their Riesz formulas and the rank-two bound for their differences; derivatives of P_0 are minus their sums. Thus $K_b = \sum_{j \in \{+, -, 0\}} P_j' P_j$ is smooth in trace norm and skew-adjoint. The projection identities give $[P_j, K_b] = -P_j'$, so $U_b' = K_b U_b$, $U_b(0) = I$, has a unitary solution satisfying

$$U_b(\theta)^* P_j(\theta) U_b(\theta) = P_j(0), \quad U_b(\theta) - I = \int_0^\theta K_b(t) U_b(t) dt.$$

The Bochner integral and differential equation make $U_b - I$ smooth in trace norm. Put $S = U_\sigma$, $T = U_a$, $M = T_1^* S_1$. Endpoint agreement makes M commute with $X_{1,0}$, and $M - I$ is trace class. Compact-normal spectral calculus gives a commuting skew-adjoint logarithm L : choose eigenangles $\phi \in (-\pi, \pi]$, with zero on the eigenspace of one, and use $|\phi| \leq (\pi/2)|e^{i\phi} - 1|$ for trace summability. Then $V_\theta = S_\theta e^{-\theta L} T_\theta^*$ conjugates $a(\theta) = T_\theta X_{1,0} T_\theta^*$ to $\sigma(\theta)$ and has $V_0 = V_1 = I$. The construction includes the infinite zero eigenspace. $\square$

For a bounded operator-valued symbol b , use the conventions

$$\widehat{b}(k) = \int_0^1 b(\theta) e^{-2\pi i k \theta} d\theta, \quad (\mathcal{L}(b)x)_j = \sum_l \widehat{b}(j-l)x_l, \quad T_n(b) = P_n \mathcal{L}(b) P_n, \quad (3.34)$$

where P_n selects cells $0, \dots, n-1$ and acts as identity on each fiber. Laurent multiplication is multiplicative and preserves adjoints.

Lemma 3.22 (A continuous seam). *Suppose $F \in C^3([0, 1]; \mathfrak{S}_1)$ and $F(0) = F(1)$. Define*

$$\mathfrak{c}(F) = \frac{\|F'(1) - F'(0)\|_1}{8} + \frac{\|F''(1) - F''(0)\|_1 + \int_0^1 \|F'''(\theta)\|_1 d\theta}{12\pi}. \quad (3.35)$$

Then $\|[P_n, \mathcal{L}(F)]\|_1 \leq \mathfrak{c}(F)$ for every n . The same assertion applies to scalar symbols, using absolute values in (3.35).

Proof. For $k \neq 0$, three integrations by parts give

$$\hat{F}(k) = \frac{D}{4\pi^2 k^2} + R_k, \quad D = F'(1) - F'(0), \quad R_k = \frac{-D_2 + \int_0^1 F'''(\theta) e^{-2\pi i k \theta} d\theta}{(2\pi i k)^3},$$

where $D_2 = F''(1) - F''(0)$. Thus

$$\sum_{k \neq 0} |k| \|R_k\|_1 \leq \frac{\|D_2\|_1 + \int_0^1 \|F'''(\theta)\|_1 d\theta}{24\pi}.$$

For the principal term use the positive Hankel matrix

$$H_{jl}^{(2)} = (j + l + 1)^{-2} = \int_0^1 t^{j+l} (-\log t) dt, \quad \|H^{(2)}\|_1 = \sum_{j \geq 0} (2j + 1)^{-2} = \pi^2/8.$$

The rank-one integral proves positivity and finite trace. Each end block of the Laurent sequence k^{-2} compresses $H^{(2)}$, with row reversal at the right end. The commutator therefore has norm at most $4\|H^{(2)}\|_1$, giving $\|D\|_1/8$ after tensoring. For the remainder use the bilateral shift S and $\|[P_n, S^k \otimes R_k]\|_1 = 2 \min(n, |k|) \|R_k\|_1$. Summation proves (3.35); the zeroth coefficient commutes with P_n and the Fourier series converges in operator norm. $\square$

Apply the lemma to $F = V - I$ and put $C_V = \mathfrak{c}(V - I)$. The Laurent unitary $\mathcal{V} = \mathcal{L}(V)$ satisfies $\mathcal{L}(a) = \mathcal{V}^* \mathcal{L}(\sigma) \mathcal{V}$. The exact identity, with $B = \mathcal{L}(\sigma)$,

$$\mathcal{V}^* P_n B P_n \mathcal{V} - P_n \mathcal{V}^* B \mathcal{V} P_n = (\mathcal{V}^* P_n - P_n \mathcal{V}^*) B P_n \mathcal{V} + P_n \mathcal{V}^* B (P_n \mathcal{V} - \mathcal{V} P_n)$$

therefore has trace norm at most $2C_V$, since $\|B\| = 1$. Both compressions are trace class: for σ this follows from the original finite-interval operator and Proposition 2.2; for a the compression acts on E^n and vanishes on its complement. Lemma 3.18 gives

$$|\operatorname{tr} f(T_n(\sigma)) - \operatorname{tr}_{E^n} f(T_n(a))| \leq 2C_V \operatorname{Lip}(f). \quad (3.36)$$

The gauge compressed to P_n was not assumed unitary.

Lemma 3.23 (A numerical bound for the full-fiber gauge). *For the path of Lemma 3.21, with the specified eigenangle logarithm, the constant in (3.36) satisfies $C_V < 65$.*

Proof. On the whole fiber set

$$D = M_{2\pi i(t-1/2)}, \quad P = |e_0\rangle\langle e_0| + |e_1\rangle\langle e_1|, \quad Q = I - P, \quad R = QDQ.$$

Both D and R are skew-adjoint, with norms at most π . The scalar phase introduced by centering does not change the spectral projections of σ , so $P_j(\theta) = e^{-\theta D} P_j(0) e^{\theta D}$ and

$$K_\sigma(\theta) = e^{-\theta D} K_0 e^{\theta D}, \quad K_0 = \sum_j P_j(0) D P_j(0) - D.$$

Indeed $P'_j = P_j D - D P_j$. In the basis (e_0, e_1) , $PDP = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; its expectations on $(e_0 \pm e_1)/\sqrt{2}$ vanish. Thus

$$K_0 = QDQ - D = -PDP - PDQ - QDP.$$

Its range lies in $P\mathcal{H} + \text{ran}(QDP)$, of dimension at most four. The three displayed blocks are orthogonal in Hilbert–Schmidt inner product. Direct integration gives $\|PDP\|_2^2 = 2$ and $\text{tr}(PD^*DP) = 2\pi^2/3$, whence

$$\|K_0\|_2^2 = 4\pi^2/3 - 2, \quad k_0 := \|K_0\|_1 \leq 2\sqrt{4\pi^2/3 - 2} < 67/10.$$

The original transport is exactly $S_\theta = e^{-\theta D} e^{\theta R}$: its derivative is $e^{-\theta D} K_0 e^{\theta R} = K_\sigma(\theta) S_\theta$, and $S_0 = I$. Differentiating this formula further yields

$$\|S_\theta^{(j)}\|_1 \leq (2\pi)^{j-1} k_0, \quad j = 1, 2, 3.$$

Only this differentiated product is estimated in trace norm; its two exponential factors need not separately be trace-class perturbations of the identity.

For the star symbol put $J = (\pi/2)(|e_{-1}\rangle\langle e_1| - |e_1\rangle\langle e_{-1}|)$. The unitary $e^{\theta J}$ fixes e_0 and rotates e_1 to w_θ . All three diagonal spectral blocks $P_j(\theta)JP_j(\theta)$ vanish, including the zero block, so $K_a = J$ and $T_\theta = e^{\theta J}$. Since J has rank two,

$$\|(T_\theta^*)^{(j)}\| \leq (\pi/2)^j, \quad \|(T_\theta^*)^{(j)}\|_1 = \pi(\pi/2)^{j-1}, \quad \|T_1 - I\|_1 = 2\sqrt{2}.$$

For $M = T_1^* S_1$ and the logarithm L used in Lemma 3.21, the eigenangle inequality in that proof gives

$$\|L\| \leq \pi, \quad \ell_0 := \|L\|_1 \leq \frac{\pi}{2}\|M - I\|_1 \leq \frac{\pi}{2}(k_0 + 2\sqrt{2}) < 15.$$

Here $\|S_1 - I\|_1 \leq k_0$ follows by integration. No finite-rank assumption on $M - I$ or L is used. The numerical bounds follow already from $\pi < 22/7$ and $\sqrt{2} < 99/70$.

Apply Leibniz to $V_\theta = S_\theta e^{-\theta L} T_\theta^*$. In each term use the trace norm of the first differentiated factor and operator norms for the other factors. The resulting uniform bounds on the first three derivatives are

$$\begin{aligned} v_1 &= k_0 + \ell_0 + \pi, \\ v_2 &= 5\pi k_0 + 2\pi\ell_0 + \pi^2/2, \\ v_3 &= (79/4)\pi^2 k_0 + (13/4)\pi^2\ell_0 + \pi^3/4. \end{aligned} \tag{3.37}$$

Explicitly, for $j = 1, 2, 3$ the product sum is bounded by

$$\frac{k_0}{2\pi} [(7\pi/2)^j - (3\pi/2)^j] + \frac{\ell_0}{\pi} [(3\pi/2)^j - (\pi/2)^j] + \pi(\pi/2)^{j-1},$$

which gives (3.37). The two endpoint differences in (3.35) are bounded by $2v_1, 2v_2$, and the third-derivative integral by v_3 . Consequently

$$\begin{aligned} C_V &\leq \frac{v_1}{4} + \frac{2v_2 + v_3}{12\pi} \\ &\leq k_0 \left(\frac{13}{12} + \frac{79\pi}{48} \right) + \ell_0 \left(\frac{7}{12} + \frac{13\pi}{48} \right) + \frac{\pi}{3} + \frac{\pi^2}{48} \\ &< \frac{760709}{11760} < 65. \end{aligned}$$

□

3.4.3 The scalar defect and the test function

Grouping the center and leaves of (3.32) gives

$$T_n(a) \simeq \begin{pmatrix} 0 & B_n \\ B_n^* & 0 \end{pmatrix}, \quad B_n = (T_n(s), T_n(c)), \quad H_n = B_n B_n^* = T_n(c)^2 + T_n(s)^2.$$

Restricted to E^n its eigenvalues are $\pm s_j(B_n)$, $1 \leq j \leq n$, and n additional zeros. For a real scalar symbol w write

$$D_n(w) = P_n \mathcal{L}(w)(I - P_n) \mathcal{L}(w) P_n = T_n(w^2) - T_n(w)^2 \geq 0.$$

The functions $u = (c + s)/\sqrt{2}$, $v = (c - s)/\sqrt{2}$ satisfy

$$H_n = I - D_n(u) - D_n(v). \quad (3.38)$$

Indeed $c^2 + s^2 = 1$ gives $H_n = I - D_n(c) - D_n(s)$, and the parallelogram law for $A_w = P_n \mathcal{L}(w)(I - P_n)$ gives $D_n(u) + D_n(v) = D_n(c) + D_n(s)$. This rotation leaves only v discontinuous at the periodic seam. Since $T_n(a)$ is a contraction, $0 \leq H_n \leq I$.

Set $v_0(\theta) = (1 - 2\theta)/\sqrt{2}$ and $z = v - v_0$. The endpoint values of u agree; those of z are both zero. These functions are smooth on $[0, 1]$, so Lemma 3.22 applies. In particular, $\|P_n \mathcal{L}(w)(I - P_n)\|_1 \leq \mathfrak{c}(w)$ for $w = u, z$. Expanding $D_n(v) - D_n(v_0)$ and using boundedness of the other factors gives

$$\|D_n(u) + D_n(v) - D_n(v_0)\|_1 \leq C_*, \quad C_* := \|u\|_\infty \mathfrak{c}(u) + (\|v\|_\infty + \|v_0\|_\infty) \mathfrak{c}(z). \quad (3.39)$$

The constant is independent of n , f and τ . For example, if $A_w = P_n \mathcal{L}(w)(I - P_n)$, then $D_n(v) - D_n(v_0) = A_z A_v^* + A_{v_0} A_z^*$, which proves the stated bound. Also

$$0 \leq D_n(v_0) \leq T_n(v_0^2) \leq \frac{1}{2}I. \quad (3.40)$$

Lemma 3.24 (Paired test function). *Let $f \in C^2([-1, 1])$ be real, with $f(0) = 0$, and put*

$$h(x) = f(\sqrt{1-x}) + f(-\sqrt{1-x}) - f(1) - f(-1), \quad 0 \leq x \leq 1. \quad (3.41)$$

Then $h(0) = 0$, $\text{Lip}_{[0,1]}(h) \leq \|f''\|_\infty$, and $h \in C^2([0, 1/2])$ with $\|h''\|_{[0,1/2]} \leq 2\|f''\|_\infty$. Moreover

$$|\text{tr } f(T_n(a)) - n[f(1) + f(-1)] - \text{tr } h(D_n(v_0))| \leq C_* \|f''\|_\infty. \quad (3.42)$$

Proof. For $r = \sqrt{1-x} > 0$,

$$h'(x) = \frac{f'(-r) - f'(r)}{2r}, \quad h''(x) = \frac{f''(r) + f''(-r)}{4r^2} - \frac{f'(r) - f'(-r)}{4r^3}.$$

Integrating f'' from $-r$ to r proves the first bound, including the limit at $r = 0$. The second expression is bounded by $\|f''\|_\infty/r^2$, hence by $2\|f''\|_\infty$ when $x \leq 1/2$. Signed singular-value pairing gives exactly

$$\text{tr } f(T_n(a)) = n[f(1) + f(-1)] + \text{tr } h(D_n(u) + D_n(v)).$$

Possible zero singular values cause no extra term because $f(0) = 0$. Apply Lemma 3.18 and (3.39). Only the global Lipschitz bound is used in this first comparison. All later model spectra lie in $[0, 1/2]$, where the second derivative is bounded. $\square$

3.4.4 The Hilbert and Carleman operators

We establish boundedness before using the square of the infinite Hilbert matrix. Let $\mathcal{C}$ denote the Carleman operator with kernel $(u + v)^{-1}$ on $L^2(0, \infty)$. The unitary map

$$(\mathcal{M}f)(s) = e^{s/2} f(e^s)$$

turns it into convolution with $[2 \cosh(s/2)]^{-1}$. For the angular Fourier convention $\widehat{k}(\omega) = \int_{\mathbb{R}} k(s) e^{-i\omega s} ds$, Euler's integral at $z = 1/2 - i\omega$ gives

$$\int_{\mathbb{R}} \frac{e^{-i\omega s}}{2 \cosh(s/2)} ds = \int_0^\infty \frac{t^{-1/2-i\omega}}{1+t} dt = \pi \text{sech}(\pi\omega). \quad (3.43)$$

Thus $\mathcal{C}$ is bounded, positive, and has norm π . The operator $\mathcal{C}_{\text{sh}}$ with kernel $(x + y + 1)^{-1}$ is unitarily its compression to $u \geq 1/2$, by translation.

Embed $\ell^2(\mathbb{N}_0)$ as unit-cell constants in $L^2(0, \infty)$. The Hilbert matrix $\mathbf{H}_{jl} = (j + l + 1)^{-1}$, initially on finitely supported sequences, has embedded kernel $(\lfloor x \rfloor + \lfloor y \rfloor + 1)^{-1}$ and is zero on the orthogonal complement. Write $\tilde{\mathbf{H}}$ for this embedding.

Lemma 3.25 (A nuclear comparison). *The Hilbert matrix extends to a bounded operator, and*

$$\|\tilde{\mathbf{H}} - \mathcal{C}_{\text{sh}}\|_1 \leq \Delta, \quad \Delta := \frac{2}{1 - e^{-2}}. \quad (3.44)$$

Proof. Put $r_\lambda(x) = e^{-\lambda \lfloor x \rfloor}$, $s_\lambda(x) = e^{-\lambda x}$. The difference of the two kernels is represented by

$$\int_0^\infty e^{-\lambda} (|r_\lambda\rangle\langle r_\lambda| - |s_\lambda\rangle\langle s_\lambda|) d\lambda.$$

Here $\|r_\lambda\|^2 = (1 - e^{-2\lambda})^{-1}$, $\|s_\lambda\|^2 = (2\lambda)^{-1} \leq \|r_\lambda\|^2$, and $\|r_\lambda - s_\lambda\| \leq \min(\lambda, 1)\|r_\lambda\|$. The integrand's trace norm is therefore at most $2e^{-\lambda} \min(\lambda, 1)/(1 - e^{-2\lambda})$. For $0 < \lambda \leq 1$, concavity gives $1 - e^{-2\lambda} \geq \lambda(1 - e^{-2})$; for $\lambda \geq 1$ use $1 - e^{-2\lambda} \geq 1 - e^{-2}$. Integration bounds the nuclear norm by Δ . Testing against compactly supported functions identifies the kernel of the nuclear integral with the displayed difference. Since $\mathcal{C}_{\text{sh}}$ is bounded, this also proves boundedness of the Hilbert matrix. Only the difference of the two Laplace integrals has been integrated in trace norm. $\square$

The Fourier coefficients of v_0 are $\hat{v}_0(k) = (\sqrt{2\pi}ik)^{-1}$ for $k \neq 0$. Multiplication of the left and right exterior blocks in $D_n(v_0)$ now gives the exact identity

$$D_n(v_0) = \alpha(P_n \mathbf{H}^2 P_n + J_n P_n \mathbf{H}^2 P_n J_n), \quad \alpha = \frac{1}{2\pi^2}, \quad (3.45)$$

where J_n reverses the n coordinates. Indeed, each exterior index produces the denominator $j + l + 1$, and the opposite Fourier signs multiply to the positive coefficient α . By positivity of both terms and (3.40), $\alpha \|\mathbf{H} P_n x\|^2 \leq \|P_n x\|^2 / 2$. Testing finitely supported vectors proves $\|\mathbf{H}\| \leq \pi$. Consequently (3.44) also gives

$$\|\tilde{\mathbf{H}}^2 - \mathcal{C}_{\text{sh}}^2\|_1 \leq 2\pi\Delta. \quad (3.46)$$

In particular all compressions of $\alpha \mathbf{H}^2$, $\alpha \mathcal{C}_{\text{sh}}^2$ and $\alpha \mathcal{C}^2$ have spectrum in $[0, 1/2]$.

3.4.5 Separating the two endpoints

Lemma 3.26 (Smooth compression). *Let X be bounded and positive, P an orthogonal projection, PXP trace class, and $(I - P)XP$ Hilbert–Schmidt. If h is real and C^2 on $[0, \|X\|]$, with $h(0) = 0$, then*

$$|\text{tr } h(PXP) - \text{tr } Ph(X)P| \leq \frac{1}{2} \|h''\|_\infty \|(I - P)XP\|_2^2. \quad (3.47)$$

The first trace is taken on $P\mathcal{H}$. This is the positive case of the compression estimate in [32, Theorem 1.2].

Proof. Since $|h(X)| \leq \text{Lip}(h)X$, the compressed traces exist. Choose an eigenbasis v_j of the compact compression PXP , including its kernel, with eigenvalues λ_j . Scalar Taylor expansion in the spectral measure of X gives

$$|\langle v_j, h(X)v_j \rangle - h(\lambda_j)| \leq \frac{1}{2} \|h''\|_\infty \langle v_j, (X - \lambda_j)^2 v_j \rangle = \frac{1}{2} \|h''\|_\infty \|(I - P)Xv_j\|^2.$$

The linear term vanishes because $\langle v_j, Xv_j \rangle = \lambda_j$. Summing proves the estimate. $\square$

Take $n = 2m$, and split its cells into two equal halves with projections P, Q . With $A = \alpha P_n H^2 P_n$ and $A' = J_n A J_n$, the trace of A on the second half and that of A' on the first half obey

$$\operatorname{tr} Q A Q = \alpha \sum_{j=m}^{2m-1} \sum_{l \geq 0} (j+l+1)^{-2} \leq 2\alpha, \quad \operatorname{tr} P A' P \leq 2\alpha. \quad (3.48)$$

The inner sum is at most $(j+1)^{-1} + (j+1)^{-2}$, and its sum over the m indices is at most $m/(m+1) + m/(m+1)^2 < 1$. Since $\|A\|, \|A'\| \leq 1/2$, positivity gives

$$\|P A Q\|_2^2 \leq \|A\| \operatorname{tr} Q A Q \leq \alpha, \quad \|P A' Q\|_2^2 \leq \alpha.$$

Thus $\|P(A+A')Q\|_2^2 \leq 4\alpha$. Apply Lemma 3.26 to both complementary blocks of $A+A' = D_n(v_0)$; their errors sum to at most $4\alpha\|h''\|_{[0,1/2]}$. Remove the other corner in each diagonal half using Lemma 3.18 and (3.48). The remaining two corners are identical. We obtain the quantitative estimate

$$\left| \operatorname{tr} h(D_{2m}(v_0)) - 2 \operatorname{tr} h(\alpha P_m H^2 P_m) \right| \leq 4\alpha(\|h'\|_{[0,1/2]} + \|h''\|_{[0,1/2]}). \quad (3.49)$$

Under the step-function embedding, P_m is restriction to $(0, m)$. Its added zero eigenvalues do not affect the trace. By (3.46) and Lemma 3.18, replacing its Hilbert-square corner by the shifted Carleman-square corner costs at most $2\pi\alpha\Delta \operatorname{Lip}(h)$. After translation the latter interval is $I = [a, m+a]$, $a = 1/2$. The difference between the compression of the full square and the square of the half-line compression is the positive operator

$$P_I \mathcal{C}^2 P_I - P_I \mathcal{C} P_{[a,\infty)} \mathcal{C} P_I = P_I \mathcal{C} P_{(0,a)} \mathcal{C} P_I,$$

whose trace is

$$\int_a^{m+a} \int_0^a \frac{dv du}{(u+v)^2} = \log \frac{2(m+a)}{m+2a} \leq \log 2. \quad (3.50)$$

Thus, for one corner,

$$\left| \operatorname{tr} h(\alpha P_m H^2 P_m) - \operatorname{tr} h(\alpha P_I \mathcal{C}^2 P_I) \right| \leq \alpha(2\pi\Delta + \log 2) \operatorname{Lip}(h). \quad (3.51)$$

3.4.6 The Mellin trace and completion of the proof

By (3.43), the Mellin transform of $\alpha \mathcal{C}^2$ is convolution on $L^2(\mathbb{R})$ with multiplier and kernel

$$d(\omega) = \frac{1}{2} \operatorname{sech}^2(\pi\omega), \quad \ell(s) = \frac{s}{4\pi^2 \sinh(s/2)}, \quad \ell(0) = \frac{1}{2\pi^2}. \quad (3.52)$$

Here inverse angular Fourier transformation has the factor $1/(2\pi)$. Write D for this convolution operator. The interval I becomes an interval of length $L = \log(2m+1)$. If R_L restricts to such an interval, then

$$\|(I - R_L) D R_L\|_2^2 = \int_{\mathbb{R}} \min(L, |s|) |\ell(s)|^2 ds \leq M_D := \frac{3\zeta(3)}{\pi^4}. \quad (3.53)$$

To evaluate M_D , expand $\sinh^{-2}(s/2) = 4 \sum_{j \geq 1} j e^{-js}$ for $s > 0$ and integrate s^3 term by term; the first moment $\int |s| |\ell(s)|^2 ds$ converges at both zero and infinity.

The positive compression $R_L D R_L$ is trace class, of trace $L\ell(0)$. Since $h(0) = 0$ and $|h(d(\omega))| \leq \operatorname{Lip}(h)d(\omega)$, the integral of $|h(d)|$ is finite. Fourier factorization of its positive and negative parts gives

$$\operatorname{tr} R_L h(D) R_L = \frac{L}{2\pi} \int_{\mathbb{R}} h(d(\omega)) d\omega.$$

Lemma 3.26 and (3.53) consequently yield

$$\left| \operatorname{tr} h(R_L D R_L) - \frac{L}{2\pi} \int_{\mathbb{R}} h(d(\omega)) d\omega \right| \leq \frac{1}{2} M_D \|h''\|_{[0,1/2]}. \quad (3.54)$$

Combining (3.49), (3.51) and (3.54) gives two copies of the Mellin bulk term. Since $\int_{\mathbb{R}} d(\omega) d\omega = 1/\pi$, replacing $\log(2m+1)$ by $\log n$, $n = 2m$, costs at most $\text{Lip}(h)/\pi^2$. In particular, with

$$B_1 = 4\alpha + 4\pi\alpha\Delta + 2\alpha \log 2 + \pi^{-2}, \quad B_2 = 4\alpha + M_D, \quad (3.55)$$

we have

$$\left| \text{tr } h(D_n(v_0)) - \frac{\log n}{\pi} \int_{\mathbb{R}} h(d(\omega)) d\omega \right| \leq B_1 \text{Lip}_{[0,1]}(h) + B_2 \|h''\|_{[0,1/2]} \quad (n \geq 2 \text{ even}). \quad (3.56)$$

For the paired test (3.41), substitute

$$t = \frac{1 + \tanh(\pi\omega)}{2}, \quad d(\omega) = 2t(1-t), \quad d\omega = \frac{dt}{2\pi t(1-t)}.$$

The coefficient in (3.56) is exactly

$$\frac{1}{\pi} \int_{\mathbb{R}} h(d(\omega)) d\omega = \frac{1}{2\pi^2} \int_0^1 \frac{f(r_t) + f(-r_t) - f(1) - f(-1)}{t(1-t)} dt = \varkappa(f). \quad (3.57)$$

Here $r_t = \sqrt{1 - 2t(1-t)} = \sqrt{t^2 + (1-t)^2}$ agrees with the r_t of (1.3). The integral converges; for instance, $1 - r_t = 2t(1-t)/(1+r_t)$ gives

$$|\varkappa(f)| \leq \frac{2}{\pi^2} \|f'\|_{\infty}. \quad (3.58)$$

We now finish Theorem 1.1. At an integer parameter $q \geq 2$, the interval $(-q, q)$ has $n = 2q$ complete cells; integer translation does not change the compression's unitary class. Equations (3.36), (3.42), (3.56), and Lemma 3.24 prove

$$\begin{aligned} & |\text{tr } f(\mathcal{K}_q) - n[f(1) + f(-1)] - \varkappa(f) \log n| \\ & \leq 2C_V \|f'\|_{\infty} + (C_* + B_1 + 2B_2) \|f''\|_{\infty}. \end{aligned} \quad (3.59)$$

For real $\tau \geq 2$ choose a nearest integer q and apply (3.31). The bulk changes by at most $2\|f'\|_{\infty}$, since $|2q - 2\tau| \leq 1$ and $f(0) = 0$. Also $1 < 2q/\tau \leq 5/2$, so replacing $\log(2q)$ by $\log \tau$ costs at most $2\pi^{-2} \log(5/2) \|f'\|_{\infty}$, by (3.58). Thus the estimate holds for every constant satisfying

$$C_{\text{tr}} \geq 2C_V + 4b(1/2) + 2 + \frac{2 \log(5/2)}{\pi^2} + C_* + B_1 + 2B_2. \quad (3.60)$$

All quantities on the right are fixed independently of f and τ , proving the uniform C^2 bound. No optimal constant is claimed.

Remark 3.27 (An admissible numerical trace constant). One may take $C_{\text{tr}} = 143$ in Theorem 1.1. Here are rational bounds for the remaining scalar terms in (3.60). Direct differentiation of u and $z = v - v_0$ gives

$$\mathfrak{c}(u) = \frac{\pi}{8\sqrt{2}} + \frac{\pi(2 - \sqrt{2})}{48}, \quad \mathfrak{c}(z) = \frac{\pi}{12\sqrt{2}}, \quad C_* = \frac{\pi(3 + \sqrt{2})}{24} < \frac{3}{5}.$$

For example, the derivative jumps of u have absolute values $\pi/\sqrt{2}$ and zero, and $\int_0^1 |u'''| = (\pi/2)^2(2 - \sqrt{2})$. The first-derivative jump of z vanishes, while its second-derivative jump and $\int_0^1 |z'''|$ both equal $\sqrt{2}(\pi/2)^2$.

For the strip term, set $r = 11/7 > \pi/2$ and use its positive power series. The ratio of successive terms from index five onwards is at most $r/6$, so

$$b(1/2) \leq \sum_{j=0}^4 \frac{r^j}{j!(2j+1)} + \frac{r^5}{5!11} \frac{1}{1-r/6} = \frac{152829401}{80385480} < \frac{191}{100}.$$

The elementary bounds $\pi > 31/10$, $e^2 > 7$, $\log 2 < 7/10$, $\log(5/2) < 1$ and $\zeta(3) < 5/4$ give

$$\Delta < \frac{7}{3}, \quad B_1 = \frac{3 + \log 2}{\pi^2} + \frac{2\Delta}{\pi} < \frac{5450}{2883} < \frac{19}{10}, \quad B_2 < \frac{229700}{923521} < \frac{1}{4}.$$

The exponential comparisons follow from positive finite Taylor sums, and the zeta bound from $1 + 1/8 + \int_2^\infty x^{-3} dx = 5/4$. Thus the scalar portion of (3.60) is less than

$$4\frac{191}{100} + 2 + \frac{200}{961} + \frac{3}{5} + \frac{19}{10} + \frac{1}{2} = \frac{308676}{24025} < 13.$$

Lemma 3.23 now gives $2C_V + \text{scalar portion} < 143$. The same estimates give $A_{\text{band}} < 144$, $C_0 < 138$ and $B_2 < 1/4$; these upper bounds may replace the corresponding constants in the band-local and gap estimates. These estimates concern the whole-fiber transport and are conservative; no optimal constant is asserted.

Finally $\int_{\mathbb{R}} d(\omega)^j d\omega = 2^{-j} B(1/2, j)/\pi$ checks the quadratic and quartic coefficients, while (3.30) independently controls every odd Lipschitz test.

3.5 An independent fourth-moment bound

Theorem 3.28. *For every $\tau \geq 2$,*

$$\left| \text{tr}(\mathcal{K}_\tau^2 - \mathcal{K}_\tau^4) - \frac{4}{3\pi^2} \log \tau \right| \leq 32. \quad (3.61)$$

Proof. This proof uses the exact quadratic formula, but not the general trace law. Let P multiply by $\mathbf{1}_{[-1,1]}$ and let $\mathcal{G}_\tau$ be the whole-line operator with kernel K_τ . Its square is the Fourier projection onto $(-\tau, \tau)$. Put $C = P\mathcal{G}_\tau^2 P = C_{2\tau}$ and $W = C - \mathcal{K}_\tau^2 = P\mathcal{G}_\tau(1 - P)\mathcal{G}_\tau P$, so $0 \leq W \leq C \leq I$. For $k_z(x) = K_\tau(z, x)$,

$$|k_z(x)| \leq \min(2\tau, 2/(\pi|z - x|)), \quad \int_{|z|>1} \|k_z\|^2 dz \leq 32\tau/\pi.$$

Thus $W = \int_{|z|>1} |k_z\rangle\langle k_z| dz$ converges in trace norm; Cauchy–Schwarz justifies products and the double integrals below. Reflection gives

$$\text{tr}(CW) = 2 \int_0^\infty \langle k_{1+u}, Ck_{1+u} \rangle du, \quad \text{tr} W^2 = 2T_s + 2T_o, \quad (3.62)$$

where $T_s = \iint |\langle k_{1+u}, k_{1+v} \rangle|^2 du dv$ and $T_o = \iint |\langle k_{1+u}, k_{-1-v} \rangle|^2 du dv$ on $(0, \infty)^2$. Write $A_u = \sin(2\pi\tau(1 + u))$, $\sigma(s) = \sin(2\pi\tau(1 - s))$, $g_u(x) = 1/[\pi(1 + u - x)]$. Direct integration gives

$$\begin{aligned} k_{1+u}(1 - s) &= \frac{A_u - \sigma(s)}{\pi(u + s)}, \quad \|g_u\|^2 = \frac{2}{\pi^2 u(2 + u)}, \\ \|k_{1+u}\|^2 &\leq \min \left\{ \frac{8\tau}{\pi}, \frac{8}{\pi^2 u(2 + u)} \right\}. \end{aligned} \quad (3.63)$$

We repeatedly use

$$\left| \int_a^b w(t) e^{i\lambda t} dt \right| \leq 2w(a)/|\lambda| \quad (w \geq 0 \text{ nonincreasing, } \lambda \neq 0), \quad (3.64)$$

which follows by integrating against the oscillatory primitive and using the total variation $w(a) - w(b)$.

The mixed trace ($\tau \geq 1$). Extend g_u by zero. Integration by parts gives $|\widehat{g}_u(\xi)| \leq 1/(\pi^2 u|\xi|)$ and $\int_{|\xi| \geq b} |\widehat{g}_u|^2 \leq 2/(\pi^4 u^2 b)$. Each one-sided tail is half this bound because $|\widehat{g}_u|^2$ is even. If

$G_0 = \widehat{g}_u$, $G_\pm = \widehat{g}_u(\cdot \pm \tau)$, then $\widehat{k}_{1+u} = A_u G_0 + iG_+/2 - iG_-/2$. On $[-\tau, \tau]$ the diagonal squared terms differ from $(A_u^2 + 1/4)\|g_u\|^2$ by at most $(2A_u^2 + 1/4)/(\pi^4 u^2 \tau)$. Split the $G_0, G_\pm$ pairs at $\mp\tau/2$ and the G_+, G_- pair at zero; on each piece one factor lies in the indicated Fourier tail. Cauchy–Schwarz gives, for the first two pairs, $4|A_u|\|g_u\|/(\pi^2 u \sqrt{\tau})$ each, and for the last pair, including its coefficient, $\sqrt{2}\|g_u\|/(\pi^2 u \sqrt{\tau})$. Consequently

$$\left| \langle k_{1+u}, Ck_{1+u} \rangle - (A_u^2 + 1/4)\|g_u\|^2 \right| \leq \frac{9}{4\pi^4 u^2 \tau} + \frac{8 + \sqrt{2}}{\pi^3 u^{3/2} \sqrt{\tau}}. \quad (3.65)$$

Split (3.62) at τ^{-1} and 1. The two exterior integrals before reflection are bounded by $8/\pi$ and $4 \log 3/\pi^2$ using (3.63). On the middle interval the integrated error in (3.65) is at most $9/(4\pi^4) + 2(8 + \sqrt{2})/\pi^3$. The main term equals

$$\frac{3}{2\pi^2} \int_{1/\tau}^1 \frac{du}{u(2+u)} - \frac{1}{\pi^2} \int_{1/\tau}^1 \frac{\cos(4\pi\tau(1+u))}{u(2+u)} du.$$

The first integral is $\frac{1}{2} \log \tau + \frac{1}{2} \log((2 + \tau^{-1})/3)$; the oscillatory integral has modulus at most $1/(4\pi)$ by (3.64). Including reflection proves

$$\left| \text{tr}(CW) - \frac{3}{2\pi^2} \log \tau \right| \leq E_C := 2 \left[\frac{8}{\pi} + \frac{4 \log 3}{\pi^2} + \frac{9}{4\pi^4} + \frac{2(8 + \sqrt{2})}{\pi^3} + \frac{3}{4\pi^2} \log \frac{3}{2} + \frac{1}{4\pi^3} \right] < 7.33. \quad (3.66)$$

The squared boundary operator. Put $a = \tau^{-1}$, $\Omega = [a, 1]^2$. The operator $W_R = \int_0^\infty |k_{1+v}\rangle \langle k_{1+v}| dv$ satisfies $W_R \leq I$; hence integrating a row of T_s gives at most $\|k_{1+u}\|^2$. A union bound on the four strips outside Ω and (3.63) give

$$0 \leq T_s - \iint_{\Omega} |\langle k_{1+u}, k_{1+v} \rangle|^2 \leq C_{\text{cut}} := 16/\pi + 8 \log 3/\pi^2. \quad (3.67)$$

Define the continuous diagonal extensions $\ell(u, v) = (\log u - \log v)/(u - v) = \int_0^\infty [(u+s)(v+s)]^{-1} ds$ and $m(u, v) = \int_0^2 [(u+s)(v+s)]^{-1} ds$. Then $0 \leq \ell - m \leq 1/2$, $\ell \leq (uv)^{-1/2}$, and $0 \leq \ell^2 - m^2 \leq \ell$; the bound on ℓ follows from $2 \sinh y \geq 2y$ with $y = \frac{1}{2} |\log(u/v)|$. Expanding (3.63) gives

$$\pi^2 \langle k_{1+u}, k_{1+v} \rangle = qm - \delta, \quad q = A_u A_v + 1/2,$$

where $\delta = (A_u + A_v) \int_0^2 \sigma(s)/[(u+s)(v+s)] ds + \frac{1}{2} \int_0^2 \cos(4\pi\tau(1-s))/[(u+s)(v+s)] ds$. Equation (3.64) gives $|\delta| \leq 9/(4\pi\tau uv) =: \Delta$. Since $|q| \leq 3/2$, $|(qm - \delta)^2 - q^2 m^2| \leq 3m\Delta + \Delta^2$. Integrate using $\int_a^1 u^{-3/2} du \leq 2\sqrt{\tau}$ and $\int_a^1 u^{-2} du \leq \tau$, then replace m^2 by ℓ^2 :

$$\left| \pi^4 \iint_{\Omega} |\langle k_{1+u}, k_{1+v} \rangle|^2 - \iint_{\Omega} q^2 \ell^2 \right| \leq C_1 := 27/\pi + 81/(16\pi^2) + 9. \quad (3.68)$$

The final 9 uses $(9/4) \iint_{[0,1]^2} (uv)^{-1/2} = 9$.

For $\Psi(t) = (\log t)^2/(1-t)^2$, $\Psi(1) = 1$, one has

$$\Psi(1/t) = t^2 \Psi(t), \quad \int_0^1 \Psi = \int_1^\infty \Psi = \pi^2/3, \quad \int_1^\infty \Psi(t) \log t dt = 6\zeta(3).$$

Expand $(1-t)^{-2}$ on $(0, 1)$ and integrate $t^{n-1}(-\log t)^j$ to obtain $j!/n^{j+1}$; positivity justifies the series, and reciprocity gives the upper-half integrals. For $L(u) = \int_a^1 \ell(u, v)^2 dv$, $L(a) = \tau \int_1^\tau \Psi \leq \pi^2 \tau/3$ and L decreases. Applying (3.64) at fixed v therefore bounds every integral of $\cos(\lambda u + \phi(v))\ell^2$ over Ω by $2\pi^2 \tau/(3|\lambda|)$. Product-to-sum expands $q^2 - 1/2$ into

$$-\frac{1}{4}(c_u + c_v) + \frac{1}{8} \cos(4\pi\tau(u-v)) + \frac{1}{8} \cos(4\pi\tau(2+u+v)) + \frac{1}{2} \cos(2\pi\tau(u-v)) - \frac{1}{2} \cos(2\pi\tau(2+u+v)),$$

where $c_u = \cos(4\pi\tau(1+u))$. The absolute coefficients at frequencies $4\pi\tau$ and $2\pi\tau$ sum to $3/4$ and 1. Thus $|\iint_{\Omega}(q^2 - 1/2)\ell^2| \leq C_2 := 11\pi/24$, including difference phases, whose derivative in u is nonzero. The substitution $v = ut$ gives, with

$$E_1 = \int_a^1 \frac{du}{u} \int_{1/u}^{\infty} \Psi(t) dt, \quad E_2 = \int_a^1 \frac{du}{u} \int_0^{a/u} \Psi(t) dt,$$

$$\iint_{\Omega} \ell^2 = \frac{2\pi^2}{3} \log \tau - E_1 - E_2, \quad 0 \leq E_1, E_2 \leq 6\zeta(3).$$

For the upper tail interchange the integrals to obtain the bound $\int_1^{\infty} y^{-1} \int_y^{\infty} \Psi(t) dt dy = \int_1^{\infty} \Psi(t) \log t dt$; the lower tail has the same bound by reciprocity after $u = y/\tau$. Combining gives a local squared-pair error at most $C_1 + C_2 + 6\zeta(3)$ after multiplication by π^4 .

For opposite endpoints partial fractions give

$$|\langle k_{1+u}, k_{-1-v} \rangle| \leq \frac{4}{\pi^2} \frac{\log(1+2/u) + \log(1+2/v)}{2+u+v}.$$

Squaring and integrating first in v gives $T_o \leq 128\zeta(3)/\pi^4$: the remaining integral $\int_0^{\infty} \log^2(1+2/u)/(2+u) du = 2\zeta(3)$ follows by $t = u/(u+2)$ and the geometric series. Together with (3.67), this proves

$$\left| \text{tr } W^2 - \frac{2}{3\pi^2} \log \tau \right| \leq E_W := 2 \left[C_{\text{cut}} + \frac{C_1 + C_2 + 6\zeta(3)}{\pi^4} \right] + \frac{256\zeta(3)}{\pi^4} < 15.676. \quad (3.69)$$

Finally cyclicity and $\mathcal{K}_{\tau}^2 = C - W$ give $\text{tr}(\mathcal{K}_{\tau}^2 - \mathcal{K}_{\tau}^4) = D_R - D_J + 2 \text{tr}(CW) - \text{tr } W^2$. The exact quadratic estimates for $\tau \geq 2$ give

$$\left| D_R - \pi^{-2} \log \tau \right| \leq \frac{\log(8\pi) + 1 + \gamma}{\pi^2} + \frac{1}{4\pi^3} < 0.4946,$$

$$\left| D_J - 2\pi^{-2} \log \tau \right| \leq \frac{2}{\pi^2} [\log(4\pi) + 1 + \gamma + \log 2] + \frac{2}{\pi^3} < 1.0375.$$

The logarithmic coefficients combine as $1 - 2 + 3 - 2/3 = 4/3$, and the errors sum to $0.4946 + 1.0375 + 2(7.33) + 15.676 = 31.8681 < 32$. $\square$

Corollary 3.29 (Interior windows and small eigenvalues). *For $0 < a \leq b < 1$, put $m_j = \min\{a^j(1-a^j), b^j(1-b^j)\}$, $j = 1, 2$. For $\tau \geq 2$,*

$$\#\{k : a \leq |\lambda_k(\mathcal{K}_{\tau})| \leq b\} \leq \min \left\{ \frac{4\tau - \text{tr } \mathcal{K}_{\tau}^2}{m_1}, \frac{4(3\pi^2)^{-1} \log \tau + 32}{m_2} \right\}. \quad (3.70)$$

The first bound holds for every $\tau > 0$. For $0 < \beta < 1$, either $X = \mathcal{K}_{\tau}$, $s = 4\tau$, or $X = A_N$, $s = 2d\varepsilon$, also satisfies $\sum_{|\lambda_k(X)| \leq \beta} \lambda_k(X)^2 \leq \beta(s - \text{tr } X^2)/(1 - \beta)$.

Proof. $\text{tr } |X| \leq s$ gives $\sum |\lambda|(1 - |\lambda|) \leq s - \text{tr } X^2$. The other nonnegative sum is $\sum \lambda^2(1 - \lambda^2) = \text{tr}(X^2 - X^4)$. Their minima on the window are m_1, m_2 ; discard all other terms. For the small-eigenvalue bound use $r^2 \leq \beta r(1-r)/(1-\beta)$ when $r \leq \beta$. $\square$

3.6 Uniform phase stability

Proposition 3.30 (Uniform phase law and exact pairing). *For $\phi \in \mathbb{R}$, let $\mathcal{K}_{\tau, \phi}$ have kernel*

$$\frac{\sin(2\pi\tau x + \phi) - \sin(2\pi\tau y + \phi)}{\pi(x - y)}.$$

For every real Lipschitz f with $f(0) = 0$ and every $\tau > 0$,

$$|\operatorname{tr} f(\mathcal{K}_{\tau,\phi}) - \operatorname{tr} f(\mathcal{K}_\tau)| \leq 4b(1/2) \operatorname{Lip}(f),$$

where b is defined in (3.27). Consequently Theorem 1.1 holds uniformly in ϕ , with the same $\varkappa(f)$ and remainder constant $C_{\text{tr}} + 4b(1/2)$. Moreover

$$\begin{aligned} \mathcal{R}\mathcal{K}_{\tau,\phi}\mathcal{R} &= \mathcal{K}_{\tau,-\phi}, & \mathcal{K}_{\tau,\phi+\pi} &= -\mathcal{K}_{\tau,\phi}, & \mathcal{K}_{\tau,\phi+2\pi} &= \mathcal{K}_{\tau,\phi}, \\ \operatorname{tr} \mathcal{K}_{\tau,\phi} &= \frac{2}{\pi} \sin(2\pi\tau) \cos \phi. \end{aligned} \tag{3.71}$$

At $\phi = \pi/2 \pmod{\pi}$ its spectrum has exact $\pm$ pairing with multiplicity, and every odd Lipschitz test has zero trace. Each eigenvector for a nonzero eigenvalue has even and odd parts of equal norm.

Proof. The operator is unitarily equivalent to the compression of $\mathcal{G}$ to $(-\tau, \tau) + \phi/(2\pi)$. Reduce ϕ modulo 2π so that $|\phi/(2\pi)| \leq 1/2$, and apply Lemmas 3.19 and 3.18. The trace law follows since $\operatorname{Lip}(f) \leq \|f'\|_\infty$. Reflection and phase shifts in the kernel give the first three identities; integration of its diagonal gives the fourth. At the indicated phases, reflection anticommutes with the operator and therefore pairs eigenvalues of opposite signs. It exchanges the signs, rather than defining commuting parity sectors there. If $\mathcal{K}_{\tau,\phi}u = \lambda u$ with $\lambda \neq 0$, then u and $\mathcal{R}u$ belong to distinct eigenspaces and are orthogonal. Thus the two reflection components of u have squared norm $\|u\|^2/2$. $\square$

At the centered phase $\phi = 0$, neither parity spectrum is sign-symmetric, for any $\tau > 0$. Indeed its trace in (2.5) is nonzero because $\operatorname{Si}(t) > |\sin t|$ for $t > 0$. On $(0, \pi]$, differentiate $\operatorname{Si}(t) - \sin t$ and use $\sin t - t \cos t > 0$. On $[\pi, 2\pi]$, use monotonicity and $\operatorname{Si}(2\pi) \geq \pi/2 - 1/\pi > 1$; for $t \geq 2\pi$ the same inequality follows from $|\operatorname{Si}(t) - \pi/2| \leq 2/t$. A sign-symmetric trace-class spectrum would have zero trace. The full centered spectrum is also asymmetric whenever $\sin(2\pi\tau) \neq 0$.

4 Finite matrices and endpoint geometry

4.1 A finite trace law with a growing scale

The fixed-parameter critical limit alone does not transfer a large- τ trace asymptotic to growing matrices. The midpoint realization gives a quantitative sampling estimate; the frequency-space comparison below additionally bounds trace errors independently of the dimension. Its natural parameter is $\tau = d\varepsilon/2$, where $d = 2N + 1$; it differs by $\varepsilon/2$ from the scale $\tau_N = N\varepsilon$ used in Theorem 2.5.

Theorem 4.1 (Uniform sampling and trace transfer). *Let $h = 2/d$, $x_m = mh$, and let ι_N send the m th coordinate vector to $h^{-1/2}\mathbf{1}_{[x_m-h/2, x_m+h/2]}$ in $L^2[-1, 1]$. For $0 < \varepsilon < 1/2$ and $\tau = d\varepsilon/2$,*

$$\|\iota_N A_N(\varepsilon) \iota_N^* - \mathcal{K}_\tau\|_1 \leq \frac{16\pi\tau^2}{3d}. \tag{4.1}$$

More sharply, for every real $f \in C^2([-1, 1])$ with $f(0) = 0$,

$$|\operatorname{tr} f(A_N(\varepsilon)) - \operatorname{tr} f(\mathcal{K}_\tau) - f'(0)\Delta_{N,\varepsilon}| \leq \frac{5\pi^2}{9} d\varepsilon^3 \|f''\|_\infty, \tag{4.2}$$

where the exact linear correction is

$$\Delta_{N,\varepsilon} = 2 \sin(\pi d\varepsilon) \left(\frac{\varepsilon}{\sin(\pi\varepsilon)} - \frac{1}{\pi} \right), \quad |\Delta_{N,\varepsilon}| \leq \frac{\pi^2 \varepsilon^2}{6}. \tag{4.3}$$

All estimates are uniform in N and ε on this range.

Proof. Put $P = \iota_N \iota_N^*$, $Q = I - P$, and let $U_\tau : L^2[-1, 1] \rightarrow L^2(-\tau, \tau)$ have kernel $e^{2\pi i \xi x}$. Let B have kernel $e^{2\pi i \xi x_m}$ for x in the m th cell. The exchange J_τ from (2.2) gives

$$\mathcal{K}_\tau = U_\tau^* J_\tau U_\tau, \quad \iota_N A_N(\varepsilon) \iota_N^* = B^* J_\tau B, \quad \|U_\tau\|_2 = \|B\|_2 = 2\sqrt{\tau}.$$

Indeed $h\mathcal{K}_\tau(x_m, x_n) = A_N(\varepsilon)_{mn}$, including the diagonal. Since $|e^{ia} - e^{ib}| \leq |a - b|$,

$$\|U_\tau - B\|_2^2 \leq 4\pi^2 \int_{-\tau}^{\tau} \xi^2 d\xi \sum_m \int_{x_m-h/2}^{x_m+h/2} (x - x_m)^2 dx = \frac{4\pi^2 h^2 \tau^3}{9}.$$

Factoring the difference of the two quadratic forms proves (4.1) by the Hilbert–Schmidt product inequality.

For the sharper estimate let S multiply by $s(\xi) = \sin(\pi h \xi)/(\pi h \xi)$, with $s(0) = 1$. Exact integration over a cell gives $U_\tau P = SB$, and therefore $PK_\tau P = B^* S J_\tau S B$. Write $C = U_\tau^* U_\tau = C_{2\tau}$ and

$$\eta = \text{tr } QCQ = \|U_\tau Q\|_2^2 = 2 \int_{-\tau}^{\tau} (1 - s(\xi)^2) d\xi \leq \bar{\eta} := \frac{4\pi^2 h^2 \tau^3}{9} = \frac{2\pi^2}{9} d\varepsilon^3.$$

Here $|\pi h \xi| \leq \pi h \tau = \pi \varepsilon < \pi/2$ for $|\xi| \leq \tau$, so $0 < s \leq 1$ on that range, and $1 - \sin z/z \leq z^2/6$ gives $1 - s(\xi)^2 \leq 2(1 - s(\xi)) \leq (\pi h \xi)^2/3$. On each exchanged frequency pair $\xi, \xi - \tau$, $0 < \xi < \tau$,

$$0 \leq 1 - s(\xi)s(\xi - \tau) \leq \frac{\pi^2 h^2}{6} (\xi^2 + (\xi - \tau)^2) \leq \frac{\pi^2 h^2 \tau^2}{6}.$$

It follows that

$$\|B^* J_\tau B - PK_\tau P\|_1 \leq \|B\|_2^2 \|J_\tau - S J_\tau S\| \leq \frac{3}{2} \bar{\eta}. \quad (4.4)$$

Set $M = \|f''\|_\infty$ and $r(x) = f(x) - f'(0)x$. Then $\text{Lip}(r) \leq M$ and $|r(x)| \leq Mx^2/2$. Scalar Taylor expansion in the spectral measure of $\mathcal{K}_\tau$, for an eigenbasis of $PK_\tau P$ on $PL^2[-1, 1]$, yields

$$|\text{tr } r(PK_\tau P) - \text{tr } Pr(\mathcal{K}_\tau)P| \leq \frac{1}{2} M \|Q\mathcal{K}_\tau P\|_2^2 \leq \frac{1}{2} M \eta.$$

The linear Taylor term vanishes because each basis vector is an eigenvector of the compression. Positivity of the operator is not needed for this scalar argument. The last inequality uses $\|Q\mathcal{K}_\tau P\|_2 \leq \|Q\mathcal{K}_\tau\|_2$ and $\mathcal{K}_\tau^2 \leq C$ from (2.4). The same order and $|r(\mathcal{K}_\tau)| \leq M\mathcal{K}_\tau^2/2$ give $|\text{tr } Qr(\mathcal{K}_\tau)Q| \leq M\eta/2$. Together with (4.4) and the scalar trace inequality of Lemma 3.18, this bounds the difference of the two r -traces by $5M\bar{\eta}/2$. The remaining linear traces are exactly (4.3), by the finite cosine sum and (2.5). Finally $0 \leq \pi\varepsilon - \sin(\pi\varepsilon) \leq (\pi\varepsilon)^3/6$ and $\sin(\pi\varepsilon) \geq 2\varepsilon$ prove its bound. $\square$

4.1.1 A comparison independent of the matrix dimension

The frequency-space comparison gives finite trace bounds independent of the matrix dimension. For $0 < \varepsilon < 1/2$, define

$$L_\varepsilon = 2\varepsilon \csc(2\pi\varepsilon) - \frac{1}{\pi}, \quad \Gamma_\varepsilon = \frac{16\sqrt{2/\pi}\varepsilon}{(1 - 2\varepsilon)\sqrt{1 - 4\varepsilon^2}}. \quad (4.5)$$

These are $O(\varepsilon^2)$ and $O(\varepsilon)$ respectively as $\varepsilon \downarrow 0$, and are bounded on $0 < \varepsilon \leq 1/2 - \delta$ for each fixed $\delta > 0$.

Theorem 4.2 (Direct finite trace comparison). *Let $d = 2N + 1$, $0 < \varepsilon < 1/2$ and $\tau = d\varepsilon/2$. For every real $f \in C^2([-1, 1])$, $f(0) = 0$,*

$$|\text{tr } f(A_N(\varepsilon)) - \text{tr } f(\mathcal{K}_\tau) - f'(0)\Delta_{N,\varepsilon}| \leq L_\varepsilon \|f''\|_\infty, \quad (4.6)$$

where $\Delta_{N,\varepsilon}$ is (4.3). Every real Lipschitz f on $[-1, 1]$ with $f(0) = 0$ satisfies

$$|\mathrm{tr} f(A_N(\varepsilon)) - \mathrm{tr} f(\mathcal{K}_\tau)| \leq \Gamma_\varepsilon \mathrm{Lip}(f). \quad (4.7)$$

Both estimates are independent of N ; no lower bound on τ is needed. Together with the nuclear sampling estimate they give

$$\mathbf{E}_{N,\varepsilon} = \min \left\{ \Gamma_\varepsilon, \frac{16\pi\tau^2}{3d} \right\}, \quad (4.8)$$

$$|\mathrm{tr} f(A_N(\varepsilon)) - \mathrm{tr} f(\mathcal{K}_\tau)| \leq \mathbf{E}_{N,\varepsilon} \mathrm{Lip}(f). \quad (4.9)$$

Proof. On $I = (-\varepsilon, \varepsilon)$ let J exchange its two halves by translation through ε . The two positive trace-class contractions

$$C_d(x, y) = \frac{\sin(\pi d(x - y))}{\sin(\pi(x - y))}, \quad C_d^0(x, y) = \frac{\sin(\pi d(x - y))}{\pi(x - y)}$$

have trace $2d\varepsilon$. They are the restrictions of the Fourier projections onto modes $-N, \dots, N$ on the unit circle and onto $(-d/2, d/2)$ on the real line, respectively. For $C \geq 0$ put $A(C) = C^{1/2}JC^{1/2}$. Polar decomposition of the corresponding restriction maps identifies the nonzero spectra of $A(C_d)$ and $A(C_d^0)$ with those of $A_N(\varepsilon)$ and $\mathcal{K}_\tau$, respectively, including multiplicities. Indeed the first exchange matrix element is

$$2 \cos(\pi\varepsilon(m + n)) \frac{\sin(\pi\varepsilon(m - n))}{\pi(m - n)},$$

with continuous diagonal; the same formula in continuous frequencies, followed by dilation from $(-d/2, d/2)$ to $(-1, 1)$, gives K_τ .

We first record a covariance trace estimate. If C_0, C_1 are positive trace-class contractions and J is a self-adjoint contraction, then

$$|\mathrm{tr} f(A(C_1)) - \mathrm{tr} f(A(C_0)) - f'(0) \mathrm{tr} J(C_1 - C_0)| \leq \|f''\|_\infty \|C_1 - C_0\|_1. \quad (4.10)$$

For a polynomial p with $p(0) = 0$, cyclicity gives $\mathrm{tr} p(A(C)) = \mathrm{tr} p(JC)$. Along $C_t = C_0 + t(C_1 - C_0)$ the derivative after subtracting the affine part has gradient

$$p'(JC_t)J - p'(0)J = JC_t^{1/2}g(A(C_t))C_t^{1/2}J, \quad g(x) = \frac{p'(x) - p'(0)}{x}.$$

The continuous value at zero is understood. Its norm is at most $\|p''\|_\infty$. Integrate against $C_1 - C_0$ in trace duality. Approximate f'' uniformly by polynomials and integrate twice, preserving $f(0)$ and $f'(0)$, to obtain (4.10). The trace passage follows from $\|A(C)\|_1 \leq \mathrm{tr} C$ and $|p(x) - f(x)| \leq \|p' - f'\|_\infty |x|$.

Put $h(u) = \csc(\pi u) - 1/(\pi u)$, with $h(0) = 0$. Pairing opposite poles in the partial fraction expansion of the cosecant and then expanding geometrically gives

$$h(u) = \frac{2}{\pi} \sum_{k \geq 0} \eta(2k + 2) u^{2k+1}, \quad |u| < 1, \quad \eta(s) = \sum_{j \geq 1} (-1)^{j+1} j^{-s} \in (0, 1).$$

The rank-one binomial expansion of $(x - y)^m$ on I gives nuclear norm at most $2\varepsilon(2\varepsilon)^m$. Positivity of the displayed Taylor coefficients therefore bounds the operator H_h with kernel $h(x - y)$ by

$$\|H_h\|_1 \leq 2\varepsilon h(2\varepsilon) = L_\varepsilon.$$

Now $C_d - C_d^0$ has kernel $\sin(\pi d(x - y))h(x - y)$, half the difference of two unitary modulations of H_h . Hence $\|C_d - C_d^0\|_1 \leq L_\varepsilon$. Moreover

$$\mathrm{tr} J(C_d - C_d^0) = 2\varepsilon \sin(\pi d\varepsilon)h(\varepsilon) = \Delta_{N,\varepsilon}.$$

Equation (4.10) proves (4.6).

For the Lipschitz estimate set $q = 2\varepsilon$ and $\alpha = 8\varepsilon^2/[\pi(1 - 4\varepsilon^2)]$. Truncating the series for h before $k = K$ gives a kernel of rank at most $2K$; the two sine modulations have combined rank at most $4K$. The remaining nuclear norm is at most αq^{2K} . The singular values of $D = C_d - C_d^0$ thus satisfy

$$s_{4K+1}(D) \leq \alpha q^{2K}, \quad \text{tr } |D|^{1/2} \leq \frac{4\sqrt{\alpha}}{1-q}.$$

We use the following square-root inequality, a special case of the operator-function norm comparisons in [3]:

$$\|\sqrt{X} - \sqrt{Y}\|_1 \leq \text{tr } |X - Y|^{1/2} \quad (X, Y \geq 0). \quad (4.11)$$

Here is a proof of the precise compact-operator form needed. In finite dimension write $X - Y = D_+ - D_-$ and $Z = X + D_- = Y + D_+$. Operator monotonicity of the square root gives $\sqrt{Z} - \sqrt{X} \geq 0$ and $\sqrt{Z} - \sqrt{Y} \geq 0$. For positive U, V , the triangle inequality for the nuclear norm of the row operator $[\sqrt{U} \ \sqrt{V}]$ gives $\text{tr } \sqrt{U+V} \leq \text{tr } \sqrt{U} + \text{tr } \sqrt{V}$. Consequently

$$\|\sqrt{X} - \sqrt{Y}\|_1 \leq \text{tr}(\sqrt{Z} - \sqrt{X}) + \text{tr}(\sqrt{Z} - \sqrt{Y}) \leq \text{tr } \sqrt{D_-} + \text{tr } \sqrt{D_+}.$$

For compact X, Y , compress both by common increasing finite-rank projections. The square roots converge in operator norm, and $s_j(P(X - Y)P) \leq s_j(X - Y)$. Lower semicontinuity of each finite Ky Fan sum, followed by passage to the number of singular values, proves (4.11) whenever its right side is finite. The square-root monotonicity used here also follows directly by integrating $x/(t + x)$ in the standard positive integral formula for $\sqrt{x}$.

Since the two covariances are contractions,

$$\|A(C_d) - A(C_d^0)\|_1 \leq 2\|\sqrt{C_d} - \sqrt{C_d^0}\|_1 \leq \frac{8\sqrt{\alpha}}{1 - 2\varepsilon} = \Gamma_\varepsilon.$$

The scalar trace inequality of Lemma 3.18, together with the exact spectral identifications, proves (4.7). Finally, (4.1) and the scalar trace inequality of Lemma 3.18, together with (4.7), prove (4.9). $\square$

Corollary 4.3 (Finite trace, Schatten and gap bounds). *Let $0 < \varepsilon < 1/2$, $d = 2N + 1$, $\tau = d\varepsilon/2 \geq 2$, $g = 2^{-1/2}$ and $C_0 = 2C_V + 4b(1/2)$. For real $f \in C^2([-1, 1])$ with $f(0) = 0$,*

$$\text{tr } f(A_N(\varepsilon)) = d\varepsilon[f(1) + f(-1)] + \varkappa(f) \log \tau + R_{N,\varepsilon}(f), \quad (4.12)$$

where, writing $M_j = \|f^{(j)}\|_\infty$,

$$|R_{N,\varepsilon}(f)| \leq C_{\text{tr}}(M_1 + M_2) + \frac{\pi^2 \varepsilon^2}{6} |f'(0)| + \min\{L_\varepsilon, \frac{5\pi^2}{9} d\varepsilon^3\} M_2.$$

More generally, if $f(0) = 0$, f is real L -Lipschitz and has $C^{1,1}$ restrictions to both closed outer bands, let M be the larger Lipschitz constant of its derivatives there. Then

$$|\text{tr } f(A_N(\varepsilon)) - d\varepsilon[f(1) + f(-1)] - \varkappa(f) \log \tau| \leq (A_{\text{band}} + E_{N,\varepsilon})L + B_2 M. \quad (4.13)$$

In particular, for $p \geq 1$ and $D_p = p(p - 1) \max\{1, g^{p-2}\}$,

$$\left| \|A_N(\varepsilon)\|_p^p - 2d\varepsilon - \varkappa_p \log \tau \right| \leq p(A_{\text{band}} + E_{N,\varepsilon}) + B_2 D_p. \quad (4.14)$$

All these remainders are bounded uniformly for $0 < \varepsilon \leq 1/2 - \delta$ with fixed $\delta > 0$, including fixed ε as $N \rightarrow \infty$.

For the gap bounds put $C_{\text{fin}} = C_0 + E_{N,\varepsilon}$ and $w_g(x) = \min\{|x|, (g - |x|)_+\}$. Choose a nearest integer q to τ and set $n = 2q$. For $0 < a < g$ and either sign, one has

$$\sum_j w_g(\lambda_j(A_N(\varepsilon))) \leq C_{\text{fin}},$$

$$n - \frac{C_{\text{fin}}}{g-a} \leq N_{\pm}^{(N)}(a; \varepsilon) \leq n + \frac{C_{\text{fin}}}{a}, \quad |n - 2\tau| \leq 1.$$

where $N_{\pm}^{(N)}(a; \varepsilon) = \#\{j : \pm\lambda_j(A_N(\varepsilon)) > a\}$.

Proof. Combine Theorem 1.1 with the two smooth comparisons (4.2) and (4.6), retaining (4.3). The band estimate follows from Theorem 3.2 and (4.9). For $f(x) = |x|^p$, the two derivative constants are $L = p$ and $M = D_p$ (including $D_1 = 0$), and Corollary 3.3 gives $\varkappa(f) = \varkappa_p$.

The function w_g is nonnegative and 1-Lipschitz, so (4.9) and Proposition 3.4 prove the weighted bound. The comparison with the star matrix M_n in Proposition 3.4, followed by (4.9), bounds every Lipschitz trace difference between $A_N(\varepsilon)$ and M_n by $C_{\text{fin}} \text{Lip}(f)$. Applying the same two ramp functions as in that proposition gives the strict counts. $\square$

The signed transition counts also transfer with error $O((\log \tau)^{2/3})$, uniformly on compact subsets of the outer bands, with the constant also depending on δ : the finite smooth error has the same $M_1 + M_2$ dependence used in the smoothing proof of Theorem 3.5. The nuclear term in (4.8) retains the vanishing fixed- τ comparison. The smooth minimum additionally retains its exact linear correction, with no additive C_0 term.

4.2 Finite onset, endpoint packets and concentration

We retain the matrix, Fourier and parity conventions of Section 2.1. The following results describe the individual edges and eigenvectors, in addition to the trace laws.

Theorem 4.4 (Onset and the four small-scale edges). *Put $a = \min\{1, (4\tau)^{-1}\}$, $p_\tau = 16a\tau/\pi^2$ and $m_\tau = 128a^3\tau^3/(3\pi^2)$. For every $\tau > 0$,*

$$\lambda_+^+(\tau) \geq p_\tau > 0, \quad \lambda_-^-(\tau) \leq -m_\tau < 0. \quad (4.15)$$

If $\varepsilon = \tau/N$ and $N > 2\tau$, $\nu_N = (2\tau + 8\pi\tau^2)/N < \min(p_\tau, m_\tau)$, then $\lambda_{N,+}^+ \geq p_\tau - \nu_N > 0$ and $\lambda_{N,-}^- \leq -m_\tau + \nu_N < 0$. For arbitrary $0 < \varepsilon_N < 1/2$, the three regimes are

$$\left. \begin{array}{l} N\varepsilon_N \rightarrow 0 \\ N\varepsilon_N \rightarrow \tau \in (0, \infty) \\ N\varepsilon_N \rightarrow \infty \end{array} \right| \begin{array}{l} \|A_N(\varepsilon_N)\| \rightarrow 0 \\ \tilde{A}_{N,N\varepsilon_N} \rightarrow \mathcal{K}_\tau \text{ in norm} \\ \lambda_{N,p}^{\pm}(\varepsilon_N) \rightarrow \pm 1 \quad (p = \pm 1). \end{array} \quad (4.16)$$

The middle regime transfers every isolated nonzero parity spectral cluster. In the last regime ε_N need not converge. As $\tau \downarrow 0$, the four continuum edges satisfy

$$\lambda_+^+ = 4\tau - \frac{16\pi^2}{9}\tau^3 + \frac{872\pi^4}{2025}\tau^5 + O(\tau^7), \quad \lambda_+^- = -\frac{224\pi^4}{2025}\tau^5 + O(\tau^7), \quad (4.17)$$

$$\lambda_-^- = -\frac{8\pi^2}{9}\tau^3 + \frac{16\pi^4}{75}\tau^5 - \frac{6352\pi^6}{275625}\tau^7 + O(\tau^9), \quad \lambda_-^+ = \frac{352\pi^6}{275625}\tau^7 + O(\tau^9). \quad (4.18)$$

All other even eigenvalues are $O(\tau^7)$ and all other odd eigenvalues are $O(\tau^9)$. The displayed edges are simple for sufficiently small τ , when $F_+ = \lambda_+^+$ and $F_- = -\lambda_-^-$.

Proof. For real even f , the exchange identity gives $\langle f, \mathcal{K}_\tau f \rangle = 2 \int_0^\tau C_f(s)C_f(\tau-s)ds$, where $C_f(s) = \int f(x) \cos(2\pi sx) dx$; for odd f it gives the negative of the same expression with $S_f(s) = \int f(x) \sin(2\pi sx) dx$. Test $e_a = (2a)^{-1/2} \mathbf{1}_{[-a,a]}$ and $o_a = \text{sgn}(x)e_a$. Their transforms are

$$C_{e_a}(s) = \frac{\sin(2\pi as)}{\pi s \sqrt{2a}} \geq \frac{2\sqrt{2a}}{\pi}, \quad S_{o_a}(s) = \frac{1 - \cos(2\pi as)}{\pi s \sqrt{2a}} \geq \frac{8\sqrt{2}a^{3/2}s}{\pi}.$$

Here $0 \leq 2\pi as \leq \pi/2$, and the inequalities follow from $\sin z \geq 2z/\pi$ and $1 - \cos z \geq 4z^2/\pi^2$ on that interval. Integration and minimax prove (4.15); Theorem 2.5 gives the finite certificate and first two regimes, using $\|\mathcal{K}_\tau\| \leq 4\tau$ for norm collapse. The last regime follows from the concentration bounds proved below.

For the expansions put $\beta_k = (-1)^k (2\pi\tau)^{2k+1} / [\pi(2k+1)!]$. The even and odd kernels are respectively $\sum_{i,l \geq 0} \beta_{i+l} x^{2i} y^{2l}$ and $\sum_{i,l \geq 0} \beta_{i+l+1} x^{2i+1} y^{2l+1}$; absolute uniform convergence follows from $\sum_k (2k+1) |\beta_k| = 2\tau \cosh(2\pi\tau)$. Truncation at $i+l \leq 2$ has matrix $C_r G_r$, $r = 0, 1$, with

$$C_r = \begin{pmatrix} \beta_r & \beta_{r+1} & \beta_{r+2} \\ \beta_{r+1} & \beta_{r+2} & 0 \\ \beta_{r+2} & 0 & 0 \end{pmatrix}, \quad (G_r)_{il} = \frac{2}{2i+2l+2r+1} \quad (0 \leq i, l \leq 2).$$

These matrices are similar to self-adjoint matrices $G_r^{1/2} C_r G_r^{1/2}$. With $u = 2\pi\tau$, the even kernel tail is at most $2\tau u \sum_{j \geq 0} u^{2j+5} / (2j+5)! \leq \tau u^6 \cosh u/60$; the odd tail uses $7!$ instead of $5!$. Twice the kernel bounds therefore give

$$\|\mathcal{K}_\tau|_{\mathcal{H}_+} - T_0\| \leq \frac{32\pi^6}{15} \tau^7 \cosh(2\pi\tau), \quad \|\mathcal{K}_\tau|_{\mathcal{H}_-} - T_1\| \leq \frac{64\pi^8}{315} \tau^9 \cosh(2\pi\tau). \quad (4.19)$$

For $r = 0, 1$, the coefficients of $z^3 - e_1 z^2 + e_2 z - e_3$ are

$$\begin{aligned} e_{1,0} &= 4\tau - \frac{16\pi^2}{9} \tau^3 + \frac{8\pi^4}{25} \tau^5, & e_{2,0} &= -\frac{896\pi^4}{2025} \tau^6 + O(\tau^8), \\ e_{1,1} &= -\frac{8\pi^2}{9} \tau^3 + \frac{16\pi^4}{75} \tau^5 - \frac{16\pi^6}{735} \tau^7, & e_{2,1} &= -\frac{2816\pi^8}{2480625} \tau^{10} + O(\tau^{12}), \end{aligned}$$

with $e_{3,0} = O(\tau^{15})$ and $e_{3,1} = O(\tau^{21})$. The leading rank-one part gives one root $z = 4\tau + O(\tau^3)$ for $r = 0$, respectively $z = -8\pi^2 \tau^3/9 + O(\tau^5)$ for $r = 1$. The identity $z = e_1 - e_2/z + e_3/z^2$ gives the two dominant expansions. The remaining roots have sums $-224\pi^4 \tau^5/2025 + O(\tau^7)$ and $352\pi^6 \tau^7/275625 + O(\tau^9)$, and products $O(\tau^{14})$ and $O(\tau^{18})$, respectively. The quadratic formula gives the other displayed edges and residual roots $O(\tau^9)$, $O(\tau^{11})$. Minimax with zero-completed signed lists and (4.19) transfers these statements to the true operators, including simplicity. $\square$

Theorem 4.5 (Endpoint packets). *Let a_N be unit vectors whose step embeddings $U_N a_N$ converge strongly to ϕ , and put $f_N(t) = \sum_{|m| \leq N} a_{N,m} e^{2\pi i m t}$, $\Phi(s) = \int_{-1}^1 \phi(x) e^{2\pi i x s} dx$. Zero-extending the rescaled function from $[-N/2, N/2]$ gives*

$$N^{-1/2} f_N(s/N) \longrightarrow \Phi(s) \quad \text{in } L^2(\mathbb{R}). \quad (4.20)$$

Consequently $f_N \rightharpoonup 0$ in $L^2(0, 1)$, and

$$|f_N(t)|^2 dt \longrightarrow w_0 \delta_0 + w_1 \delta_1, \quad w_0 = \int_0^\infty |\Phi|^2, \quad w_1 = \int_{-\infty}^0 |\Phi|^2. \quad (4.21)$$

For vectors in one fixed reflection sector, $w_0 = w_1 = 1/2$. The required strong convergence holds for phase-aligned eigenvectors at a simple parity eigenvalue λ of $\mathcal{K}_\tau$ separated within that sector by $g > 0$: at $\varepsilon = \tau/N$, when $\nu_N = (2\tau + 8\pi\tau^2)/N < g/4$, the corresponding Riesz projections satisfy

$$\|\tilde{P}_{N,\tau,\lambda} - P_{\tau,\lambda}\| \leq 4\nu_N/g. \quad (4.22)$$

For a finite isolated cluster, the projections and suitably aligned orthonormal cluster bases converge; at crossings one follows the cluster.

Proof. Set $h = 2/(2N+1)$, $\beta_N = Nh$. Exact cell integration gives

$$N^{-1/2} f_N(s/N) = \beta_N^{-1/2} \frac{\widehat{U_N a_N}(s/\beta_N)}{\text{sinc}(s/N)}, \quad |s| \leq N/2,$$

with $\text{sinc}(z) = \sin(\pi z)/(\pi z)$. Plancherel and strong continuity of dilations handle the numerator. The zero-extended multipliers $\mathbf{1}_{|s| \leq N/2} / \text{sinc}(s/N)$ are bounded by $\pi/2$ and converge pointwise to one, hence strongly on L^2 . This proves (4.20); its squared moduli converge in L^1 . Positive and negative s correspond to $0+$ and $1-$, giving the endpoint measure and tightness on scale N^{-1} . Concentration in shrinking neighborhoods and absolute continuity of the L^2 integral give weak convergence to zero. Parity makes $|\Phi|^2$ even. Finally the resolvent identity on $|z - \lambda| = g/2$, with resolvent bounds $2/g$ and $4/g$, gives (4.22). Projections at distance less than one have equal rank; normalizing $\tilde{P}_N \phi$ gives the simple eigenvector. For clusters, the same contour argument followed by the near-identity unitary between the two projections gives aligned bases. $\square$

4.2.1 One exchange estimate for both finite and continuum edges

Let P, Q be orthogonal projections, S unitary, and ϱ a self-adjoint unitary, satisfying

$$[S, P] = 0, \quad QS^2Q = 0, \quad [\varrho, P] = [\varrho, Q] = 0, \quad \varrho S = S^* \varrho, \quad PQP \text{ compact.}$$

Put $\Sigma = S^2$, $R_- = SQS^*$, $R_+ = S^*QS$, $R = R_- + R_+$, $J = R_+ \Sigma^* R_- + R_- \Sigma R_+$, $\Pi_\pm = (R \pm J)/2$, and, on $\text{ran } P$,

$$\mathcal{A} = PJP, \quad \mathcal{B} = PRP, \quad \mathcal{C} = PQP, \quad \mathcal{C}_\pm = P\Pi_\pm P, \quad \mathcal{E}_r = \ker(\varrho - rI).$$

Assume $\text{ran } P \cap \mathcal{E}_r$ and $\text{ran } Q \cap \mathcal{E}_r$ are nonzero in each sector used; both concrete frames below have this property. In this common frame, $\lambda_p^\pm$ denote the spectral extrema of $\mathcal{A}|_{\text{ran } P \cap \mathcal{E}_p}$. On $L^2(\mathbb{R})$ take $P = UU^*$, $Q = \mathbf{1}_{(-\tau/2, \tau/2)}$, $Sg(\xi) = g(\xi + \tau/2)$; on $L^2(\mathbb{T})$ take $P = P_N$, $Q = \mathbf{1}_{(-\varepsilon/2, \varepsilon/2)}$, $Sf(t) = f(t + \varepsilon/2)$. Then

$$(\mathcal{A}, \mathcal{B}, \mathcal{C}) = \begin{cases} (\mathcal{K}_\tau, C_{2\tau}, C_\tau) & \text{on the line,} \\ (A_N, C_N(2\varepsilon), C_N(\varepsilon)) & \text{on the circle.} \end{cases} \quad (4.23)$$

Here ϱ is reflection, centered arcs are read modulo one, and $C_N(\ell)_{mn} = \sin(\pi \ell(m-n))/[\pi(m-n)]$, with diagonal ℓ . Translation commutes with P in either Fourier representation; the shifted half-bands are disjoint. Fourier integration gives the three operators in the two cases, with $(\text{tr } \mathcal{C}, \text{tr } \mathcal{B}) = (2\tau, 4\tau)$ and $(d\varepsilon, 2d\varepsilon)$.

Lemma 4.6 (Exchange and dual concentration). *One has $J^2 = R$, $\Pi_\pm$ are orthogonal projections, $\mathcal{C}_+ + \mathcal{C}_- = \mathcal{B}$, $\mathcal{C}_+ - \mathcal{C}_- = \mathcal{A}$, and $\mathcal{A}^2 \leq \mathcal{B}$. The maps $\Gamma_\pm v = (Sv \pm S^*v)/\sqrt{2}$ are isometries from $\text{ran } Q$ onto $\text{ran } \Pi_\pm$ and send parity r to $\pm r$. For $v \in \text{ran } Q$, define*

$$\Theta_\pm(v) = \frac{1}{2} \|(1 \pm \Sigma)(1 - P)v\|^2, \quad \Theta_0(v) = \|(1 - P)v\|^2, \quad \mu_r = \|\mathcal{C}|_{\mathcal{E}_r}\|, \quad \eta_r^2 = 1 - \mu_r,$$

and let $\theta_\pm^r$ be the infimum of $\Theta_\pm$ over unit vectors of $\text{ran } Q \cap \mathcal{E}_r$. Then

$$1 - \|\mathcal{C}_\pm|_{\mathcal{E}_p}\| = \theta_\pm^{\pm p}, \quad \inf_{\text{ran } Q \cap \mathcal{E}_r, \|v\|=1} \Theta_0(v) = \eta_r^2, \quad \theta_+^r + \theta_-^r \leq 2\eta_r^2. \quad (4.24)$$

For $\mu_r > 1/2$,

$$\theta_+^r \leq 1 - \lambda_r^+ \leq \frac{2\mu_r}{2\mu_r - 1} \eta_r^2, \quad \theta_-^r \leq 1 + \lambda_{-r}^- \leq \frac{2\mu_r}{2\mu_r - 1} \eta_r^2. \quad (4.25)$$

The lower bounds have no restriction on μ_r ; the upper factor is at most 3 when $\mu_r \geq 3/4$.

Proof. $QS^{\pm 2}Q = 0$ gives $R_-R_+ = 0$ and $J^2 = R$, hence the projection identities; $\mathcal{A}^2 \leq PJ^2P = \mathcal{B}$. The vectors Sv, S^*v are orthogonal, and $J\Gamma_\pm v = \pm \Gamma_\pm v$. Conversely $JF = \pm F$, $RF = F$ give $F = \Gamma_\pm(\sqrt{2}S^*R_-F)$. The adjoints are $\Gamma_\pm^* = Q(S^* \pm S)/\sqrt{2}$. Commuting P past S gives $\|(1 - P)\Gamma_\pm v\|^2 = \Theta_\pm(v)$ and $\Theta_+ + \Theta_- = 2\Theta_0$. The variational formula for $\|P\Pi P\|$ over unit

vectors of $\text{ran } \Pi$ proves (4.24); $\mathcal{A} \leq \mathcal{C}_+$ and $-\mathcal{A} \leq \mathcal{C}_-$ give the lower comparisons. Compactness of QP , hence of $\Pi_\pm P$, ensures attainment of the required extrema.

Choose a unit $V \in \text{ran } P \cap \mathcal{E}_r$ with $PQV = \mu V$, $\mu = \mu_r > 1/2$, and put $u = QV/\sqrt{\mu}$, $w = (1 - Q)V$, $\eta^2 = 1 - \mu$, $\rho = \Re\langle V, \Sigma V \rangle$. Since $\langle u, \Sigma u \rangle = 0$ and Σ preserves $P \oplus (1 - P)$, $|\rho| \leq \eta^2/\mu$. Thus $g_\pm = (S \pm S^*)V/2$ have

$$\|g_\pm\|^2 = \frac{1}{2}(1 \pm \rho) \geq (2\mu - 1)/(2\mu).$$

Write $Y_\pm = \|Q\Sigma^{\pm 1}V\|^2 = \|Q\Sigma^{\pm 1}w\|^2$. The identities $S^*RS = Q + \Sigma^*Q\Sigma$, $SRS^* = Q + \Sigma Q\Sigma^*$ and $S^*RS^* = Q\Sigma^* + \Sigma^*Q$ give

$$\|(1 - R)g_\pm\|^2 = \frac{1}{2}\eta^2 \pm \frac{1}{2}\Re\langle w, \Sigma w \rangle - \frac{1}{4}(Y_+ + Y_-).$$

Applying the projection SRS^* to $(1 \mp \Sigma)w$, using $Qw = 0$, gives $Y_+ + Y_- \leq 2\eta^2 \mp 2\Re\langle w, \Sigma w \rangle$. Hence

$$\|(1 - R)g_\pm\|^2 \leq \eta^2 - \frac{1}{2}(Y_+ + Y_-), \quad 2\|\Pi_\mp g_\pm\|^2 = \frac{1}{4}\|Q(\Sigma^* - \Sigma)V\|^2 \leq \frac{1}{4}(Y_+ + Y_-).$$

Their sum is $\|g_\pm\|^2 \mp \langle g_\pm, Jg_\pm \rangle \leq \eta^2$. Division by the displayed norm lower bound and minimax prove (4.25). $\square$

4.2.2 Uniform rates for all four edges

Theorem 4.7 (Concentration bounds). *The half-band leakages in (4.25) satisfy*

$\text{continuum, } \tau \geq 3/2$	$\frac{\eta_{+1}^2}{27/(64\pi^2\tau^3)}$	$\frac{\eta_{-1}^2}{108/(25\pi^2\tau^3)}$	as upper bounds.
$\text{finite, } 0 < \varepsilon < 1/2$	$1/(d\varepsilon)$	$3/(N\varepsilon) + 1/[4(N\varepsilon)^3]$	

(4.26)

Both continuum leakages are at most $1/4$ when $\tau \geq 3/2$; both finite leakages are at most $1/4$ when $d\varepsilon \geq 25$. For the finite even and odd bounds, respectively, $\mu_r > 1/2$ already holds when $d\varepsilon > 2$ and $d\varepsilon \geq 13$. Thus the factor in (4.25) is at most 3 in the first two stated ranges. More strongly, put $\tau_N = N\varepsilon$ and $c_* = (2/5)\log 16$. For $\tau_N \geq 20$,

$$\eta_r^2 \leq 283000\sqrt{\tau_N}e^{-c_*\tau_N}, \quad 1 - \lambda_{N,p}^+, \quad 1 + \lambda_{N,p}^- \leq 850000\sqrt{\tau_N}e^{-c_*\tau_N} \quad (r, p = \pm 1). \quad (4.27)$$

All finite constants are independent of N, ε .

Proof. In the continuum use the normalized even and odd trials $\cos(\pi x/2)$ and $\sin(\pi x)$, whose Fourier transforms are $4\cos(2\pi\xi)/[\pi(1 - 16\xi^2)]$ and $2i\sin(2\pi\xi)/[\pi(1 - 4\xi^2)]$. For $|\xi| \geq \tau/2 \geq 3/4$, their squares are at most $81/(1024\pi^2\xi^4)$ and $81/(100\pi^2\xi^4)$. Integrating the two tails, using $2\int_{\tau/2}^\infty \xi^{-4}d\xi = 16/(3\tau^3)$, gives the first row. In the circle use $D_N/\sqrt{d}$ and $D'_N/\|D'_N\|_2$, where

$$D_N(x) = \frac{\sin(\pi dx)}{\sin(\pi x)}, \quad \|D'_N\|_2^2 = \frac{4\pi^2 N(N+1)d}{3}, \quad |D_N| \leq \frac{1}{2|x|}, \quad |D'_N| \leq \frac{\pi d}{2|x|} + \frac{\pi}{4x^2}.$$

Integrating outside $|x| < \varepsilon/2$, and using $(a+b)^2 \leq 2a^2 + 2b^2$, gives the second row: the odd bound before simplification is $3d/[2N(N+1)\varepsilon] + 1/[2N(N+1)d\varepsilon^3]$. Here $d \leq 2(N+1)$ and $(N+1)d \geq 2N^2$. The threshold assertions follow from $2N\varepsilon > d\varepsilon - 1/2$: the decreasing function $3/s + 1/(4s^3)$ is below $1/2$ at $s = 25/4$, and below $1/4$ at $s = 49/4$. The continuum values are below $1/4$ at $\tau = 3/2$.

For the exponential estimate set $m = \lceil 2/\varepsilon \rceil$, $h = 2m + 1$, $a = h\varepsilon$, $k = \lfloor (N-1)/m \rfloor$ and $q(x) = D_m(x)/h$. Then

$$4 \leq a \leq 11/2, \quad (2/5)\tau_N - 6/5 < k \leq \tau_N/2, \quad k \geq 2.$$

These inequalities use $2/\varepsilon \leq m < 2/\varepsilon + 1$ and $\varepsilon < 1/2$. The even and odd polynomials $F_+ = q^k$, $F_- = \sin(2\pi x)q^k$ have degree at most $mk + 1 \leq N$. Put $\delta = (\pi h \sqrt{k})^{-1} < 1/4$. On $|x| \leq \delta$, sine's Taylor lower bound and Bernoulli's inequality give $q^{2k} \geq (1 - 1/(6k))^{2k} \geq 2/3$, while $|\sin(2\pi x)| \geq 4|x|$. Thus

$$\|F_+\|_2^2 \geq 4\delta/3, \quad \|F_-\|_2^2 \geq 64\delta^3/9.$$

On $0 < |x| \leq 1/2$, $|q(x)| \leq (2h|x|)^{-1}$. Using also $|\sin(2\pi x)| \leq 2\pi|x|$ and extending the tail to infinity,

$$\int_{|x| > \varepsilon/2} |F_+|^2 \leq \frac{\varepsilon a^{-2k}}{2k-1}, \quad \int_{|x| > \varepsilon/2} |F_-|^2 \leq \frac{\pi^2 \varepsilon^3 a^{-2k}}{2k-3}.$$

The ratios bound the respective leakages:

$$B_+ = \frac{3\pi a \sqrt{k}}{4(2k-1)} a^{-2k} \leq B_- = \frac{9\pi^5 a^3 k^{3/2}}{64(2k-3)} a^{-2k}.$$

Since $k/(2k-3) \leq 2$ and $4 \leq a \leq 11/2$, both are at most

$$\frac{9\pi^5}{32\sqrt{2}} \left(\frac{11}{2}\right)^3 16^{6/5} \sqrt{\tau_N} e^{-c_* \tau_N} < 283000 \sqrt{\tau_N} e^{-c_* \tau_N}.$$

Finally $d\varepsilon \geq 40$ ensures the factor 3 in (4.25), and $3 \cdot 283000 < 850000$. The algebraic leakages tend to zero when $N\varepsilon_N \rightarrow \infty$, proving the last regime in (4.16). $\square$

Corollary 4.8 (Lower comparisons and the continuum exponent). *The weighted leakages have the concrete forms*

$$\Theta_{\pm}(v) = \begin{cases} \int_{|x| > 1} (1 \pm \cos(2\pi\tau x)) |\mathcal{F}^* v(x)|^2 dx, & \text{on the line,} \\ \sum_{|m| > N} (1 \pm \cos(2\pi m\varepsilon)) |\hat{v}(m)|^2, & \text{on the circle.} \end{cases} \quad (4.28)$$

Removing the cosine weight gives Θ_0 . If $\varepsilon = a/q$ in lowest terms and q is odd, then

$$1 - \lambda_{N,r}^+ \geq 2 \sin^2(\pi/(2q)) \eta_r^2 \geq 2\eta_r^2/q^2. \quad (4.29)$$

If $d\varepsilon$ is an integer, then $q \mid d$ and q can be replaced by d in the last denominator. In the continuum,

$$\begin{aligned} 1 - \lambda_{+1}^+, \quad 1 + \lambda_{-1}^- &\leq (8\pi + o(1)) \sqrt{\tau} e^{-2\pi\tau}, \\ 1 - \lambda_{-1}^+, \quad 1 + \lambda_{+1}^- &\leq (64\pi^2 + o(1)) \tau^{3/2} e^{-2\pi\tau}. \end{aligned} \quad (4.30)$$

Both defects in physical parity p are at least $1 - \lambda_0(2\pi\tau)$ if $p = +1$, and $1 - \lambda_1(2\pi\tau)$ if $p = -1$, where $\lambda_n(c)$ are the classical prolate eigenvalues for $\sin(c(x-y))/[\pi(x-y)]$.

Proof. Fourier transformation of Σ gives (4.28). For odd q , the distance of ma/q from $1/2$ modulo one is at least $1/(2q)$; hence $1 + \cos(2\pi m\varepsilon) \geq 2 \sin^2(\pi/(2q)) \geq 2/q^2$. Infimizing and using (4.25) proves (4.29). Prolate eigenfunctions alternate parity [45], and the fixed- n law [19, 43]

$$1 - \lambda_n(c) \sim \frac{\sqrt{\pi} 2^{3n+2}}{n!} c^{n+1/2} e^{-2c}$$

gives $\eta_{+1}^2 \sim 4\pi \sqrt{\tau} e^{-2\pi\tau}$ and $\eta_{-1}^2 \sim 32\pi^2 \tau^{3/2} e^{-2\pi\tau}$. The factor in (4.25) tends to 2. The lower bounds follow from $-C_{2\tau} \leq \mathcal{K}_{\tau} \leq C_{2\tau}$ in each parity. $\square$

These are one-sided physical rates: the lower comparison has exponent 4π , the upper comparison 2π . A uniform inequality $\theta_{\pm}^r \geq c_0 \eta_r^2$ is not proved; the cosine weights vanish, or have infimum zero for irrational ε , so pointwise comparison cannot establish it. The finite exponent is not asserted sharp. At $\varepsilon = \tau/N$, the argument of Theorem 2.5 applied to c_{τ} gives embedded error $(\tau + 2\pi\tau^2)/N$ for $C_N(\varepsilon) \rightarrow C_{\tau}$, and $(2\tau + 8\pi\tau^2)/N$ for $C_N(2\varepsilon) \rightarrow C_{2\tau}$. Indeed $|c_{\tau}| \leq \tau$, $|c'_{\tau}| \leq \pi\tau^2/2$. Sector eigenvalues, η_r^2 and the dual quantities therefore converge; for the last use $\mathcal{C}_{\pm} = (\mathcal{B} \pm \mathcal{A})/2$ and (4.24).

4.2.3 Global finite defects and Lipschitz statistics

Put $\Delta(\theta) = [3 + 2\log^+(2\pi\theta)]/\pi^2$, where $\log^+ x = \max(\log x, 0)$.

Theorem 4.9 (Defect, trace and Wasserstein bounds). *For $x = d\varepsilon$, $N \geq 1$ and $0 < \varepsilon < 1/2$, put*

$$B(x) = \min\{2x, 3 + 2\log^+(2x), 8\pi^{-2}[3 + \log^+(\pi x)]\}. \quad (4.31)$$

Then $0 \leq 2d\varepsilon - \text{tr } A_N^2 \leq B(x)$, and

$$t_N := \text{tr } A_N = \frac{2\varepsilon \sin(\pi d\varepsilon)}{\sin(\pi\varepsilon)}, \quad |t_N| \leq \min(2x, 1). \quad (4.32)$$

Every real L -Lipschitz f with $f(0) = 0$ satisfies

$$|\text{tr } f(A_N) - x[f(1) + f(-1)]| \leq L[2B(x) + |t_N|] \leq L[7 + 4\log^+(2x)], \quad (4.33)$$

$$|\text{tr } f(\mathcal{K}_\tau) - 2\tau[f(1) + f(-1)]| \leq L[8\Delta(\tau) + 2/\pi] \quad (\tau > 0). \quad (4.34)$$

For $\tau \geq 2$, the last upper bound can be replaced by $L[4\pi^{-2}\log \tau + 2.712]$. For the empirical probability measure $\mu_N = d^{-1} \sum_j \delta_{\lambda_j(A_N)}$ and $\nu_\varepsilon = (1 - 2\varepsilon)\delta_0 + \varepsilon(\delta_1 + \delta_{-1})$,

$$W_1(\mu_N, \nu_\varepsilon) \leq \min\{4\varepsilon, [7 + 4\log^+(2d\varepsilon)]/d\}. \quad (4.35)$$

In particular μ_N converges weakly to ν_ε at fixed ε .

Proof. For $X = C_+ - C_-$ with positive trace-class $C_\pm$, $\|X\| \leq 1$ and $s = \text{tr}(C_+ + C_-)$, scalar interpolation gives

$$|\text{tr } f(X) - \frac{s}{2}[f(1) + f(-1)]| \leq 2L(s - \text{tr } X^2) + L|\text{tr } X|. \quad (4.36)$$

Indeed put $a = [f(1) + f(-1)]/2$, $b = [f(1) - f(-1)]/2$. On each half interval, $|f(t) - a|t| - bt| \leq 2L|t|(1 - |t|)$, by writing, for $t \geq 0$, $f(t) - tf(1) = (1 - t)f(t) + t(f(t) - f(1))$. The spectral projection onto the positive half-line gives $\text{tr } X_+ \leq \text{tr } C_+$, and likewise $\text{tr } X_- \leq \text{tr } C_-$; hence $T = \text{tr } |X| \leq s$. Absolute summability follows from $|f(t)| \leq L|t|$. The trace error is at most $2L(T - \text{tr } X^2) + L(s - T) + L|\text{tr } X|$, proving (4.36).

The half-band concentration defect is at most $\Delta(\tau)$ on the line and $3/4 + \frac{1}{2}\log^+(2d\varepsilon)$ on the circle. Indeed on the line it equals $2 \int_0^\infty c_\tau(u)^2 \min(u, 2) du$; splitting the bound $c_\tau^2 \leq \min(\tau^2, (\pi u)^{-2})$ at $(\pi\tau)^{-1}$, 2 gives $\Delta(\tau)$ (for $2\pi\tau < 1$, use $8\tau/\pi - 4\tau^2 \leq 3/\pi^2$). On the circle it equals $2 \int_0^{1/2} |D_N(t)|^2 \min(\varepsilon, t) dt$. Splitting $|D_N(t)|^2 \leq \min(d^2, (2t)^{-2})$ at $1/(2d)$, ε gives $3/4 + \frac{1}{2}\log(2d\varepsilon)$ when $d\varepsilon \geq 1/2$; otherwise it gives $2d\varepsilon - \varepsilon - d^2\varepsilon^2 \leq 3/4$.

In the exchange frame,

$$0 \leq \text{tr } \mathcal{B} - \text{tr } \mathcal{A}^2 = \|(1 - P)JP\|_2^2 \leq 4\text{tr}(\mathcal{C} - \mathcal{C}^2).$$

For the last inequality use $J = \Sigma^* R_- + \Sigma R_+$: each $(1 - P)R_\pm P$ has squared Hilbert–Schmidt norm $\text{tr}(\mathcal{C} - \mathcal{C}^2)$ by unitary translation. This proves the middle bound in (4.31) and the continuum estimate; the first bound $2x$ is immediate. For the remaining finite bound, precisely $\min(|k|, d)$ pairs with difference $k \neq 0$ cross the Fourier cutoff. The divided-difference entries of the full exchange therefore give

$$2x - \text{tr } A_N^2 \leq \frac{8}{\pi^2} \sum_{k \geq 1} \frac{\min(k, d) \min(1, \pi^2 \varepsilon^2 k^2)}{k^2}.$$

If $1 \leq k_0 = (\pi\varepsilon)^{-1} \leq d$, the ranges $k \leq k_0$, $k_0 < k \leq d$, $k > d$ contribute at most 1, $1 + \log(d/k_0)$ and 1. If $k_0 < 1$, the sum is at most $2 + \log d < 3 + \log(\pi x)$. If $k_0 > d$, use $2x < 2/\pi < 24/\pi^2$ instead. This proves $B(x)$. The finite cosine sum gives (4.32), using $\sin(\pi\varepsilon) \geq 2\varepsilon$. Apply (4.36) with $s = 2x$ or 4τ . For the sharper continuum bound the exact quadratic formula gives $4\tau - \text{tr } \mathcal{K}_\tau^2 \leq 2\pi^{-2}\log \tau + 1.0375$ for $\tau \geq 2$; thus the remaining constant is at most $2.075 + 2/\pi < 2.712$. Finally apply the finite estimate to $f = \varphi - \varphi(0)$ and use the Kantorovich dual formula. Transport through δ_0 costs at most 2ε for each measure, because $\text{tr } |A_N| \leq 2d\varepsilon$, proving the other Wasserstein bound. $\square$

4.2.4 Counts and finite inertia

Write $\mathcal{N}_{p,\sigma}(\alpha)$ for the number of eigenvalues in parity p with $\sigma\lambda > \alpha$; omission of p sums both parities. The trace budget gives the following direct alternative to counting concentrated subspaces.

Theorem 4.10 (All-scale signed counts). *For $0 < \alpha < 1$ put $q_\alpha = \max\{\alpha^{-1}, (1-\alpha)^{-1}\}$. At every $\tau > 0$ and every $N \geq 1$, $0 < \varepsilon < 1/2$,*

$$\begin{aligned} |\mathcal{N}_{p,\sigma}(\alpha; \tau) - \tau| &\leq 2 + 4q_\alpha \Delta(\tau), \\ |\mathcal{N}_{N,p,\sigma}(\alpha; \varepsilon) - d\varepsilon/2| &\leq 2 + q_\alpha B(d\varepsilon). \end{aligned} \quad (4.37)$$

Without parity the corresponding bounds are

$$|\mathcal{N}_\sigma(\alpha; \tau) - 2\tau| \leq \pi^{-1} + 4q_\alpha \Delta(\tau), \quad |\mathcal{N}_{N,\sigma}(\alpha; \varepsilon) - d\varepsilon| \leq \frac{1}{2} + q_\alpha B(d\varepsilon). \quad (4.38)$$

More precisely, in each displayed estimate $|N-m| \leq c+q_\alpha D_*$, one has $-c-D_*/(1-\alpha) \leq N-m \leq c+D_*/\alpha$. Thus each parity count is $\tau + O_\alpha(\log(2+\tau))$, respectively $d\varepsilon/2 + O_\alpha(1+\log(2+d\varepsilon))$, uniformly in N, ε . The finite inertia in each parity satisfies $n_\sigma \geq d\varepsilon/2 - 2 - B(d\varepsilon)/2$, and without parity $n_\sigma \geq d\varepsilon - 1/2 - B(d\varepsilon)/2$. For every $N \geq 1$ the even block has both signs and the odd block has a negative eigenvalue; for $N = 1$ the odd block has no positive eigenvalue.

Proof. For a contraction $X = C_+ - C_-$ with positive trace-class summands, put $s = \text{tr}(C_+ + C_-)$, $T = \text{tr}|X| \leq s$ and $D = s - \text{tr} X^2$. For $0 \leq r \leq 1$, $-r(1-r)/(1-\alpha) \leq \mathbf{1}_{r>\alpha} - r \leq r(1-r)/\alpha$. Sum over the sign- σ eigenvalues. With $E = T - \text{tr} X^2$, $F = s - T$, and $\text{tr} X_\sigma = (T + \sigma \text{tr} X)/2$, subtracting $F/2$ from the error gives, since $E + F = D$ and $(1-\alpha)^{-1} \geq 1/2$,

$$-\frac{D}{1-\alpha} \leq N_\sigma(\alpha) - \frac{s}{2} - \frac{\sigma \text{tr} X}{2} \leq \frac{D}{\alpha}. \quad (4.39)$$

This also implies the symmetric estimate with q_α . The same inequalities hold for the nonstrict count $\sigma\lambda \geq \alpha$; at $r = \alpha$ the scalar upper bound is an equality. Also $n_\sigma \geq \text{tr} X_\sigma \geq (s - D + \sigma \text{tr} X)/2$. Apply the inequalities separately in every parity sector. For our exchange compressions, $B = C_+ + C_-$ satisfies $B - X^2 \geq 0$ by the compression identity in the preceding proof; hence each sector defect is at most the full defect.

The full traces are $s = 4\tau, 2d\varepsilon$ and $|\text{tr} X| \leq 2/\pi, 1$. The reflected continuum traces are $\text{tr}(\mathcal{R}C_{2\tau}) = \text{Si}(4\pi\tau)/\pi$ and $\text{tr}(\mathcal{R}C_\tau) = 2\text{Si}(2\pi\tau)/\pi$. Since $0 < \text{Si}(u) \leq \text{Si}(\pi) < \pi$ by decreasing alternating humps, the parity correction $(|\text{tr} \mathcal{R}B| + |\text{tr} X| + |\text{tr} \mathcal{R}X|)/4$ is less than two. For the finite traces, write $S_N(\theta) = \sum_{j=1}^N \sin(j\theta)/j$. Exact reflected diagonals give

$$\text{tr}(\mathcal{R}_N C_N(2\varepsilon)) = 2\varepsilon + S_N(4\pi\varepsilon)/\pi, \quad \text{tr}(\mathcal{R}_N A_N) = 2\varepsilon + 2S_N(2\pi\varepsilon)/\pi.$$

One has $|S_N(\theta)| \leq 1 + \pi$: reduce to $0 < \theta \leq \pi$, bound the terms $j \leq 1/\theta$ by a total of one, and Abel-sum the remaining terms, using the geometric-sum bound $1/\sin(\theta/2)$. Their modulus is at most $\theta/\sin(\theta/2) \leq \pi$. Thus the finite parity correction is at most $(6 + 3/\pi)/4 < 2$. Apply the preceding scalar inequalities and the defect bounds already proved to obtain every stated estimate.

For the exact finite signs put $\vartheta = 2\pi\varepsilon \in (0, \pi)$. The even vector e_0 has form $2\varepsilon > 0$; the odd vector $e_1 - e_{-1}$ has Rayleigh quotient $(\vartheta \cos \vartheta - \sin \vartheta)/\pi < 0$, since the numerator has derivative $-\vartheta \sin \vartheta$. The even compression to $e_0, (e_1 + e_{-1})/\sqrt{2}$ is

$$\pi^{-1} \begin{pmatrix} \vartheta & \sqrt{2} \sin \vartheta \\ \sqrt{2} \sin \vartheta & \vartheta \cos \vartheta + \sin \vartheta \end{pmatrix}.$$

Its determinant has the sign of $q = \vartheta^2 \cos \vartheta + \vartheta \sin \vartheta - 2 \sin^2 \vartheta$. Writing $u = \sin \vartheta / \vartheta$, for $0 < \vartheta \leq 2$ use $u \geq 1 - \vartheta^2/6 \geq 1/3$, $\cos \vartheta \leq 1 - \vartheta^2/2 + \vartheta^4/24$, and monotonicity of $2u^2 - u$ for $u \geq 1/3$ to obtain $-q/\vartheta^2 = 2u^2 - u - \cos \vartheta \geq \vartheta^4/72 > 0$. For $2 \leq \vartheta < \pi$, use $2u^2 - u \geq -1/8$ and $\cos \vartheta \leq \cos 2 \leq -1/3$. Thus the even compression is indefinite. At $N = 1$ the odd space is the one-dimensional span already tested. $\square$

4.3 Signed quadratic products and finite oscillations

Put $D_R = 4\tau - \text{tr } C_{2\tau}^2$, $D_J = 4\tau - \text{tr } \mathcal{K}_\tau^2$ and $S = \text{tr}(\mathcal{K}_\tau - \mathcal{K}_\tau C_{2\tau})$. The operators $C_\pm = (C_{2\tau} \pm \mathcal{K}_\tau)/2$ are positive concentrations, not the positive/negative spectral parts of $\mathcal{K}_\tau$.

Proposition 4.11 (Signed quadratic constants). *Exactly,*

$$\text{tr}(C_\pm - C_\pm^2) = \frac{1}{4}(D_R + D_J \pm 2S), \quad \text{tr}(C_+ C_-) = \frac{1}{4}(D_J - D_R). \quad (4.40)$$

For $P = UU^*$, $A = (1 - P)R_\tau P$ and $B = (1 - P)J_\tau P$, $D_R = \|A\|_2^2$, $D_J = \|B\|_2^2$ and $S = \Re\langle A, B \rangle_{\text{HS}}$. For every $\tau > 0$,

$$S = \frac{4 \log 2}{\pi^2} \cos(2\pi\tau) + R_2, \quad |R_2| \leq \frac{3}{\pi^3 \tau}. \quad (4.41)$$

Writing $c_+ = \cos(\pi\tau)$, $c_- = \sin(\pi\tau)$, it follows that

$$\text{tr}(C_\pm - C_\pm^2) = \frac{3 \log \tau + \log(2\pi^3) + 3(1 + \gamma) + 16(\log 2)c_\pm^4}{4\pi^2} + r_\pm, \quad (4.42)$$

$$\text{tr}(C_+ C_-) = \frac{\log(2\pi\tau) + 1 + \gamma + 2(\log 2) \cos(4\pi\tau)}{4\pi^2} + r_0, \quad (4.43)$$

where $|r_\pm| \leq 21/(8\pi^3\tau)$ and $|r_0| \leq 9/(8\pi^3\tau)$.

Proof. Expand $C_\pm^2$ and use trace cyclicity. Since $\Pi_\pm = (R_\tau \pm J_\tau)/2$ are projections, their compressed defects are $\|(1 - P)\Pi_\pm P\|_2^2 = \|A \pm B\|_2^2/4$, which also proves the Hilbert–Schmidt identities. For (4.41), put $t = 2\pi\tau$ and use $u = x - y, v = x + y$ in the product-kernel trace. Integrating first in v gives

$$\text{tr}(\mathcal{K}_\tau C_{2\tau}) = \frac{4}{\pi^2} \left[\sin t \int_0^t \frac{\sin^2(2w)}{2w^2} dw - \cos t \int_0^t \frac{\sin^2 w \sin 2w}{w^2} dw \right].$$

The infinite integrals are $\pi/2$ and $\log 2$ (Lemma 3.14); their tails have modulus at most $1/(2t)$ and $1/t$. Subtracting from $\text{tr } \mathcal{K}_\tau = 2 \sin t/\pi$ proves (4.41). Insert the exact quadratic and prolate defect formulas in (4.40). With $c = \cos(2\pi\tau)$, the oscillating numerator is $2(\log 2)(2c^2 - 1) \pm 8(\log 2)c = 4(\log 2)(1 \pm c)^2 - 6 \log 2$. The error sums are $(1/2 + 4 + 6)/4 = 21/8$ and $(1/2 + 4)/4 = 9/8$ in units $(\pi^3\tau)^{-1}$. In particular $D_R = \text{tr}(C_+ - C_+^2) + \text{tr}(C_- - C_-^2) - 2 \text{tr}(C_+ C_-)$. $\square$

Theorem 4.12 (Finite quadratic traces). *Let $s_k = \sin^2(\pi\epsilon k)/(\pi^2 k^2)$, $s_0 = \epsilon^2$. For $N \geq 1$, $d = 2N + 1$, $0 < \epsilon < 1/2$, exactly*

$$\text{tr } A_N^2 = \sum_{|k| \leq 2N} s_k \left[2(d - |k|) + \frac{2 \sin(2\pi\epsilon(d - |k|))}{\sin(2\pi\epsilon)} \right]. \quad (4.44)$$

For each $\delta \in (0, 1/2)$, uniformly in $0 < \epsilon \leq 1/2 - \delta$ and $N \geq 1$,

$$2d\epsilon - \text{tr } A_N^2 = \frac{2}{\pi^2} [\log(4N \sin(\pi\epsilon)) + 1 + \gamma + (\log 2) \cos(2\pi\epsilon d)] + O_\delta \left(\epsilon^2 + \frac{1}{N\epsilon} \right). \quad (4.45)$$

For $\Pi_W = C_N(2W)$, uniformly in $N \geq 1$, $0 < W \leq 1/4$,

$$\text{tr}(\Pi_W - \Pi_W^2) = \frac{\log(4N \sin(2\pi W)) + 1 + \gamma}{\pi^2} + O((NW)^{-1}), \quad (4.46)$$

with an absolute constant, including the endpoint $W = 1/4$.

Proof. Put $K = 2N$, $\theta = 2\pi\varepsilon$, $h = \pi\varepsilon$. The entrywise square is $4s_{m-n}\cos^2(h(m+n))$. For fixed $k = m - n$, the $M = d - |k|$ sums $m + n$ form the progression $-(M-1), -(M-3), \dots, M-1$; its cosine sum is $\sin(M\theta)/\sin\theta$, proving (4.44). Write its unmodulated and modulated parts as Σ_1, Σ_2 . Parseval for an arc gives $\sum_{k \in \mathbb{Z}} s_k = \varepsilon$, so

$$2d\varepsilon - \Sigma_1 = 4d \sum_{k > K} s_k + 4 \sum_{k=1}^K k s_k.$$

For decreasing $b_k \downarrow 0$, geometric and Abel summation give $|\sum_{k > K} b_k \cos(k\theta)| \leq b_{K+1}/\sin(\theta/2)$. Using $b_k = k^{-2}, k^{-1}$, $\sin(\pi\varepsilon) \geq 2\varepsilon$, $\sum_{k > K} k^{-2} = K^{-1} + O(K^{-2})$ and $H_K = \log K + \gamma + O(K^{-1})$, one obtains

$$4d \sum_{k > K} s_k = \frac{2}{\pi^2} + O((N\varepsilon)^{-1}), \quad 4 \sum_{k=1}^K k s_k = \frac{2}{\pi^2} [\log(4N \sin(\pi\varepsilon)) + \gamma] + O((N\varepsilon)^{-1}).$$

Here $\sum_{k \geq 1} \cos(k\theta)/k = -\log(2 \sin(\theta/2))$ follows by Abel limiting the power series for $-\log(1 - re^{i\theta})$; the displayed tail estimate justifies that limit.

Set $g_c(w) = \sin^2 w \cos(2w)/w^2$, $g_s(w) = \sin^2 w \sin(2w)/w^2$, with values 1, 0 at zero, and $X = Kh$. The modulation is

$$\Sigma_2 = \frac{2}{\sin \theta} [\sin(\theta d) \Sigma_c - \cos(\theta d) \Sigma_s], \quad \Sigma_\bullet = \frac{2\varepsilon}{\pi} \left[\mathcal{T}_\bullet + \frac{h}{2} g_\bullet(X) \right],$$

where $\mathcal{T}_\bullet$ is the trapezoidal rule on $[0, X]$ with mesh h . Its half-weight at zero is the exact diagonal contribution; it cannot be discarded. Twice integrating the interpolation error on each cell gives $|\mathcal{T}_\bullet - \int_0^X g_\bullet| \leq h^2 \int_0^\infty |g''|/8 = O(h^2)$. For finiteness, write $g_\bullet = \phi_\bullet/w^2$, where $\phi_c = \frac{1}{2} \cos 2w - \frac{1}{4} - \frac{1}{4} \cos 4w$ and $\phi_s = \frac{1}{2} \sin 2w - \frac{1}{4} \sin 4w$: the origin is removable and, for $w \geq 1$, $|g''| \leq 6w^{-2} + 8w^{-3} + 6w^{-4}$. Also $h|g_\bullet(X)|/2 \leq 1/(2K)$. Lemma 3.14 gives $\int_0^\infty g_c = 0$, $\int_0^\infty g_s = \log 2$ with tails at most $1/X$. Consequently $\Sigma_2 = -2\pi^{-2}(\log 2) \cos(\theta d) + O_\delta(\varepsilon^2 + (N\varepsilon)^{-1})$, using $\theta/\sin \theta = 1 + O_\delta(\varepsilon^2)$. This proves (4.45). Finally $\Sigma_1 = 2 \operatorname{tr} C_N(\varepsilon)^2$; divide the unmodulated calculation by two and set $\varepsilon = 2W$ to obtain (4.46). That argument never divides by $\sin \theta$. $\square$

The prolate constant is classical; [1, eq. (D.6)] gives a sharper fixed-band expansion. At $\varepsilon = \tau/N$, (4.45) matches the continuum logarithmic and oscillating terms with error $O(\tau^2/N^2 + 1/\tau)$; this error does not vanish at fixed τ , whose convergence follows separately from the operator limit.

4.3.1 Adjacent bands and cyclic words

Put $C'_\tau = M_{e^{2\pi i \tau x}} C_\tau M_{e^{-2\pi i \tau x}}$, $\mathcal{N} = M_{e^{-i\pi \tau x}}$ and $\mathcal{V} = \mathcal{N}^* C_\tau \mathcal{N}^*$. With the positive-sign Fourier convention, the frequency bands of C_τ, C'_τ are $(-\tau/2, \tau/2)$ and $(-3\tau/2, -\tau/2)$.

Proposition 4.13 (Adjacent products). *Write $D(a) = \operatorname{tr}(C^{(a)} - (C^{(a)})^2)$, $X = \operatorname{tr}(C_\tau^2 C_\tau'^2)$, $\Omega_2 = \operatorname{tr}(\mathcal{V}^3 \mathcal{V}^*)$, $\Omega_3 = \operatorname{tr} \mathcal{V}^4$. Then $\mathcal{K}_\tau = \mathcal{V} + \mathcal{V}^*$, the Ω_j are real, and*

$$\operatorname{tr} \mathcal{K}_\tau^4 = 2 \operatorname{tr} C_\tau^4 + 4X + 8\Omega_2 + 2\Omega_3. \quad (4.47)$$

For every $\tau > 0$,

$$0 \leq X \leq \operatorname{tr}(C_\tau C'_\tau) = D(\pi\tau) - \frac{1}{2}D(2\pi\tau) = \frac{\log(2\pi\tau) + 1 + \gamma}{2\pi^2} + r, \quad |r| \leq \frac{5}{4\pi^3\tau},$$

and $|\Omega_2| \leq b(1/4)$, $|\Omega_3| \leq 4b(1/8)$. Moreover $X = (12\pi^2)^{-1} \log \tau + O(1)$.

Proof. The kernel factorization is product-to-sum. In degree four the cyclic word classes have sizes 2, 4, 8, 2: alternating, contiguous two-and-two, three-and-one, and pure. They give (4.47); for example $\text{tr}(\mathcal{V}^2\mathcal{V}^{*2}) = X$. The antiunitary $f(x) \mapsto \overline{f(-x)}$ fixes $C_\tau, \mathcal{N}$ and $\mathcal{V}$, proving reality of the word traces. The sum $C_\tau + C'_\tau$ is unitarily equivalent to $C_{2\tau}$, so its defect gives the quadratic product identity. Apply Lemma 3.12 at $\pi\tau, 2\pi\tau$ for the error $1 + 1/4$ in units $(\pi^3\tau)^{-1}$. The inequalities follow successively from $C_\tau^2 \leq C_\tau$ and $C_\tau'^2 \leq C'_\tau$ in positive trace pairings.

For $F(\phi) = \text{tr}(e^{i\phi}\mathcal{V} + e^{-i\phi}\mathcal{V}^*)^4$, the strip and trace comparisons give $|F(\phi) - F(\psi)| \leq 16b(|\phi - \psi|/(2\pi))$. The word expansion is $F(\phi) = 2\text{tr} C_\tau^4 + 4X + 8\Omega_2 \cos 2\phi + 2\Omega_3 \cos 4\phi$. Thus $16\Omega_2 = F(0) - F(\pi/2)$ and $8\Omega_3 = F(0) + F(\pi/2) - 2F(\pi/4)$, proving the bounds. In degree two the same argument gives

$$\Omega_1 := \text{tr} \mathcal{V}^2 = -\pi^{-2}(\log 2) \cos(4\pi\tau) + \rho_1, \quad |\rho_1| \leq 3/(\pi^3\tau). \quad (4.48)$$

Indeed $\text{tr} \mathcal{K}_\tau^2 = 2\text{tr} C_\tau^2 + 2\Omega_1$ and the two quadratic errors sum to $6/(\pi^3\tau)$.

For the remaining coefficient, [27, Theorem 2.1], with $q = 1, p(\lambda) = \lambda, F = 1, g = 0, x = 2\pi\tau$, gives $\log \det(I + \gamma C_\tau) = 2\tau \log(1 + \gamma) + (2\pi^2)^{-1} \log^2(1 + \gamma) \log \tau + c(\gamma) + o(1)$. The constant is analytic near zero; the fixed n th derivative of the remainder there is $O((\log \tau)^n/\tau)$ by that paper's eqs. (8.40)–(8.43). The γ^4 coefficient $11/12$ therefore gives $\text{tr} C_\tau^4 = 2\tau - 11(6\pi^2)^{-1} \log \tau + O(1)$, also consistent with the Landau–Widom integral $\int_0^1 (t^4 - t)/[t(1 - t)] dt = -11/6$ [31, p. 478]. Subtract twice this formula from the quartic trace law for $\mathcal{K}_\tau$ in (4.47) and divide by four. $\square$

This computes one adjacent word, not arbitrary cyclic words, and does not supply an independent proof of the quartic coefficient used.

4.4 Prolate coordinates and a prime-power application

Proposition 4.14 (Full-band coordinates). *Let $C_{2\tau}\psi_n = \ell_n\psi_n$ be the real orthonormal prolate basis, with $\ell_n = \lambda_n(2\pi\tau) > 0$ and parity $(-1)^n$. Then $\chi_n = \ell_n^{-1/2}R_\tau U\psi_n = i^n\varphi_n$ form an orthonormal basis of $L^2(-\tau, \tau)$, with φ_n real of parity $(-1)^n$. The matrix $\mathcal{J}_{mn} = \langle \chi_m, J_\tau \chi_n \rangle$ is a real symmetric involution, vanishes for opposite parities, and obeys*

$$(\mathcal{K}_\tau)_{mn} = \sqrt{\ell_m \ell_n} \mathcal{J}_{mn}, \quad (C_\pm)_{mn} = \sqrt{\ell_m \ell_n} (\delta_{mn} \pm \mathcal{J}_{mn})/2. \quad (4.49)$$

For equal parities, $\mathcal{J}_{mn} = 2(-1)^{(n-m)/2} \int_0^\tau \varphi_m(\xi) \varphi_n(\xi - \tau) d\xi$. All series in

$$\begin{aligned} \text{tr} \mathcal{K}_\tau &= \sum_n \ell_n \mathcal{J}_{nn}, & \text{tr} \mathcal{K}_\tau^2 &= \sum_{m,n} \ell_m \ell_n \mathcal{J}_{mn}^2, \\ D_J &= \sum_{m,n} \ell_n (1 - \ell_m) \mathcal{J}_{mn}^2, & D_R &= \sum_n \ell_n (1 - \ell_n) \end{aligned} \quad (4.50)$$

converge absolutely.

Proof. For $A = R_\tau U$, $A^*A = C_{2\tau}$ and both A, A^* are injective by the entire-function argument in Theorem 2.3. Thus the χ_n are orthonormal and complete: orthogonality of g to all χ_n implies $A^*g = 0$. Separating the sine and cosine parts of $U\psi_n$ gives $i^n\varphi_n$. The exchange is a self-adjoint involution commuting with reflection; its matrix entries are $i^{n-m}\langle \varphi_m, J_\tau \varphi_n \rangle$, real for equal parities and zero otherwise. Splitting the exchange integral into two halves gives the stated formula. Now $\mathcal{K}_\tau = A^*J_\tau A$ and $C_\pm = A^*(I \pm J_\tau)A/2$ give (4.49). Parseval yields $\sum_m \mathcal{J}_{mn}^2 = 1$; together with $\sum_n \ell_n = 4\tau$, this proves the trace formulas and all absolute-convergence assertions. $\square$

4.4.1 One prime-power summand

For a prime power q , put $a_q = \Lambda(q)/\sqrt{q}$, where Λ is the von Mangoldt function, and $\varepsilon_q(c) = 1 - \log q / \log c$, $c > 1$. With the localized Weil-form basis $L^{-1/2}e^{2\pi i n x/L}$, $L = \log c$, $|n| \leq N$, and

prime distribution $-\sum_{q \leq c} a_q \delta_{\log q}$, the individual summand is

$$-a_q A_N(\varepsilon_q(c)), \quad \tau = N\varepsilon_q(c), \quad \log(c/q) = \frac{\tau \log q}{N - \tau}. \quad (4.51)$$

The matrix formula is [15, Proposition 4.1], specialized as in [41]; the last equality is exact inversion.

Corollary 4.15 (Finite-section design rule). *For fixed $q, T, \eta > 0$, take $N > 2T$, $N \geq (2T + 8\pi T^2)/\eta$ and $c(\tau) = q^{N/(N-\tau)}$. Uniformly on $0 < \tau \leq T$, the embedded normalized block is within η of $\mathcal{K}_\tau$, and the physical summand within $a_q \eta$ of $-a_q \mathcal{K}_\tau$. If $\eta < \min(p_\tau, m_\tau)$ the normalized block has a positive even and negative odd eigenvalue; multiplication by $-a_q$ reverses them.*

Proof. The inversion gives $\varepsilon_q = \tau/N < 1/2$; apply Theorem 2.5 and (4.15). $\square$

The cutoff layer has width $T \log q / (N - T) = O_{q,T}(N^{-1})$. For phase-aligned eigenvectors at a fixed simple isolated parity eigenvalue, Theorem 4.5 gives weights $1/2, 1/2$ at the exchanged-arc endpoints: for every $\zeta > 0$, a sufficiently large fixed C_ζ makes each C_ζ/N neighborhood's limiting mass at least $1/2 - \zeta$. The disjoint-arc range is exactly $\sqrt{c} < q < c$, or $0 < \tau < N/2$. At $q = \sqrt{c}$ the matrix is $\text{diag}((-1)^m)$; entrywise,

$$A_N(1 - \varepsilon) = -A_N(\varepsilon) + 2 \text{diag}(\cos(2\pi m \varepsilon))_{m=-N}^N. \quad (4.52)$$

Also $0 < a_q \leq 2/e$, so signed count estimates transfer by reversing signs and replacing a threshold $0 < \alpha < a_q$ by α/a_q . This describes one moving summand, with no sign conclusion for the complete Weil form. Its other prime and archimedean terms also move; fixed-mode continuity alone gives no dimension-uniform right neighborhood. The complete-form estimates [50, 4] and activation-wall analysis [48] concern different questions.

5 Commensurate frequencies and odd logarithmic terms

5.1 Commensurate frequencies: symmetric bulk and an odd logarithm

For $M \geq 1$ and real $\mathbf{c} = (c_1, \dots, c_M)$, set $c_j = 0$ for $j > M$ and

$$\mathcal{K}_{\tau, \mathbf{c}} = \sum_{m=1}^M c_m \mathcal{K}_{m\tau}, \quad K_{\tau, \mathbf{c}}(x, y) = \frac{\sum_m c_m [\sin(2\pi m \tau x) - \sin(2\pi m \tau y)]}{\pi(x - y)}. \quad (5.1)$$

Write $A_{\mathbf{c}} = \sum_m |c_m|$, $D_{\mathbf{c}} = \sum_m m |c_m|$, and $\mathcal{A}_3(\mathbf{c}) = \sum_{j,k \geq 1, j+k \leq M} c_j c_k c_{j+k}$; the last sum counts ordered pairs. Each frequency exchanges positive and negative Fourier intervals. The bulk graph is therefore bipartite and has a sign-symmetric spectrum. At the jump, however, the two exchanges share a vertex and create triangles. Their weights give an odd logarithmic term even though the odd bulk term vanishes.

Theorem 5.1 (Finite-frequency trace laws). *For fixed M , uniformly in $\tau \geq 2$ and $\mathbf{c} \in \mathbb{R}^M$,*

$$\text{tr } \mathcal{K}_{\tau, \mathbf{c}}^2 = 4\tau \sum_m m c_m^2 - \frac{2}{\pi^2} \sum_m c_m^2 \log \tau + O_M(A_{\mathbf{c}}^2), \quad (5.2)$$

$$\text{tr } \mathcal{K}_{\tau, \mathbf{c}}^3 = \frac{3}{\pi^2} \mathcal{A}_3(\mathbf{c}) \log \tau + O_M(A_{\mathbf{c}}^3). \quad (5.3)$$

The exact Bloch fiber has nonzero eigenvalues $\pm s_j(B_{\mathbf{c}})$, with multiplicity, where

$$B_{\mathbf{c}} = (c_{j+k-1})_{j,k=1}^M. \quad (5.4)$$

Zero singular values are omitted; if $c_M \neq 0$ all $2M$ values are nonzero.

For $(1, a)$ the cubic coefficient is $3a/\pi^2$; for $(1, 1, 1)$ it is $9/\pi^2$. A sum-free occupied frequency set forces $\mathcal{A}_3 = 0$, while signed coefficients can also cancel. This alone says nothing about higher odd traces. The matrix-jump precedents are [12, 8]; we prove the stated bounded polynomial remainders directly.

5.1.1 The graph and its jump

Let $\mathcal{G}_m$ have kernel $G_m(X, Y) = [\sin(2\pi mX) - \sin(2\pi mY)]/[\pi(X - Y)]$ on the line. Fourier transformation gives

$$(J_m g)(\xi) = \mathbf{1}_{(0, m)}(\xi)g(\xi - m) + \mathbf{1}_{(-m, 0)}(\xi)g(\xi + m). \quad (5.5)$$

Indeed its inverse kernel is $2 \cos(\pi m(X + Y)) \sin(\pi m(X - Y))/[\pi(X - Y)]$. Thus $\mathcal{G}_{\mathbf{c}} = \sum c_m \mathcal{G}_m$ is self-adjoint, $\|\mathcal{G}_{\mathbf{c}}\| \leq A_{\mathbf{c}}$, and dilation identifies $\mathcal{K}_{\tau, \mathbf{c}}$ with its compression to $(-\tau, \tau)$. On $\mathcal{H} = L^2(0, 1)$ put $e_j(t) = e^{2\pi i j t}$, $W_\theta = M_{e^{2\pi i \theta t}}$ and $X_{j,k} = |e_j\rangle\langle e_k| + |e_k\rangle\langle e_j|$. Then

$$\sigma_{\mathbf{c}}(\theta) = W_\theta^* H_{\mathbf{c}} W_\theta, \quad H_{\mathbf{c}} = \sum_m c_m \sum_{j=1}^m X_{j, j-m}. \quad (5.6)$$

To verify the cell coefficients, set $d = t - s$. The kernel of the inner sum is $2e^{\pi i d} \cos(\pi m(t + s)) \sin(\pi m d) / \sin(\pi d)$. Multiplication by $\int_0^1 e^{-2\pi i(k+d)\theta} d\theta = e^{-\pi i d} \sin(\pi d) / [\pi(k+d)]$ gives $G_m(k + t, s)$, including removable singularities. Ordering the vertices as $j = 1, \dots, M$ and $1 - k$, $k = 1, \dots, M$, gives

$$H_{\mathbf{c}} = \begin{pmatrix} 0 & B_{\mathbf{c}} \\ B_{\mathbf{c}} & 0 \end{pmatrix}. \quad (5.7)$$

Its signed singular-value pairing proves the spectral assertion; column reversal in $B_{\mathbf{c}}$ gives a triangular matrix, so $\det B_{\mathbf{c}} = (-1)^{M(M-1)/2} c_M^M$. In particular

$$\mathrm{tr} \sigma_{\mathbf{c}}^3 = 0, \quad \mathrm{tr} \sigma_{\mathbf{c}}^2 = 2 \sum_m m c_m^2. \quad (5.8)$$

Set $H_i = \sigma_{\mathbf{c}}(i)$, $i = 0, 1$, and $C_t = (1 - t)H_0 + tH_1$ on $\mathrm{span}\{e_{-M}, \dots, e_M\}$. Their common edges join $+j$ to $-k$ with weight c_{j+k} . The other edges join zero to $+j$ in H_0 and to $-j$ in H_1 , with weight c_j . Hence every triangle is $0, +j, -k$; its product contributes six times to the cubic trace. Also $\mathrm{tr}(H_1 - H_0)^2 = 4 \sum_m c_m^2$. Therefore

$$\mathrm{tr} C_t^3 = 6t(1 - t) \mathcal{A}_3(\mathbf{c}), \quad (5.9)$$

$$\mathrm{tr} C_t^2 = 2 \sum_m m c_m^2 - 4t(1 - t) \sum_m c_m^2. \quad (5.10)$$

5.1.2 From the jump to the trace

Use the Laurent conventions (3.34), with P_n selecting n cells and acting on the entire fiber.

Lemma 5.2 (Quadratic and cubic localization). *If $b \in C^3([0, 1]; \mathfrak{S}_1)$ is self-adjoint and its two endpoint ranges span a finite-dimensional space, put $\ell(t) = (1 - t)b(0) + tb(1)$. For $p = 2, 3$,*

$$\mathrm{tr} T_n(b)^p - n \int_0^1 \mathrm{tr} b^p = \mathrm{tr} T_n(\ell)^p - n \int_0^1 \mathrm{tr} \ell^p + O(1), \quad (5.11)$$

uniformly in n .

Proof. Put $r = b - \ell$, $b_u = \ell + ur$, $B_u = \mathcal{L}(b_u)$, $R = \mathcal{L}(r)$, $P = P_n$, $Q = I - P$ and $A_* = \sup_u \|B_u\|$. Both r and rb_u have zero endpoints. Lemma 3.22 therefore bounds $\|[P, R]\|_1$ and $\|[P, RB_u]\|_1$ by $\mathbf{c}(r)$ and $\mathbf{c}(rb_u)$, uniformly in n, u . For $\Delta_p(u) = \mathrm{tr}(PB_u^p P) - \mathrm{tr}(PB_u P)^p$, cyclicity and $P + Q = I$ give

$$\Delta'_2(u)/2 = \mathrm{tr}(PRQB_u P),$$

$$\Delta'_3(u)/3 = \mathrm{tr}(PRQB_u^2 P) + \mathrm{tr}(P(RB_u)QB_u P) - \mathrm{tr}(PRQB_u QB_u P).$$

Cyclicity in the first trace is fiberwise; each compression has finitely many trace-class blocks. Thus the derivatives have respective bounds $2A_* \mathbf{c}(r)$ and $3[2A_*^2 \mathbf{c}(r) + A_* \mathbf{c}(rb_u)]$. Integration in u , and $\mathrm{tr}(P\mathcal{L}(b^p)P) = n \int_0^1 \mathrm{tr} b^p$, prove the claim. $\square$

For the scalar sawtooth $S_n = T_n(t)$, its Fourier coefficients are $1/2$ at zero and $i/(2\pi k)$ otherwise, whence

$$\operatorname{tr} S_n = n/2, \quad \operatorname{tr} S_n^2 = \frac{n}{4} + \frac{1}{2\pi^2} \sum_{k=1}^{n-1} \frac{n-k}{k^2} = \frac{n}{3} - \frac{\log n}{2\pi^2} + O(1). \quad (5.12)$$

The error is bounded by $H_{n-1} - \log n = O(1)$ and $n \sum_{k \geq n} k^{-2} = O(1)$. Now $T_n(\ell) = I_n \otimes H_0 + S_n \otimes (H_1 - H_0)$. Diagonalizing only S_n and using (5.9)–(5.12) gives the cubic trace $n\mathcal{A}_3 + 3\pi^{-2}\mathcal{A}_3 \log n + O(A_{\mathbf{c}}^3)$; the integral of the cubic interpolant is $\mathcal{A}_3$. Subtract that bulk term in (5.11) and use (5.8). The quadratic calculation gives $2n \sum_m m c_m^2 - 2\pi^{-2} \sum_m c_m^2 \log n + O_M(A_{\mathbf{c}}^2)$. The constants are uniform in $\mathbf{c}$: the bounded generator of W_θ and $\|H_{\mathbf{c}}\|_1 \leq 2MA_{\mathbf{c}}$ bound $\|r\|_{C^3(\mathfrak{S}_1)}$ by $C_M A_{\mathbf{c}}$, $\|rb_u\|_{C^3(\mathfrak{S}_1)}$ by $C_M A_{\mathbf{c}}^2$, and $A_* \leq A_{\mathbf{c}}$. No spectral gap or eigenvector choice is used.

For a nearest integer q to $\tau \geq 2$, set $n = 2q$. Dilation and Lemma 3.19 give, with b from (3.27),

$$\|P_{(-\tau, \tau)} \mathcal{G}_{\mathbf{c}} P_{(-\tau, \tau)} - P_{(-q, q)} \mathcal{G}_{\mathbf{c}} P_{(-q, q)}\|_1 \leq 4 \sum_m |c_m| b(m/2). \quad (5.13)$$

Indeed each endpoint moves by at most $1/2$, and a strip of length h for $\mathcal{G}_m$ has nuclear norm at most $b(mh)$. The trace perturbation inequality for x^2, x^3 gives errors $O_M(A_{\mathbf{c}}^2), O_M(A_{\mathbf{c}}^3)$. Replacing $\log(2q)$ by $\log \tau$ and q by τ in the bulk has the same orders, completing Theorem 5.1.

Proposition 5.3 (General first-order law). *For $\mathbf{c} \neq 0$ and real $f \in C^2([-A_{\mathbf{c}}, A_{\mathbf{c}}])$, $f(0) = 0$,*

$$\begin{aligned} \operatorname{tr} f(\mathcal{K}_{\tau, \mathbf{c}}) &= 2\tau \sum_{j=1}^M [f(s_j(B_{\mathbf{c}})) + f(-s_j(B_{\mathbf{c}}))] \\ &\quad + O_M(A_{\mathbf{c}} \|f'\|_{\infty} + A_{\mathbf{c}}^2 \|f''\|_{\infty} (1 + \log \tau)), \quad \tau \geq 2. \end{aligned} \quad (5.14)$$

For $\mathbf{c} = 0$ the trace is zero.

Proof. The kernel is bounded by $\min\{2D_{\mathbf{c}}, 2A_{\mathbf{c}}/(\pi|X - Y|)\}$. For $P = P_I$, $|I| = 2q$, the measure of points lost by displacement u is $\min(2q, |u|)$. Splitting the squared-kernel integral at $u_0 = A_{\mathbf{c}}/(\pi D_{\mathbf{c}})$ and $2q$ gives

$$\|P \mathcal{G}_{\mathbf{c}} (I - P)\|_2^2 \leq \frac{4A_{\mathbf{c}}^2}{\pi^2} [3 + 2 \log(2\pi q D_{\mathbf{c}}/A_{\mathbf{c}})].$$

Scalar Taylor expansion in the spectral measure of each compression eigenvector has zero linear term. Summing its quadratic remainder yields

$$|\operatorname{tr} f(P \mathcal{G}_{\mathbf{c}} P) - \operatorname{tr} P f(\mathcal{G}_{\mathbf{c}}) P| \leq \frac{1}{2} \|f''\|_{\infty} \|P \mathcal{G}_{\mathbf{c}} (I - P)\|_2^2.$$

The traces exist because $f(0) = 0$ bounds $|f(\mathcal{G}_{\mathbf{c}})|$ by a constant times the frequency projection onto $(-M, M)$, whose bounded interval compression is trace class. On $I = (-q, q)$ the latter trace equals $2q \operatorname{tr}_{\mathcal{H}} f(H_{\mathbf{c}})$ by the Bloch formula. Use (5.13), $D_{\mathbf{c}}/A_{\mathbf{c}} \leq M$ and $\|H_{\mathbf{c}}\|_1 \leq 2MA_{\mathbf{c}}$ to replace q by τ . $\square$

5.1.3 Two frequencies

For $\mathbf{c} = (1, a)$ the bulk graph is the path $(-1, 1, 0, 2)$, with weights $a, 1, a$. Its polynomial is $\lambda^4 - (1 + 2a^2)\lambda^2 + a^4$; hence its spectrum is $\pm \rho_+, \pm \rho_-$, where $\rho_{\pm} = (\sqrt{1 + 4a^2} \pm 1)/2$, and $\|\mathcal{G}_a\| = \rho_+$. The quadratic and cubic traces are $4\tau(1 + 2a^2) - 2\pi^{-2}(1 + a^2) \log \tau + O_a(1)$ and $3a\pi^{-2} \log \tau + O_a(1)$. At the jump,

$$C_t = aX_{-1,1} + t(aX_{-2,0} + X_{-1,0}) + (1 - t)(X_{0,1} + aX_{0,2}). \quad (5.15)$$

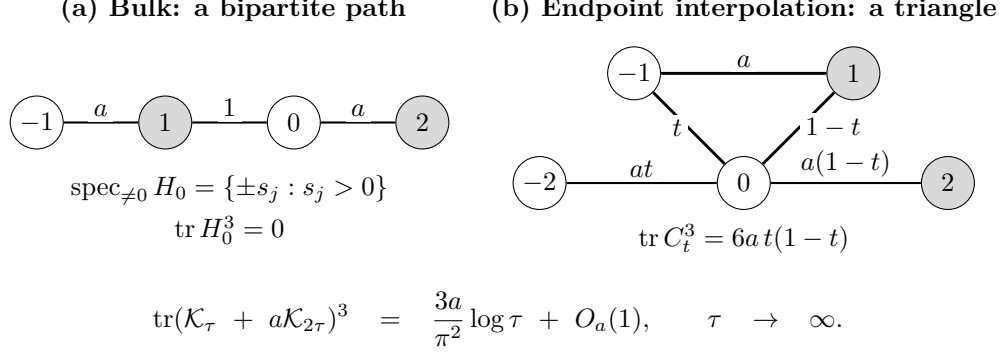


Figure 2: Exact weighted graphs for the two-harmonic family. Vertices are Fourier indices; shaded vertices have positive index. Isolated zero vertices are omitted in (a). Every bulk edge joins the two index classes, so the bulk spectrum is sign-symmetric. For $C_t = (1-t)H_0 + tH_1$, the three edges on $\{-1, 0, 1\}$ have product $at(1-t)$. The six oriented closed walks around this triangle give the cubic trace. The logarithmic law also uses the polynomial localization and scalar sawtooth calculation proved in the text. Here $a \in \mathbb{R}$ is fixed; the triangle is present when $a \neq 0$ and $0 < t < 1$.

Put $s = t(1-t)$, $r^2 = 1 - 2s$. The vector $(1-t)e_{-2} - te_2$ is null. Combining the other two leaves into $(te_{-2} + (1-t)e_2)/r$ leaves

$$\begin{pmatrix} 0 & 0 & ar & 0 \\ 0 & 0 & t & a \\ ar & t & 0 & 1-t \\ 0 & a & 1-t & 0 \end{pmatrix},$$

whose determinant gives

$$\lambda^4 - [a^2 + (1+a^2)r^2]\lambda^2 - 2as\lambda + a^4r^2. \quad (5.16)$$

For $a \neq 0$ its constant term never vanishes; the four roots retain two positive and two negative signs. Their product and $\|C_t\| \leq \rho_+$ give $|\lambda| \geq a^4/(2\rho_+^3)$. They are not sign-paired when $0 < t < 1$, because $\text{tr } C_t^3 = 6at(1-t) \neq 0$. No disjointness of their image intervals is claimed. A general second-order C^2 law with bounded remainder, its phase profile and individual transition eigenvalues require further analysis.

6 Phase-dependent constant terms

6.1 Fixed-phase constant terms

Use the global exchange and full-fiber gauge of Sections 2.1.1 and 3.4, with scalar star entries $c = c_\theta$, $s = s_\theta$. The block Toeplitz constant [8, Theorem 2.3 and Corollary 2.4] combines with trace-norm convergence of the physical boundary corrections; boundedness alone does not give a constant term.

For $|z| < 1$ put

$$\beta(z) = \varkappa(x \mapsto \log(1 - zx)).$$

The scalar logarithm and every determinant logarithm in this section are analytic determinations equal to zero at $z = 0$.

Theorem 6.1 (Fixed-phase determinant and band-local trace constants). *For real r, ϕ there is a holomorphic function $\mathfrak{D}(z; r, \phi)$ on $|z| < 1$, with $\mathfrak{D}(0; r, \phi) = 0$, such that*

$$\log \det(I - z\mathcal{K}_{n+r, \phi}) = 2(n+r) \log(1 - z^2) + \beta(z) \log(n+r) + \mathfrak{D}(z; r, \phi) + o(1) \quad (6.1)$$

as the integer n tends to infinity. The error is locally uniform in z and uniform for (r, ϕ) in a compact set. For every real Lipschitz f on $[-1, 1]$ with $f(0) = 0$ and C^2 restrictions to $[-1, -g]$ and $[g, 1]$, where $g = 2^{-1/2}$, there is a continuous function $\mathfrak{C}_f(r, \phi)$ such that

$$\mathrm{tr} f(\mathcal{K}_{n+r, \phi}) = 2(n+r)[f(1) + f(-1)] + \varkappa(f) \log(n+r) + \mathfrak{C}_f(r, \phi) + o(1), \quad (6.2)$$

uniformly on compact sets of (r, ϕ) . In particular, for every $p \geq 1$,

$$\|\mathcal{K}_{n+r, \phi}\|_p^p = 4(n+r) + \varkappa_p \log(n+r) + \mathfrak{C}_p(r, \phi) + o(1). \quad (6.3)$$

The determinant constant is specified by (6.17).

The diagonal identity $\mathrm{tr} \mathcal{K}_{n+r, \phi} = 2\pi^{-1} \sin(2\pi r) \cos \phi$ and locally uniform differentiation give

$$\partial_z \mathfrak{D}(0; r, \phi) = -\frac{2}{\pi} \sin(2\pi r) \cos \phi. \quad (6.4)$$

The second derivative is

$$\partial_z^2 \mathfrak{D}(0; r, \phi) = \frac{2}{\pi^2} [\log(4\pi) + 1 + \gamma_E + (\log 2) \cos(4\pi r) \cos(2\phi)], \quad (6.5)$$

where γ_E is Euler's constant. Here is a direct check retaining both phases. Write $s(u) = \sin(\pi u)/(\pi u)$ and let C_τ be its compression to $(-\tau, \tau)$. Product-to-sum and the variables $u = X - Y$, $v = X + Y$ give

$$\mathrm{tr} \mathcal{K}_{\tau, \phi}^2 = 2 \mathrm{tr} C_\tau^2 + \frac{2 \cos(2\phi)}{\pi^3} \int_0^{2\tau} \frac{\sin^2(\pi u) \sin(4\pi\tau - 2\pi u)}{u^2} du.$$

Since $\int_{\mathbb{R}} s^2 = 1$,

$$2\tau - \mathrm{tr} C_\tau^2 = 4\tau \int_{2\tau}^\infty s(u)^2 du + \frac{2}{\pi^2} \int_0^{2\tau} \frac{\sin^2(\pi u)}{u} du = \frac{1 + \gamma_E + \log(4\pi\tau)}{\pi^2} + O(\tau^{-1}).$$

The last equality follows by one integration by parts in the tail and $\int_0^x (1 - \cos t) dt/t = \gamma_E + \log x - \mathrm{Ci}(x)$, where $\mathrm{Ci}(x) = -\int_x^\infty \cos t dt/t = O(x^{-1})$. Extending the oscillatory integral to infinity costs $O(\tau^{-1})$. The resulting cosine integral is zero, whereas

$$\int_0^\infty \frac{\sin^2(\pi u) \sin(2\pi u)}{u^2} du = \pi \int_0^\infty \frac{\cos(2\pi u) - \cos(4\pi u)}{u} du = \pi \log 2.$$

For the cosine integral use $\sin^2(\pi u) \cos(2\pi u) = \frac{1}{2}(\cos(2\pi u) - 1) - \frac{1}{4}(\cos(4\pi u) - 1)$ and $\int_0^\infty (1 - \cos au) du/u^2 = \pi a/2$. The sine identity follows by integration by parts and the damped Frullani integral. Thus the quadratic trace is $4\tau - 2\pi^{-2}[\log(4\pi\tau) + 1 + \gamma_E + (\log 2) \cos(4\pi\tau) \cos(2\phi)] + O(\tau^{-1})$. Twice differentiating the locally uniform limit (6.1) proves (6.5).

6.1.1 Localization of the gauge correction

Identify the physical space with $\mathcal{X} = \ell^2(\mathbb{Z}; \mathcal{H})$, where $\mathcal{H} = L^2(0, 1)$, and retain the symbols of Section 3.4. Write

$$\mathcal{B} = \mathcal{L}(\sigma), \quad \mathcal{U} = \mathcal{L}(V), \quad \mathcal{C} = \mathcal{U}^* \mathcal{B} \mathcal{U} = \mathcal{L}(a).$$

Thus $\mathcal{B}, \mathcal{C}$ are self-adjoint contractions and $\mathcal{U}$ is unitary. All three commute with the bilateral cell shift $\mathbf{S}$, defined by $(\mathbf{S}x)_j = x_{j-1}$. Let Π_+ select the cells $j \geq 0$, put $\Pi_- = I - \Pi_+$, and set $R_m = \mathbf{S}^m \Pi_- \mathbf{S}^{-m}$. The projection onto the cells $0, \dots, m-1$ satisfies

$$P_m = \Pi_+ + R_m - I. \quad (6.6)$$

Lemma 3.22 also gives $[\Pi_\pm, \mathcal{U}] \in \mathfrak{S}_1$ and the same for $\mathcal{U}^*$: each half-line has one inverse-square Hankel corner and a Fourier remainder summable with weight $|k|$.

Lemma 6.2 (Two boundary corrections). *For an orthogonal projection E put*

$$D(E) = \mathcal{U}^* E \mathcal{B} E \mathcal{U} - E \mathcal{C} E, \quad D_+ = D(\Pi_+), \quad D_- = D(\Pi_-).$$

Then D_+, D_- are trace class and

$$\|D(P_m) - D_+ - S^m D_- S^{-m}\|_1 \rightarrow 0. \quad (6.7)$$

Proof. The exact identity

$$D(E) = \mathcal{U}^* E \mathcal{B} [E, \mathcal{U}] + [\mathcal{U}^*, E] \mathcal{B} \mathcal{U} E \quad (6.8)$$

first proves the trace-class assertion. Insert (6.6) in its two commutators and subtract the identities for Π_+ and R_m . The difference consists of exactly

$$\begin{aligned} \mathcal{U}^*(P_m - \Pi_+) \mathcal{B} [\Pi_+, \mathcal{U}], & \quad \mathcal{U}^*(P_m - R_m) \mathcal{B} [R_m, \mathcal{U}], \\ [\mathcal{U}^*, \Pi_+] \mathcal{B} \mathcal{U} (P_m - \Pi_+), & \quad [\mathcal{U}^*, R_m] \mathcal{B} \mathcal{U} (P_m - R_m). \end{aligned}$$

The first and third contain the strongly null projection $P_m - \Pi_+ = -S^m \Pi_+ S^{-m}$. After translation by S^{-m} , the others contain $-S^{-m} \Pi_- S^m$. Finite-rank approximation gives $\|X_m T\|_1 \rightarrow 0$ for fixed $T \in \mathfrak{S}_1$ and uniformly bounded strongly null X_m ; it also gives $\|T X_m\|_1 \rightarrow 0$ when $X_m^* \rightarrow 0$ strongly. Applied to the fixed commutators, this proves all four limits. $\square$

Set

$$\mathcal{A}_m = \mathcal{U}^* P_m \mathcal{B} P_m \mathcal{U}, \quad \mathcal{C}_m = P_m \mathcal{C} P_m, \quad \mathcal{A}_\pm = \mathcal{U}^* \Pi_\pm \mathcal{B} \Pi_\pm \mathcal{U}, \quad \mathcal{C}_\pm = \Pi_\pm \mathcal{C} \Pi_\pm.$$

These are self-adjoint contractions; the finite compressions and $D_\pm = \mathcal{A}_\pm - \mathcal{C}_\pm$ are trace class. Define

$$e_\pm(z) = \det[I - z D_\pm (I - z \mathcal{C}_\pm)^{-1}], \quad |z| < 1. \quad (6.9)$$

These determinants are holomorphic and nonzero, because their operators equal $(I - z \mathcal{A}_\pm)(I - z \mathcal{C}_\pm)^{-1}$.

Lemma 6.3 (Relative determinant convergence). *Locally uniformly on $|z| < 1$,*

$$\frac{\det(I - z \mathcal{A}_m)}{\det(I - z \mathcal{C}_m)} \rightarrow e_+(z) e_-(z). \quad (6.10)$$

The normalized analytic logarithms converge locally uniformly as well.

Proof. Let $X_\pm(z) = D_\pm(I - z \mathcal{C}_\pm)^{-1}$. At each fixed boundary the reference resolvents and their adjoints converge strongly, uniformly for $|z| \leq \rho < 1$, by the Neumann series and its uniformly bounded tail. Lemma 6.2 and trace-class multiplication give

$$\left\| D(P_m)(I - z \mathcal{C}_m)^{-1} - X_+(z) - S^m X_-(z) S^{-m} \right\|_1 \rightarrow 0$$

uniformly on each such disk.

Weakly null shifts and finite-rank approximation give $\|T S^m W\|_1 \rightarrow 0$ for trace-class T, W , uniformly on trace-norm compact families by finite nets. Consequently

$$(I - z X_+)(I - z S^m X_- S^{-m}) = I - z X_+ - z S^m X_- S^{-m} + o_{\mathfrak{S}_1}(1)$$

uniformly on smaller disks. Multiplicativity and trace-norm continuity of the Fredholm determinant prove the limit, using the exact identity

$$I - z \mathcal{A}_m = [I - z D(P_m)(I - z \mathcal{C}_m)^{-1}](I - z \mathcal{C}_m).$$

The nonzero limits equal one at zero; Cauchy differentiation and integration of logarithmic derivatives give the normalized-log limit. $\square$

6.1.2 The model constant and fractional endpoints

On $E = \text{span}\{e_{-1}, e_0, e_1\}$ put $M_z(\theta) = I_E - za(\theta)$. For real $|z| < 1$, both M_z and M_z^{-1} are uniformly positive definite. Their half-line Toeplitz operators are therefore invertible. The symbol is piecewise smooth with one jump, so Theorem 2.1 and Corollary 2.4 of [8] apply. In their notation,

$$L(z) = \frac{1}{2\pi i} \log[(I - zX_{1,0})^{-1}(I - zX_{0,-1})]$$

has trace zero and its eigenvalues have real parts in $(-1/2, 1/2)$. Since $\det M_z \equiv 1 - z^2$, the cited result gives

$$\det T_m(M_z) = (1 - z^2)^m m^{\beta(z)} E_a(z)(1 + o(1)), \quad E_a(z) > 0 \quad (6.11)$$

for real $|z| < 1$. Here $\beta(z) = -\text{tr } L(z)^2$. Its equality with the trace coefficient follows also by comparing (6.11), Lemma 6.3, and the already established bounded-remainder law on the real interval.

The model constant has a specified Fredholm expression. Let T denote the semi-infinite block Toeplitz operator, let G_B be the Barnes G -function, and set

$$u_z(\theta) = \exp\{i(2\pi\theta - \pi)L(z)\}, \quad 0 < \theta < 1.$$

If $\ell_1(z), \ell_2(z), \ell_3(z)$ are the unordered eigenvalues of $L(z)$, the single-jump case of [8, eq. (2.11)] reads

$$E_a(z) = \prod_{j=1}^3 G_B(1 + \ell_j(z)) G_B(1 - \ell_j(z)) \det[T(M_z)T(u_z)^{-1}T(u_z^{-1})^{-1}T(M_z^{-1})]. \quad (6.12)$$

The ordered product is identity plus trace class by that theorem; no commutation of its factors is used. The jump orientation agrees because $u_z(0+)^{-1}u_z(1-) = M_z(0+)^{-1}M_z(1-)$. The inverses of the single-jump factors follow from [8, Proposition 3.11].

The cited applicability holds also for complex $|z| < 1$. Indeed $\text{Re } M_z \geq (1 - |z|)I$ and $\text{Re } M_z^{-1} \geq (1 - |z|)(1 + |z|)^{-2}I$, so both required Toeplitz operators are invertible by coercivity. The endpoint ratio has no negative real eigenvalue: otherwise a convex combination of its two endpoint matrices would be singular, although that combination is $I - z[(1 - t)X_{1,0} + tX_{0,-1}]$. Its principal matrix logarithm is thus analytic, has trace zero by continuation from zero, and has the required eigenvalue strip.

No parameter-uniform remainder is being inferred from the cited theorem. For $|z| \leq \rho < 1$, the first two x derivatives of $\log(1 - zx)$ are bounded by $\rho/(1 - \rho)$ and $\rho^2/(1 - \rho)^2$. Apply the existing uniform trace bound to its real and imaginary parts. The normalized physical logarithms are locally bounded holomorphic functions. Their convergence for real z , already supplied by (6.11) and Lemma 6.3, extends locally uniformly to the disk by Vitali's theorem. This also gives the normalized holomorphic determination of $\log E_a$. Since $\mathcal{K}_n$ has $m = 2n$ complete cells, its constant is

$$\mathfrak{D}_0(z) = \log E_a(z) + \beta(z) \log 2 + \log e_+(z) + \log e_-(z). \quad (6.13)$$

On the physical space let $L_b = P_{(b,\infty)}$ and $H_b = P_{(-\infty,b)}$. The two differences

$$F_b^L = L_b \mathcal{G} L_b - L_0 \mathcal{G} L_0, \quad F_b^R = H_b \mathcal{G} H_b - H_0 \mathcal{G} H_0$$

are trace class by Lemma 3.19. Define

$$\begin{aligned} d_L(z; b) &= \det[(I - zL_b \mathcal{G} L_b)(I - zL_0 \mathcal{G} L_0)^{-1}], \\ d_R(z; b) &= \det[(I - zH_b \mathcal{G} H_b)(I - zH_0 \mathcal{G} H_0)^{-1}]. \end{aligned} \quad (6.14)$$

Each product is identity plus trace class and invertible for $|z| < 1$.

For fixed l, h let $E_m = P_{(l, m+h)}$ and $P_m = P_{(0, m)}$ in physical coordinates. Then

$$\|E_m \mathcal{G} E_m - P_m \mathcal{G} P_m - F_l^L - S^m F_h^R S^{-m}\|_1 \rightarrow 0. \quad (6.15)$$

To check every term, set $\delta = L_l - L_0$ and $\epsilon_m = S^m (H_h - H_0) S^{-m}$. The identity $E_m = P_m + \delta + \epsilon_m$ holds once the interval is nonempty. After expanding and subtracting the two half-line corrections, the remainder is

$$\begin{aligned} & \delta \mathcal{G} (P_m - L_0) + (P_m - L_0) \mathcal{G} \delta \\ & + \epsilon_m \mathcal{G} (P_m - S^m H_0 S^{-m}) + (P_m - S^m H_0 S^{-m}) \mathcal{G} \epsilon_m \\ & + \delta \mathcal{G} \epsilon_m + \epsilon_m \mathcal{G} \delta. \end{aligned}$$

The first pair contains the escaping projection $-P_{(m, \infty)}$ and fixed trace-class strip products. Translating the second pair gives the same argument at minus infinity. The last pair vanishes since $\epsilon_m \rightarrow 0$ strongly and $\delta \mathcal{G}, \mathcal{G} \delta$ are trace class. This proves (6.15). The proof of Lemma 6.3, now with these boundary corrections and the physical baseline resolvents, yields

$$\frac{\det(I - z E_m \mathcal{G} E_m)}{\det(I - z P_m \mathcal{G} P_m)} \longrightarrow d_L(z; l) d_R(z; h) \quad (6.16)$$

locally uniformly in the disk.

For $a_0 = \phi/(2\pi)$, dilation and integer translation identify $\mathcal{K}_{n+r, \phi}$ with the compression to $(a_0 - r, 2n + a_0 + r)$. Equations (6.13) and (6.16) therefore specify the full constant:

$$\mathfrak{D}(z; r, \phi) = \mathfrak{D}_0(z) + \log d_L(z; \phi/(2\pi) - r) + \log d_R(z; \phi/(2\pi) + r) - 2r \log(1 - z^2). \quad (6.17)$$

Replacing $\log n$ by $\log(n + r)$ costs $o(1)$, uniformly for bounded r .

The fixed-strip bound gives a length-independent trace-norm endpoint modulus. For self-adjoint trace-class contractions, telescoping powers in the logarithm series gives

$$|\log \det(I - zA) - \log \det(I - zB)| \leq \frac{\rho}{1 - \rho} \|A - B\|_1, \quad |z| \leq \rho < 1.$$

The limiting half-line factors are continuous in their endpoints in trace norm. Finite nets in a compact endpoint set upgrade the proved convergence to uniform convergence there.

The C^2 trace case of Theorem 6.1. The determinant assertion has just been proved. Differentiate its locally uniform analytic limit at zero, using

$$\log \det(I - z\mathcal{K}) = - \sum_{k \geq 1} \frac{z^k}{k} \operatorname{tr} \mathcal{K}^k.$$

This proves convergence of the normalized trace remainder for every polynomial vanishing at zero. For $f \in C^2$, approximate f'' uniformly by polynomials and integrate twice with the exact values $f(0) = 0$ and $f'(0)$. The approximants converge in C^2 . The uniform trace-remainder estimate of Theorem 1.1 and Proposition 3.30 makes their limiting constants Cauchy and transfers the limit to f , uniformly on compact parameter sets. It also proves continuity of the limiting function. $\square$

Uniqueness of the limit and the exact phase symmetries give

$$\begin{aligned} \mathfrak{D}(z; r+1, \phi) &= \mathfrak{D}(z; r, \phi), & \mathfrak{D}(z; r, \phi + 2\pi) &= \mathfrak{D}(z; r, \phi), \\ \mathfrak{D}(z; r, -\phi) &= \mathfrak{D}(z; r, \phi), & \mathfrak{D}(z; r, \phi + \pi) &= \mathfrak{D}(-z; r, \phi). \end{aligned}$$

No elementary evaluation of the full Fredholm product, quantitative rate for its remainder, or boundary value on a spectral cut is asserted.

6.1.3 Lipschitz relative traces and Schatten constant terms

The exact model gap permits a stronger conclusion than global C^2 regularity. The needed relative-trace argument uses uniform finite-rank approximation of the boundary corrections; it does not use an operator-Lipschitz assertion.

Lemma 6.4 (Finite-rank tightness of relative traces). *Let A_m, B_m be self-adjoint trace-class contractions. Suppose $\sup_m \|A_m - B_m\|_1 < \infty$, and suppose that for every $\epsilon > 0$ there are self-adjoint finite-rank operators H_m such that, for all sufficiently large m ,*

$$\text{rank } H_m \leq r_\epsilon, \quad \|A_m - B_m - H_m\|_1 \leq \epsilon,$$

where r_ϵ is independent of m . If $\text{tr } p(A_m) - \text{tr } p(B_m)$ converges for every polynomial p with $p(0) = 0$, then it converges for every real Lipschitz f on $[-1, 1]$ with $f(0) = 0$.

Proof. For a self-adjoint trace-class operator A , define the signed counting function, with an immaterial value at zero, by

$$N_A(t) = \begin{cases} \#\{\lambda_j(A) > t\}, & t > 0, \\ -\#\{\lambda_j(A) < t\}, & t < 0. \end{cases}$$

It is integrable with $\int |N_A| = \|A\|_1$. Absolute continuity of a Lipschitz function and absolute summability of its scalar trace give

$$\text{tr } f(A) = \int f'(t) N_A(t) dt. \quad (6.18)$$

For $\xi_{A,B} = N_A - N_B$, two elementary bounds are

$$\int |\xi_{A,B}| \leq \|A - B\|_1, \quad |\xi_{A,B}(t)| \leq \text{rank}(A - B) \quad (t \neq 0) \quad (6.19)$$

when the rank is finite. Intersect a spectral subspace with $\ker(A - B)$ and apply the variational characterization above a positive threshold (or below a negative one) for the rank bound. For the integral bound, decompose a finite-matrix difference into signed rank-one terms. A positive increment has nonnegative counting difference by monotonicity, with integral equal to its trace by (6.18); reverse negative increments and telescope. For trace-class operators, truncate the two spectral expansions separately. Counts converge at every nonzero threshold outside the two countable spectra; Fatou proves the bound.

Put $C_m = B_m + H_m$. Its spectrum may extend outside $[-1, 1]$, which does not affect the estimates on that interval. The exact identity $\xi_{A_m, B_m} = \xi_{A_m, C_m} + \xi_{C_m, B_m}$ and (6.19) imply, for bounded integrable h supported in $[-1, 1]$,

$$\left| \int h \xi_{A_m, B_m} \right| \leq \epsilon \|h\|_\infty + r_\epsilon \|h\|_{L^1(-1,1)}. \quad (6.20)$$

Uniform polynomial approximation of derivatives and Lemma 3.18 first extend convergence to C^1 tests. For an L -Lipschitz f , mollify its zero-extended derivative and integrate from zero. Then $f_\delta(0) = 0$, $\|f'_\delta\|_\infty \leq L$ and $\|f' - f'_\delta\|_1 \rightarrow 0$. Equations (6.18) and (6.20) bound the relative-trace error by

$$2L\epsilon + r_\epsilon \|f' - f'_\delta\|_1.$$

Choose ϵ first and then δ . The bound is uniform for large m , so convergence of the smooth relative traces makes the original sequence Cauchy. $\square$

Both comparisons in the preceding subsection satisfy this lemma. Approximate D_+, D_- by self-adjoint finite-rank H_+, H_- . The sum $H_+ + S^m H_- S^{-m}$ has uniformly bounded rank and approximates the gauge correction in trace norm. The same argument applies to F_l^L, F_h^R . Polynomial relative traces converge by locally uniform determinant differentiation, so the lemma supplies both Lipschitz limits. These are limits of finite relative traces; no trace of an infinite difference $f(A_\pm) - f(B_\pm)$ is assumed.

Band-local completion of Theorem 6.1. Choose $F \in C^2([-1, 1])$ agreeing with f on the two closed bands and with $F(0) = 0$. For example, fill the central interval with the unique polynomial of degree at most six matching the value and first two derivatives at each of $-g, g$ and the value zero at zero. These are the seven Hermite conditions at nodes of multiplicities 3, 1, 3. Then $q = f - F$ is Lipschitz and vanishes at zero and on the two bands.

Lemma 7.1 gives exactly m model eigenvalues in each band, and all other eigenvalues are zero. Thus $\text{tr } q(T_m(a)) = 0$ exactly. The Lipschitz gauge comparison just established implies convergence of $\text{tr } q(\mathcal{K}_n)$. Both its bulk and logarithmic coefficients vanish, since the coefficient functional samples only the two bands and their endpoints. Add this limit to the global C^2 constant already proved for F .

The Lipschitz endpoint comparison gives the result for fixed r, ϕ , with bulk adjustment $2r[f(1) + f(-1)]$ and a vanishing change in the logarithm. The fixed-strip modulus and the scalar trace perturbation bound give compact-parameter uniformity by finite nets. Finally, $|x|^p$ is Lipschitz for $p \geq 1$ and is C^2 on both closed bands, which proves (6.3). $\square$

The band C^2 assumption in this theorem is stronger than the band $C^{1,1}$ assumption sufficient for a bounded remainder in Theorem 3.2. No convergent constant for that entire larger class is inferred here.

6.1.4 A phase constant for shrinking finite sampling scale

Use the matrix of (1.1) and the dimension-independent Lipschitz comparison of Theorem 4.2, with $\Gamma_\varepsilon = 16\sqrt{2/\pi} \varepsilon / [(1 - 2\varepsilon)\sqrt{1 - 4\varepsilon^2}]$.

Corollary 6.5 (A third term for shrinking finite scale). *Let $\varepsilon_N \rightarrow 0$ and $\tau_N = (2N+1)\varepsilon_N/2 \rightarrow \infty$. For every f satisfying Theorem 6.1,*

$$\text{tr } f(A_N(\varepsilon_N)) = (2N+1)\varepsilon_N[f(1) + f(-1)] + \kappa(f) \log \tau_N + \mathfrak{C}_f(\{\tau_N\}, 0) + o(1). \quad (6.21)$$

Here $\{\tau_N\}$ is the fractional part, and the constant profile is continuous and periodic. This includes every Schatten power $p \geq 1$, with no condition on $(2N+1)\varepsilon_N^3$.

Proof. The direct Lipschitz comparison has error at most $\Gamma_{\varepsilon_N} \text{Lip}(f) = o(1)$. Apply the phase law uniformly for $r \in [0, 1]$ with $n = \lfloor \tau_N \rfloor$ and $r = \{\tau_N\}$. Periodicity follows from uniqueness under replacing (n, r) by $(n-1, r+1)$. Thus convergence of the fractional parts is not required. $\square$

For fixed $\varepsilon > 0$ the comparison supplies a bounded remainder; it does not identify a finite-matrix constant term in that regime.

7 Physical gap states and one-period flow

7.1 Boundary spectra in the open central gaps

Retain $\mathcal{U}, \mathcal{C}_\pm, L_b, H_b$ from Section 6.1. The same localization determines every limiting eigenvalue away from zero and the band edges. Put

$$\Sigma = \{0\} \cup [-1, -g] \cup [g, 1], \quad \Omega = \mathbb{C} \setminus \Sigma, \quad g = 2^{-1/2}.$$

Lemma 7.1 (The reference resolvent gap). *The spectra of $\mathcal{C}_m, \mathcal{C}_+, \mathcal{C}_-$ are contained in Σ . The finite model has exactly m positive and m negative eigenvalues. Their resolvents, and the translated right resolvents, converge strongly to the appropriate half-line resolvents, including their adjoints, uniformly on compact subsets of Ω .*

Proof. For a finite or either half-infinite cell projection E , the scalar compressions $C_E = E\mathcal{L}(c)E$ and $S_E = E\mathcal{L}(s)E$ satisfy $C_E + S_E \geq I$ on that cell space because $c(\theta) + s(\theta) \geq 1$. Hence

$$C_E^2 + S_E^2 = \frac{1}{2}[(C_E + S_E)^2 + (C_E - S_E)^2] \geq \frac{1}{2}I.$$

The row operator $B_E = (S_E, C_E)$ has $B_E B_E^* \geq I/2$ and is onto. The polar decomposition of the star operator $\begin{pmatrix} 0 & B_E \\ B_E^* & 0 \end{pmatrix}$ therefore confines its nonzero spectrum to the two bands. Its norm is at most one by compression of $\mathcal{L}(a)$; the ambient complement adds only zero. The finite count follows from the row rank m .

At each fixed boundary the finite compressions converge strongly. Continuous functional calculus on the common spectral set Σ gives strong resolvent convergence. On compact $K \subset \Omega$ the norm bound $\text{dist}(K, \Sigma)^{-1}$, the resolvent identity and finite nets give uniformity; conjugating the parameter handles adjoints. $\square$

For fixed real l, h let $E_m = P_{(l, m+h)}$ in physical coordinates and define

$$\begin{aligned} \mathcal{A}_m(l, h) &= \mathcal{U}^* E_m \mathcal{G} E_m \mathcal{U}, \quad \mathcal{A}_l^+ = \mathcal{U}^* L_l \mathcal{G} L_l \mathcal{U}, \quad \mathcal{A}_h^- = \mathcal{U}^* H_h \mathcal{G} H_h \mathcal{U}, \\ D_l^+ &= \mathcal{A}_l^+ - \mathcal{C}_+, \quad D_h^- = \mathcal{A}_h^- - \mathcal{C}_-. \end{aligned}$$

The last two differences are trace class. Adding the gauge and strip localizations gives

$$\|\mathcal{A}_m(l, h) - \mathcal{C}_m - D_l^+ - S^m D_h^- S^{-m}\|_1 \rightarrow 0. \quad (7.1)$$

Combining the corrections before taking the resolvent is essential: the star baseline has a proved gap, whereas a physical half-line baseline could itself have a gap eigenvalue.

Define, for $\lambda \in \Omega$,

$$\begin{aligned} F_m(\lambda; l, h) &= \det[I + (\mathcal{A}_m(l, h) - \mathcal{C}_m)(\mathcal{C}_m - \lambda)^{-1}], \\ F_+(\lambda; l) &= \det[I + D_l^+(\mathcal{C}_+ - \lambda)^{-1}], \quad F_-(\lambda; h) = \det[I + D_h^-(\mathcal{C}_- - \lambda)^{-1}]. \end{aligned} \quad (7.2)$$

They are analytic Fredholm determinants of identity plus trace-class operators. The signs follow from $I + (A - C)(C - \lambda)^{-1} = (A - \lambda)(C - \lambda)^{-1}$.

Theorem 7.2 (Local gap eigenvalue convergence). *Locally uniformly on Ω ,*

$$F_m(\lambda; l, h) \longrightarrow F_+(\lambda; l) F_-(\lambda; h). \quad (7.3)$$

The two limiting factors have precisely the eigenvalues in Ω of $L_l \mathcal{G} L_l$ and $H_h \mathcal{G} H_h$ as zeros, respectively, with their finite multiplicities.

For fixed r, ϕ take $l = \phi/(2\pi) - r$, $h = \phi/(2\pi) + r$ and $m = 2n$. If J is a compact interval strictly inside $(0, g)$ or $(-g, 0)$, with endpoints outside the two boundary spectra, then every boundary eigenvalue $\alpha \in J$ attracts exactly

$$k(\alpha) = \dim \ker(L_l \mathcal{G} L_l - \alpha) + \dim \ker(H_h \mathcal{G} H_h - \alpha) \quad (7.4)$$

eigenvalues of $\mathcal{K}_{n+r, \phi}$, counting multiplicity. For sufficiently large n there are no other eigenvalues in J .

Proof. Use (7.1) and Lemma 7.1 in the proof of Lemma 6.3. Trace-class multiplication and the vanishing product between shifted boundaries give (7.3), now locally uniformly on Ω .

The limiting factors are relative determinants of the self-adjoint half-line pairs. They are nonzero for nonreal λ , so analytic Fredholm theory on the connected set Ω gives isolated zeros with finite kernels. Conjugating by $\mathcal{U}$ identifies their physical eigenspaces.

The zero order equals the eigenvalue multiplicity. Indeed, for an isolated eigenvalue α of multiplicity k , let Π be its finite-rank spectral projection and replace A by $\tilde{A} = A + \epsilon\Pi$, moving that eigenvalue outside a small disk. Factor the relative determinant through $\tilde{A}$. Its nonvanishing analytic factor is multiplied by

$$\det[(A - \lambda)(\tilde{A} - \lambda)^{-1}] = \left(\frac{\alpha - \lambda}{\alpha + \epsilon - \lambda} \right)^k.$$

This also proves the multiplicity assertion for the finite determinants.

Around the finitely many limiting zeros in J , choose disjoint disks in Ω . Rouché's theorem gives multiplicity (7.4) in each for large m ; all finite zeros are real. The limit is bounded away from zero on the remaining compact set, excluding other roots. Shrinking the disks proves convergence. $\square$

7.1.1 Exact eventual count offsets

For $\lambda \in (0, g)$ outside the two boundary spectra, define

$$\begin{aligned} \iota_+(\lambda; l) &= \text{tr}[\mathbf{1}_{(\lambda, \infty)}(\mathcal{A}_l^+) - \mathbf{1}_{(\lambda, \infty)}(\mathcal{C}_+)], \\ \iota_-(\lambda; h) &= \text{tr}[\mathbf{1}_{(\lambda, \infty)}(\mathcal{A}_h^-) - \mathbf{1}_{(\lambda, \infty)}(\mathcal{C}_-)]. \end{aligned} \quad (7.5)$$

The differences are trace class: integrate the trace-class resolvent differences over a positively oriented rectangle enclosing $[\lambda, 1]$, whose left side crosses the real axis at λ and whose right side lies beyond one. Each trace is an integer. To see this directly for a trace-class difference $D = P - Q$ of orthogonal projections, note that D anticommutes with $P + Q - I$ and $(P + Q - I)^2 = I - D^2$. Thus its eigenvalues in $(-1, 1) \setminus \{0\}$ pair with their negatives. Only the finite-dimensional $+1$ and -1 eigenspaces contribute to its trace.

Corollary 7.3 (Fixed-threshold counts inside the gap). *With l, h as in Theorem 7.2, for every $\lambda \in (0, g)$ outside the boundary spectra one has, for all sufficiently large integers n ,*

$$\#\{\lambda_j(\mathcal{K}_{n+r, \phi}) > \lambda\} = 2n + \iota_+(\lambda; l) + \iota_-(\lambda; h). \quad (7.6)$$

The analogous formula for $\lambda \in (-g, 0)$ counts eigenvalues below λ , with the negative spectral projections and the same reference count $2n$.

Proof. On the contour just described the limiting determinant is nonzero. Equation (7.3) and Cauchy differentiation imply convergence of the logarithmic derivatives there. Direct differentiation of the relative determinant, followed by the resolvent identity, gives

$$\frac{F'_m(\zeta)}{F_m(\zeta)} = \text{tr}[(\zeta - \mathcal{A}_m)^{-1} - (\zeta - \mathcal{C}_m)^{-1}],$$

and the same identity holds for each half-line factor. Contour integration consequently gives

$$\#\{\lambda_j(\mathcal{A}_m) > \lambda\} - m \longrightarrow \iota_+(\lambda; l) + \iota_-(\lambda; h).$$

Both sides are integers, so equality holds eventually. The reference count is exactly m by Lemma 7.1, and $m = 2n$. For the negative version use a contour enclosing $[-1, \lambda]$. $\square$

7.1.2 The two parity copies for a centered interval

Corollary 7.4 (Parity of an isolated gap cluster). *For $\phi = 0$, let α be an isolated eigenvalue in either open central gap of $L_{-r}\mathcal{G}L_{-r}$, of multiplicity k . Exactly k eigenvalues in each parity sector of $\mathcal{K}_{n+r}$ converge to α . The statement for $\phi = \pi$ follows by negating the eigenvalues.*

Proof. Reflection exchanges $L_{-r}\mathcal{G}L_{-r}$ and $H_r\mathcal{G}H_r$, giving total multiplicity $2k$. On $(-r, 2n+r)$ the reflection $\mathcal{R}_n u(X) = u(2n-X)$ commutes with the physical compression; no gauge symmetry is needed.

Let $v_1, \dots, v_k$ be an orthonormal basis of the left half-line eigenspace, extended by zero, and put $v_{j,n} = E_{2n}v_j$. Strong convergence gives $v_{j,n} \rightarrow v_j$ and

$$\|(E_{2n}\mathcal{G}E_{2n} - \alpha)v_{j,n}\| \rightarrow 0.$$

Reflection preserves residual norms and is weakly null as its center escapes, so left/right inner products tend to zero. Each family

$$2^{-1/2}(v_{j,n} + \mathcal{R}_n v_{j,n}), \quad 2^{-1/2}(v_{j,n} - \mathcal{R}_n v_{j,n}), \quad 1 \leq j \leq k,$$

has Gram matrix tending to identity and residual tending uniformly to zero on its span. The spectral theorem forces at least k nearby eigenvalues in each parity. The exact total $2k$ makes both counts k ; shrinking the neighborhood proves convergence. $\square$

7.1.3 One boundary phase for each gap value

Write $B_b = L_b\mathcal{G}L_b|_{L^2(b,\infty)}$. For $\sigma \in \{1, -1\}$, $0 < t < g$, and $\sigma t \notin \text{spec } B_b$, put

$$I_\sigma(t; b) = \text{tr}[\mathbf{1}_{(t,\infty)}(\sigma\mathcal{A}_b^+) - \mathbf{1}_{(t,\infty)}(\sigma\mathcal{C}_+)].$$

Thus $I_+(t; b) = \iota_+(t; b)$, whereas I_- counts negative eigenvalues of the left boundary operator by their magnitude. The one-period count is analogous to the Dirichlet spectral flow for periodic Hill operators [21, Proposition 3.21]. Here the projection index and analytic bandlimited covariance provide the crossing count for the nonlocal sine-Loewner compression.

Theorem 7.5 (One-cell boundary spectral flow). *At regular thresholds $I_\sigma(t; b)$ is nonincreasing in b , and*

$$I_\sigma(t; b+1) = I_\sigma(t; b) - 1. \quad (7.7)$$

For every $\lambda \in (-g, 0) \cup (0, g)$, precisely one phase $b \pmod{1}$ has $\lambda \in \text{spec } B_b$, and this eigenvalue is simple. Every local gap eigenbranch is real analytic and satisfies

$$\lambda'(b) = -\lambda(b)|u_b(b)|^2 \quad (7.8)$$

for a normalized physical eigenfunction u_b . Its magnitude is strictly decreasing along a connected gap branch; the derivative is allowed to vanish at isolated phases.

Proof. For $b < c$, $\delta = c - b$, and $E = P_{(b,c)}$, one has

$$L_b\mathcal{G}L_b - L_c\mathcal{G}L_c = E\mathcal{G}L_b + L_c\mathcal{G}E.$$

The Taylor expansion of the restricted Fourier kernel about b gives

$$\|E\mathcal{F}^*\mathbf{1}_{(-1,1)}\|_1 \leq \sum_{n \geq 0} \frac{(2\pi)^n}{n!} \|(x-b)^n\|_{L^2(b,c)} \|\xi^n\|_{L^2(-1,1)} \leq \sqrt{2\delta} e^{2\pi\delta}.$$

Thus $\mathcal{G} = \mathcal{F}^*J\mathcal{F}$ gives trace-norm modulus $2\sqrt{2\delta}e^{2\pi\delta}$. At a regular threshold a common Riesz contour makes the relative projection continuous in trace norm, and its integer trace locally constant.

For monotonicity work in physical space and put $P_b = \mathbf{1}_{(t,\infty)}(\sigma L_b \mathcal{G} L_b)$. For $b < c$ and $0 \neq u \in \text{ran } P_c$,

$$\langle u, \sigma L_b \mathcal{G} L_b u \rangle = \langle u, \sigma L_c \mathcal{G} L_c u \rangle > t \|u\|^2.$$

Hence $\text{ran } P_c \cap \ker P_b = \{0\}$. The trace formula for a trace-class difference of projections yields $\text{tr}(P_b - P_c) = \dim(\text{ran } P_b \cap \ker P_c) \geq 0$.

Let $Q_\sigma = \mathbf{1}_{(t,\infty)}(\sigma \mathcal{C}_+)$. We next show

$$\text{tr}(Q_\sigma - \mathbf{S} Q_\sigma \mathbf{S}^{-1}) = 1. \quad (7.9)$$

Deform the star row to a constant by $c_a = (1-a)c + a$, $s_a = (1-a)s$, $0 \leq a \leq 1$. Then $c_a + s_a \geq 1$ and $(c_a^2 + s_a^2)^{1/2} \leq 1$. The proof of Lemma 7.1 therefore preserves the gap throughout this homotopy. The difference of a half-line star and its one-cell translate is finite rank, continuously in trace norm: only the rows and columns of the removed cell in the fixed three-dimensional star fiber occur. A resolvent contour shows that $Q_{\sigma,a} - \mathbf{S} Q_{\sigma,a} \mathbf{S}^{-1}$ is trace-class continuous. Its integer trace is constant. At $a = 1$ the constant star has one state of each sign per cell, so the trace is one. Since the gauge commutes with the cell shift and $G(x+1, y+1) = G(x, y)$, $\mathcal{A}_{b+1}^+ = \mathbf{S} \mathcal{A}_b^+ \mathbf{S}^{-1}$. Equation (7.9) now proves (7.7).

We exclude a nonzero gap eigenvalue persisting on a phase interval. On $\mathcal{H} = L^2(-1, 1)$ let V_b be Fourier synthesis restricted to (b, ∞) and set $Q_b = V_b^* V_b$. Then

$$B_b = V_b J V_b^*, \quad \mathcal{T}_b(\lambda) = Q_b - \lambda J.$$

For $\lambda \neq 0$, the $I - AB/I - BA$ block factorization makes $B_b - \lambda$ invertible, respectively Fredholm, if and only if $\mathcal{T}_b(\lambda)$ is. Their kernels have the same dimension: if $\mathcal{T}_b(\lambda)u = 0$, then $V_b u \neq 0$ and $B_b V_b u = \lambda V_b u$; conversely $B_b f = \lambda f$ gives the nonzero kernel vector $J V_b^* f$. With $v_x(\xi) = e^{2\pi i x \xi}$,

$$Q_b - Q_c = \int_b^c |v_x\rangle \langle v_x| dx, \quad Q'_b = -|v_b\rangle \langle v_b|. \quad (7.10)$$

The integral is in trace norm. The bounded frequency interval makes Q_b norm analytic, and the pencil is self-adjoint Fredholm at a real nonzero central-gap parameter.

If a fixed λ persisted on an interval, analytic spectral reduction, after avoiding any isolated extra crossings, would give a nonzero analytic kernel bundle of constant finite rank. Choose an orthonormal parallel frame $u_j(b)$, so that its derivative has zero projection onto the kernel. Differentiating $\mathcal{T}_b(\lambda)u_j(b) = 0$ gives

$$0 = \langle u_j, Q'_b u_j \rangle = -|\langle v_b, u_j \rangle|^2.$$

Thus $Q'_b u_j = 0$, the differentiated equation puts u'_j in the kernel, and parallel transport forces $u'_j = 0$. The Fourier transform of this constant nonzero vector would vanish on an interval. Its bounded frequency support makes it entire, a contradiction. Analytic Fredholm reduction now shows that the crossing phases for a fixed λ are isolated and therefore finite in a period.

For the crossing direction, translate the half-line to $(0, \infty)$. The kernel $G_b(x, y) = G(x+b, y+b)$ is a norm-analytic sine/cosine combination of two fixed bounded operators. Isolated eigenclusters have real-analytic eigenbranches. A nonzero eigenfunction is the restriction of $\lambda^{-1} \mathcal{G}_b \tilde{u}$, a bandlimited function, so all its boundary traces exist and it belongs to $H^k(0, \infty)$ for every k . Integration by parts, using $\partial_b G_b = \partial_x G_b + \partial_y G_b$, gives

$$B'_b u = (\lambda - B_b) \partial_x u - G_b(\cdot, 0) u(0).$$

The boundary eigenvalue equation and Hellmann–Feynman identity imply (7.8). Each magnitude branch is therefore nonincreasing. It cannot be constant on any interval by the preceding argument, so it is strictly decreasing in value. The derivative can have isolated zeros.

Choose a regular phase b_0 for a fixed σt . At a crossing of multiplicity k , isolate the finite cluster between two nearby regular thresholds. Its k analytic branches all cross downward in magnitude, and local constancy at the lower threshold shows that $I_\sigma(t; b)$ drops by k . Its total drop on $(b_0, b_0 + 1)$ is one. Hence

$$\sum_{b \in (b_0, b_0 + 1): \sigma t \in \text{spec } B_b} \dim \ker(B_b - \sigma t) = 1.$$

This proves both the unique crossing phase and simplicity. $\square$

Corollary 7.6 (Opposite signs and centered clusters). *For $0 < t < g$, the unique phases at which t and $-t$ occur differ by $1/2$ modulo one. In particular, no B_b has both t and $-t$ as eigenvalues. Every isolated gap cluster in the centered case of Corollary 7.4 consists of exactly one even and one odd eigenvalue for all sufficiently large n .*

Proof. Translation by $1/2$ changes G to $-G$, so $B_{b+1/2}$ is unitarily equivalent to $-B_b$. Uniqueness of the crossing phase proves the first two claims. Theorem 7.5 makes the multiplicity k in Corollary 7.4 equal to one. $\square$

The flow does not evaluate the crossing phase or an absolute index, identify a floor-ranked level, or control limits toward zero or the band edges. Infinitely many levels accumulating at zero remain possible.

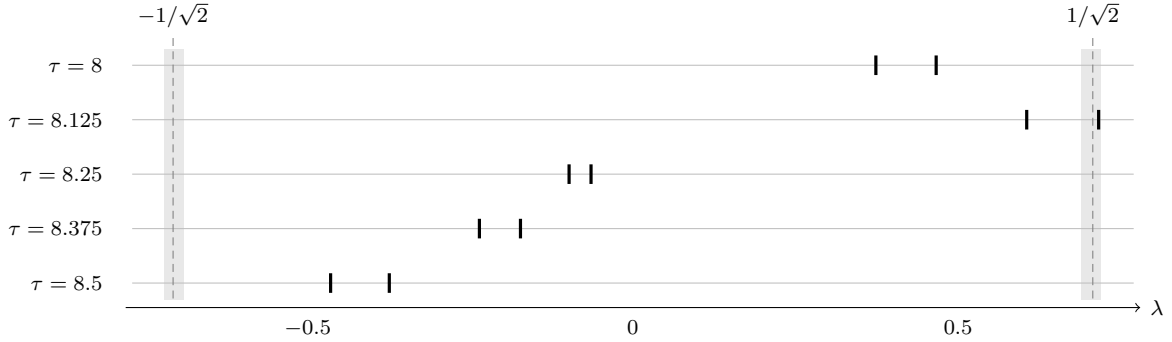


Figure 3: Fractional-scale dependence: each row shows all eigenvalues with $0.05 < |\lambda| < 0.72$; shading marks $0.69 < |\lambda| < 0.72$. The selection is not a continuation of fixed branches: the last three scales also have a positive eigenvalue outside the window, at 0.7478, 0.7836 and 0.7860. Thus neither displayed signs nor guard-band emptiness is uniform in τ . Gauss–Legendre and Clenshaw–Curtis values agree to 10^{-10} ; these are ordinary-precision samples from `anc/verify_phase_dependence.py`, not certified boundary limits.

7.2 An explicit endpoint equation and quantitative gap limits

We make the limiting spectral equation explicit and derive the physical tail exponent. Unlike the adjacent-interval transform diagonalized in [9, Sections 2–4], this pencil retains same-interval Hilbert blocks and a phased exchange, producing four endpoint channels.

For $\theta = 2\pi l$, define on $L^2(-1, 1)$

$$Q_0 = \frac{1}{2}(I + iH), \quad Hu(\xi) = \frac{1}{\pi} \text{pv} \int_{-1}^1 \frac{u(\eta)}{\xi - \eta} d\eta,$$

$$(J_\theta u)(\xi) = \begin{cases} e^{-i\theta} u(\xi - 1), & 0 < \xi < 1, \\ e^{i\theta} u(\xi + 1), & -1 < \xi < 0, \end{cases} \quad P_\lambda = Q_0 - \lambda J_\theta.$$

Proposition 7.7 (Exact frequency pencil). *For every nonzero λ , the physical operator $L_l \mathcal{G} L_l$ has eigenvalue λ precisely when P_λ has a nonzero kernel. The correspondence preserves the algebraic multiplicity of an isolated eigenvalue. If $P_\lambda u = 0$, its physical eigenfunction is the restriction to $X > l$ of*

$$h(X) = \int_{-1}^1 e^{-2\pi i(X-l)\xi} u(\xi) d\xi, \quad \mathcal{G} L_l h = \lambda h \quad \text{on } \mathbb{R}. \quad (7.11)$$

Equivalently, the frequency equation is $Hu = i(I - 2\lambda J_\theta)u$.

Proof. Let V_l restrict inverse Fourier transformation from $L^2(-1, 1)$ to $L^2(l, \infty)$, and let $M_l u(\xi) = e^{2\pi i l \xi} u(\xi)$. Direct integration gives $L_l \mathcal{G} L_l = V_l J V_l^*$ and $V_l^* V_l = M_l Q_0 M_l^*$, where $J = J_0$. The bounded-product identity used in Theorem 7.5 preserves nonzero spectra and isolated algebraic multiplicities. Conjugation by M_l gives $J_\theta Q_0$ and P_λ . The map $V_l M_l$ is injective by entire uniqueness. Finally $Q_0 u = \lambda J_\theta u$ implies $J V_l^* V_l M_l u = \lambda M_l u$; inverse transformation on the whole line gives (7.11). $\square$

The endpoint channels are $u(t), u(t-1), u(1-t), u(-t)$ as $t \downarrow 0$. For real Mellin frequency ω , put

$$q = \frac{1 + \tanh \pi \omega}{2}, \quad r = \frac{1 - \tanh \pi \omega}{2}, \quad c = \frac{1}{2} \operatorname{sech} \pi \omega, \quad D_\theta = \operatorname{diag}(1, e^{i\theta}, e^{-i\theta}, 1).$$

The endpoint pencil has the explicit multiplier

$$m_{\lambda, \theta}(\omega) = D_\theta \begin{pmatrix} q & -\lambda & 0 & ic \\ -\lambda & q & 0 & 0 \\ 0 & 0 & r & -\lambda \\ -ic & 0 & -\lambda & r \end{pmatrix} D_\theta^*, \quad \det m_{\lambda, \theta} = \lambda^2 \left(\lambda^2 - \frac{1 + \tanh^2 \pi \omega}{2} \right). \quad (7.12)$$

Its inverse is its adjugate divided by the displayed scalar determinant. In particular it is a bounded Mellin multiplier for $\lambda \in \Omega$.

For $K \in \mathfrak{S}_2$ we use the regularized determinant $\det_2(I + K) = \det((I + K)e^{-K})$; the operator inside the latter determinant differs from the identity by a trace-class operator [42, Chapter 9].

Theorem 7.8 (Explicit scalar equation in the central gaps). *The cutoff construction in Section 7.3 gives explicit operators $E, \mathcal{I}, M_\lambda, N_\lambda, \Delta_e, \Delta_i$, where M_λ^{-1} has symbol $m_{\lambda, \theta}^{-1}$ from (7.12), N_λ^{-1} is a piecewise constant Fourier multiplier, and Δ_e, Δ_i have the elementary Hilbert–Schmidt kernels (7.20) and (7.21). Define*

$$\begin{aligned} R_\lambda &= E^* M_\lambda^{-2} E + \mathcal{I}^* N_\lambda^{-2} \mathcal{I}, \\ K_\lambda &= E^* (M_\lambda^{-2} \Delta_e P_\lambda + M_\lambda^{-1} \Delta_e) + \mathcal{I}^* (N_\lambda^{-2} \Delta_i P_\lambda + N_\lambda^{-1} \Delta_i). \end{aligned} \quad (7.13)$$

Then $R_\lambda P_\lambda^2 = I + K_\lambda$, with K_λ holomorphic in $\mathfrak{S}_2$ on Ω . For every real $\lambda \in (-g, 0) \cup (0, g)$,

$$\det_2(I + K_\lambda) = 0 \iff \lambda \in \sigma_p(L_l \mathcal{G} L_l). \quad (7.14)$$

An eigenvalue of multiplicity k is a zero of order $2k$. Here $R_\lambda \geq (1 + |\lambda|)^{-2} I$ on the real gaps; it remains invertible on a neighborhood of each such point. No global complex invertibility of R_λ is asserted.

Theorem 7.9 (Endpoint exponent and physical tails). *Let $\lambda \in (-g, 0) \cup (0, g)$ and $u \in \ker P_\lambda \setminus \{0\}$, and define*

$$\beta(\lambda) = \frac{1}{\pi} \arctan \sqrt{1 - 2\lambda^2}, \quad \nu = \frac{1}{2} + \beta, \quad \gamma(\lambda) = 2\beta = \frac{1}{\pi} \arccos \frac{\lambda^2}{1 - \lambda^2}. \quad (7.15)$$

There is a possibly zero vector $v = (v_1, v_2, v_3, v_4)^T$ in the one-dimensional kernel of $m_{\lambda, \theta}(-i\beta)$ such that, for every fixed $\beta < a < 1/2$,

$$(u(t), u(t-1), u(1-t), u(-t))^T = t^{\nu-1}v + O(t^{-1/2+a}). \quad (7.16)$$

The expansion holds after any fixed number of derivatives, with the corresponding power lost from the remainder. The continuation in (7.11), with $\sigma = \operatorname{sgn} T$, satisfies

$$h(l+T) = \frac{\Gamma(\nu)}{(2\pi|T|)^\nu} [e^{-i\sigma\pi\nu/2}(v_1 + e^{2\pi iT}v_2) + e^{i\sigma\pi\nu/2}(v_4 + e^{-2\pi iT}v_3)] + O(|T|^{-1/2-a}). \quad (7.17)$$

In particular, $|h(l+T)| \leq C(1+|T|)^{-\nu}$. More precisely,

$$\int_L^\infty |h(l+\sigma T)|^2 dT = C_\sigma L^{-\gamma(\lambda)} + O(L^{-\beta-a}), \quad (7.18)$$

where

$$C_\sigma = \frac{\Gamma(\nu)^2}{(2\pi)^{2\nu}(2\beta)} \left(|e^{-i\sigma\pi\nu/2}v_1 + e^{i\sigma\pi\nu/2}v_4|^2 + |v_2|^2 + |v_3|^2 \right).$$

Both constants are positive if $v \neq 0$. Their vanishing is not excluded.

Corollary 7.10 (Rate for each isolated boundary cluster). *Fix l, h and an isolated eigenvalue α in either open central gap of one or both boundary operators in Theorem 7.2. Let k be its total boundary multiplicity. For integer $m \rightarrow \infty$, the k eigenvalues of $P_{(l,m+h)} \mathcal{G}P_{(l,m+h)}$ converging to α satisfy*

$$\max_{1 \leq j \leq k} |\lambda_j - \alpha| = O(m^{-\gamma(\alpha)}).$$

The constants may depend on the fixed endpoints, eigenvalue, and separating interval. In the centered case, each of the two parity clusters in Corollary 7.4 obeys this rate.

Section 7.3 proves the claims and supplies a finite-rank error scheme. The equation does not evaluate its roots in closed form; the tail and rate apply to every isolated root.

7.3 Proofs of the endpoint equation and gap rate

7.3.1 The explicit local models and all localization errors

In coordinates $f_1(t) = u(t)$, $f_2(t) = u(t-1)$, $0 < t < 1$, the finite Hilbert transform has kernel

$$\frac{1}{\pi} \begin{pmatrix} \operatorname{pv}(t-s)^{-1} & (t-s+1)^{-1} \\ (t-s-1)^{-1} & \operatorname{pv}(t-s)^{-1} \end{pmatrix}.$$

On $L^2(0, \infty)$ let H_∞, C_∞ have kernels $[\pi(t-s)]^{-1}$, in principal value, and $[\pi(t+s)]^{-1}$. On four copies of this space set

$$H_e = \begin{pmatrix} H_\infty & 0 & 0 & C_\infty \\ 0 & H_\infty & 0 & 0 \\ 0 & 0 & -H_\infty & 0 \\ -C_\infty & 0 & 0 & -H_\infty \end{pmatrix}, \quad Q_e = \frac{1}{2}(I + iH_e), \quad J_e = \operatorname{diag}(J_\theta, J_\theta), \quad M_\lambda = Q_e - \lambda J_e.$$

For the unitary Mellin transform $\mathcal{M}f(\omega) = (2\pi)^{-1/2} \int_0^\infty t^{-1/2-i\omega} f(t) dt$, the two scalar multipliers are $-i \tanh \pi\omega$ and $\operatorname{sech} \pi\omega$. Indeed these follow from the principal-value cotangent integral and the beta integral at exponent $1/2 + i\omega$. Thus M_λ has multiplier (7.12). Expansion of its determinant gives $\lambda^4 - \lambda^2(q^2 + r^2) + q^2r^2 - c^2qr$; using $c^2 = qr$ proves the asserted formula. The 2×2 block of Q_e on channels 1 and 4 is a projection, and its other entries are q, r . Consequently

$0 \leq Q_e \leq I$. The inverse multiplier is bounded on compact subsets of Ω : its denominator is bounded away from zero, and its adjugate is bounded.

Choose $0 < \delta < 1/4$ and a real smooth cutoff $0 \leq \chi \leq 1$, equal to one on $[0, \delta]$ and zero on $[2\delta, \infty)$, such that $\psi(t) = \sqrt{1 - \chi(t)^2 - \chi(1-t)^2}$ is smooth. A smooth sine/cosine transition has this property. Extend ψ by zero. Define

$$Ef(t) = \chi(t)(f_1(t), f_2(t), f_1(1-t), f_2(1-t))^T, \quad \mathcal{I}f = \psi f,$$

where the outputs are in $L^2((0, \infty); \mathbb{C}^4)$ and $L^2(\mathbb{R}; \mathbb{C}^2)$, respectively, with zero extension. Their adjoints are

$$E^*v(t) = \begin{pmatrix} \chi(t)v_1(t) + \chi(1-t)v_3(1-t) \\ \chi(t)v_2(t) + \chi(1-t)v_4(1-t) \end{pmatrix}, \quad \mathcal{I}^*w(t) = \psi(t)w(t).$$

Hence $E^*E + \mathcal{I}^*\mathcal{I} = 1$, $EJ_\theta = J_eE$, and $\mathcal{I}J_\theta = J_\theta\mathcal{I}$. Let Q_i be the whole-line Cauchy projection on each of the two interior components and $N_\lambda = Q_i - \lambda J_\theta$. In angular Fourier frequency,

$$n_\lambda(\omega)^{-1} = \begin{cases} -\lambda^{-1}J_\theta, & \omega < 0, \\ (I + \lambda J_\theta)/(1 - \lambda^2), & \omega > 0. \end{cases} \quad (7.19)$$

The errors $\Delta_e = EQ_0 - Q_eE$ and $\Delta_i = \mathcal{I}Q_0 - Q_i\mathcal{I}$ are independent of λ, θ . For $t > 0$, $0 < s < 1$, the first has kernel

$$\frac{i}{2\pi} \begin{pmatrix} \frac{\chi(t) - \chi(s)}{t - s} & \frac{\chi(t) - \chi(1-s)}{t + 1 - s} \\ \frac{\chi(t)}{\chi(t) - \chi(1-s)} & \frac{\chi(t) - \chi(s)}{t - s} \\ \frac{\chi(t) - \chi(1-s)}{1 - t - s} & \frac{\chi(t)}{\chi(t) - \chi(1-s)} \\ -\frac{\chi(t) - \chi(s)}{t + s} & \frac{\chi(t) - \chi(1-s)}{1 - t - s} \end{pmatrix}. \quad (7.20)$$

For $t \in \mathbb{R}$, $0 < s < 1$, the second has kernel

$$\frac{i}{2\pi} \begin{pmatrix} (\psi(t) - \psi(s))/(t - s) & \psi(t)/(t - s + 1) \\ \psi(t)/(t - s - 1) & (\psi(t) - \psi(s))/(t - s) \end{pmatrix}. \quad (7.21)$$

Divided differences take their continuous values; a term supported where its numerator is identically zero is zero. The kernels are square integrable: diagonal quotients are bounded by cutoff derivatives; corner quotients vanish below distance δ , with denominator at least δ elsewhere; unmatched cross terms are separated from their poles. Output tails are $O(|t|^{-1})$. The touching-endpoint Carleman singularity remains in the model.

7.3.2 The scalar equation and its multiplicities

Proof of Theorem 7.8. The intertwining errors satisfy $EP_\lambda = M_\lambda E + \Delta_e$ and $\mathcal{I}P_\lambda = N_\lambda \mathcal{I} + \Delta_i$. Therefore

$$EP_\lambda^2 = M_\lambda^2 E + M_\lambda \Delta_e + \Delta_e P_\lambda,$$

with the analogous interior identity. Multiplying by the inverse squares and using $E^*E + \mathcal{I}^*\mathcal{I} = 1$ gives (7.13). All inverse powers here are algebraic powers, not adjoint products. The explicit multipliers are bounded and holomorphic on Ω , so K_λ is holomorphic in $\mathfrak{S}_2$.

For real gap λ , self-adjoint model pencils of norm at most $1 + |\lambda|$ have inverse squares at least $(1 + |\lambda|)^{-2}I$. Thus R_λ is invertible and P_λ^2 is Fredholm, with $\ker P_\lambda^2 = \ker P_\lambda$. The exact frequency correspondence proves the real zero equivalence.

At a real isolated eigenvalue α of multiplicity k , the bounded-product correspondence identifies the nonzero generalized eigenspace of $J_\theta Q_0$ with that of the self-adjoint physical operator. It is therefore semisimple of dimension k . Its Riesz decomposition splits $J_\theta Q_0 - \lambda$ into $(\alpha - \lambda)I_k$ and an invertible analytic complement. The analytic Fredholm pencil $P_\lambda = J_\theta(J_\theta Q_0 - \lambda)$ has characteristic multiplicity k ; its square has multiplicity $2k$. Multiplication by the locally invertible analytic R_λ preserves this number. Finite-dimensional Schur reduction of an identity plus $\mathfrak{S}_2$ family shows that its regularized determinant has this zero order; the regularizing factor is a nonvanishing exponential. $\square$

The equation permits finite-rank error control. For $T \geq 2$, the displayed kernels give

$$\|\mathbf{1}_{t>T}\Delta_e\|_2^2 \leq \frac{3}{2\pi^2(T-1)}, \quad \|\mathbf{1}_{|t|>T}\Delta_i\|_2^2 \leq \frac{1}{\pi^2(T-1)}.$$

There are six nonzero endpoint tail entries, bounded by $(2\pi)^{-1}(t-1)^{-1}$, and two diagonal interior entries on each tail. On the remaining finite rectangles, midpoint step approximations with mesh widths h_t, h_s have Hilbert–Schmidt error at most $\sqrt{T}(B_t h_t + B_s h_s)/2$, or $\sqrt{2T}(B_t h_t + B_s h_s)/2$ for the interior, where B_t, B_s bound the Frobenius norms of the two kernel derivatives. Such bounds follow from cutoff second derivatives and the canceled corner formulas. The squared rectangle and tail errors add.

Replacing only the errors in (7.13) by these finite-rank approximations gives, for example,

$$\|K_e - K_{e,n}\|_2 \leq (\|M_\lambda^{-1}\|^2(1 + |\lambda|) + \|M_\lambda^{-1}\|)\|\Delta_e - \Delta_{e,n}\|_2.$$

Every multiplier is explicit; matrix evaluation errors must also be controlled in a certificate. A contour may certify zeros only when it bounds a domain D with $\bar{D} \Subset \Omega$ and R_λ proved invertible throughout $\bar{D}$. If model inverse norms are at most b on a disk about real λ_0 , then $\|R_\lambda - R_{\lambda_0}\| \leq 2b^3|\lambda - \lambda_0|$; a disk with $2b^3|\lambda - \lambda_0| < (1 + |\lambda_0|)^{-2}$ suffices. On the contour, a verified bound $\|(I + K_{\lambda,n})^{-1}\| \|K_\lambda - K_{\lambda,n}\| < 1$ keeps the interpolation invertible. Half the finite determinant winding then counts physical eigenvalues with multiplicity. No such numerical enclosure is claimed by the construction alone.

7.3.3 Endpoint expansions and physical tails

Proof of Theorem 7.9. For $P_\lambda u = 0$, set $f = -\Delta_e u$. The displayed kernel implies, for every integer $j \geq 0$,

$$|f^{(j)}(t)| \leq C_j \|u\|_2 \quad (0 \leq t \leq 2), \quad |f^{(j)}(t)| \leq C_j \|u\|_2 t^{-1-j} \quad (t \geq 2).$$

On bounded rectangles, derivatives of divided differences are bounded by cutoff derivatives; corner cancellation and separated denominators handle the other entries. At infinity one differentiates rational tails. These are estimates on kernels applied to $u \in L^2$, so no regularity of u has been assumed. Consequently $\hat{f} = \mathcal{M}f$ is holomorphic in $|\operatorname{Im} \omega| < 1/2$ and rapidly decreases horizontally on every closed smaller strip: in logarithmic coordinates, $e^{x/2}f(e^x)$ and all its derivatives decay exponentially at both ends, allowing repeated integration by parts.

Since $M_\lambda E u = f$, Mellin inversion gives

$$Eu(t) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} t^{-1/2+i\omega} m_{\lambda,\theta}(\omega)^{-1} \hat{f}(\omega) d\omega.$$

The determinant in (7.12) has exactly two simple zeros in this strip, at $\pm i\beta$. Away from these poles the inverse has bounded horizontal limits. Shifting down to $\operatorname{Im} \omega = -a$ crosses only $-i\beta$, giving

$$v = -i\sqrt{2\pi} \operatorname{Res}_{\omega=-i\beta} (m_{\lambda,\theta}^{-1} \hat{f}).$$

The remaining integral is $O(t^{-1/2+a})$ and may be differentiated any fixed number of times. Since $\chi = 1$ near zero, this proves (7.16). A simple determinant zero has matrix rank three, so the residue lies in the stated one-dimensional kernel; it may vanish.

The interior error has smooth derivatives and rational tails. From $N_\lambda \mathcal{I}u = -\Delta_i u$ and (7.19), the Fourier transform of $\mathcal{I}u$ decays rapidly at high frequency and is locally L^2 at zero. Thus u is smooth away from its three endpoints.

A cutoff of $t^{\nu-1}$ at zero has Fourier asymptotic $\Gamma(\nu)e^{-i\operatorname{sgn}(T)\pi\nu/2}(2\pi|T|)^{-\nu} + O(|T|^{-1})$, by the regularized Gamma integral. A remainder $r(t) = O(t^{\mu-1})$, $r'(t) = O(t^{\mu-2})$, $0 < \mu < 1$, has Fourier transform $O(|T|^{-\mu})$: split at $|T|^{-1}$ and integrate the upper part by parts. Applying these formulas to the four endpoints with $\mu = 1/2 + a$ gives (7.17), with the phases $e^{2\pi iT}$ and $e^{-2\pi iT}$ from endpoints -1 and 1 .

After squaring, the nonoscillating part integrates to $L^{1-2\nu}/(2\nu-1)$; nonzero integer-frequency cross terms are $O(L^{-2\nu})$ by integration by parts. The remainder cross term is $O(L^{1-\nu-\mu}) = O(L^{-\beta-a})$, proving (7.18). At $\omega = -i\beta$, the kernel equations and $|q| = |r| = c = \sqrt{(1-\lambda^2)}/2$ give

$$|v_4| = |v_1|, \quad |v_2| = |v_3| = \frac{\sqrt{2}|\lambda|}{\sqrt{1-\lambda^2}}|v_1|.$$

Thus a nonzero residue makes both tail constants strictly positive, even if the zero-frequency amplitude cancels. $\square$

Proof of Corollary 7.10. Theorem 8.1, proved independently from the boundary tails and the qualitative gap limit, gives the stronger $\lambda_j = \alpha + L^{-\gamma(\alpha)}\mu_j + o(L^{-\gamma(\alpha)})$ for both boundaries and $\lambda = \alpha + o(L^{-\gamma(\alpha)})$ for one boundary. Since $L = m + h - l$, these imply the stated rate, including each centered parity sector. $\square$

The following two structural results clarify the endpoint analysis: the scalar Hilbert model has an exact spectral measure, whereas the physical kernel admits no smooth scalar second-order formal commutant.

7.4 Exact endpoint spectral measure

Let T be the finite Hilbert transform on $(-1, 1)$, $Tf(x) = \pi^{-1} \text{p.v.} \int_{-1}^1 f(y)/(x-y) dy$, and put $\Pi_T = (I - iT)/2$. Thus $T^* = -T$, $\|T\| \leq 1$, and Π_T is a positive contraction. Its classical spectral representation [28] gives the following exact endpoint law.

Proposition 7.11 (Endpoint spectral measure). *For every $u > 0$, $g_u(x) = [\pi(1+u-x)]^{-1}$ has normalized iT -spectral measure $\frac{1}{2}\mathbf{1}_{[-1,1]}(y)dy$. In particular,*

$$\|g_u\|^2 = \frac{2}{\pi^2 u(2+u)}, \quad \|T^k g_u\|^2 = \frac{\|g_u\|^2}{2k+1}, \quad \langle g_u, \Pi_T^a g_u \rangle = \frac{\|g_u\|^2}{a+1} \quad (k \geq 0, a \geq 0), \quad (7.22)$$

where k is an integer.

Proof. The unitary Cayley map $\mathcal{W}f(s) = \sqrt{2}f((1-s)/(1+s))/(1+s)$ gives $\mathcal{W}T\mathcal{W}^* = -H_+$, where $H_+F(s) = \pi^{-1} \text{p.v.} \int_0^\infty F(t)/(s-t)dt$. Indeed $x(s) - x(t) = 2(t-s)/[(1+s)(1+t)]$. Also $\mathcal{W}g_u = \sqrt{2}\phi_w/(2+u)$, with $w = u/(2+u)$ and $\phi_w(s) = [\pi(w+s)]^{-1}$. For $\mathfrak{M}F(\omega) = \int_0^\infty F(s)s^{-1/2+i\omega}ds$, Parseval has factor $(2\pi)^{-1}$, and

$$\mathfrak{M}\phi_w = w^{-1/2+i\omega} \operatorname{sech}(\pi\omega), \quad \mathfrak{M}H_+\mathfrak{M}^{-1} = i \tanh(\pi\omega). \quad (7.23)$$

These follow from the beta integral $\int_0^\infty r^{z-1}/(1+r)dr = \pi/\sin \pi z$ and $\text{p.v.} \int_0^\infty r^{z-1}/(1-r)dr = \pi \cot \pi z$, $0 < \Re z < 1$. For the latter, split at one and invert the upper part; the geometric series gives $\sum_{n \geq 0} [(n+z)^{-1} - (n+1-z)^{-1}] = \pi \cot \pi z$. Substituting $s = tr$ in the transformed Hilbert kernel gives the multiplier $-\cot(\pi/2 + i\pi\omega) = i \tanh(\pi\omega)$. Regulated compactly supported

calculations extend by boundedness to L^2 . Since $\|\phi_w\|^2 = 1/(\pi^2 w)$, its normalized density is $(\pi/2) \operatorname{sech}^2(\pi\omega) d\omega$; setting $y = \tanh(\pi\omega)$ gives $dy/2$. Integrating y^{2k} and $((1-y)/2)^a$ proves the claims. $\square$

Corollary 7.12 (Endpoint leakage). *For the zero extension of g_u and the whole-line Hilbert transform H ,*

$$\int_{-1}^1 |Hg_u|^2 = \frac{1}{3} \|g_u\|^2, \quad \int_{|x|>1} |Hg_u|^2 = \frac{2}{3} \|g_u\|^2. \quad (7.24)$$

Proof. Inside the interval $Hg_u = Tg_u$; apply (7.22) with $k = 1$ and the whole-line L^2 isometry. Partial fractions also give $Hg_u(x) = [\pi^2(1+u-x)]^{-1} \log[u|1+x|/((2+u)|1-x|)]$, with the removable value at $x = 1+u$ understood by continuity. $\square$

7.5 A scalar differential commutator obstruction

The scalar second-order differential route available for several classical Hankel kernels does not apply to the present physical half-line kernel. The following obstruction is an interior identity; it is independent of endpoint-domain choices.

Proposition 7.13 (No smooth scalar second-order formal commutant). *Let $I = (a, \infty)$, $p \in C^1(I)$, $q \in C(I)$, and $\mathcal{L}u = -(pu')' + qu$. If*

$$(\mathcal{L}_x - \mathcal{L}_y) \frac{\sin x - \sin y}{x - y} = 0 \quad (x, y \in I, x \neq y),$$

then $p = 0$ and q is constant. The coefficients may be complex.

Proof. Put $d = x - y$ and $\Delta = \sin x - \sin y$. Direct differentiation gives

$$\begin{aligned} d^3(\mathcal{L}_x - \mathcal{L}_y) \frac{\Delta}{d} &= d^2[p(x) \sin x + p(y) \sin y - p'(x) \cos x - p'(y) \cos y] \\ &\quad + 2d[p(x) \cos x - p(y) \cos y] \\ &\quad + \{d^2[q(x) - q(y)] + d[p'(x) + p'(y)] - 2[p(x) - p(y)]\} \Delta. \end{aligned}$$

Choose $s = \pi/2 + n\pi > a + \pi$ and set $x = s + t$, $y = s - t$, $|t| < s - a$. Then $\Delta = 0$. With $P_s(t) = p(s + t) + p(s - t)$, the displayed identity reduces to

$$\sin t P'_s(t) + (\cos t - \sin t/t) P_s(t) = 0, \quad \left(\frac{\sin t}{t} P_s(t) \right)' = 0.$$

The second equation extends through zero by continuity. Its constant is $2p(s)$, while evaluation at $t = \pi$ makes that constant zero. Thus p is odd about every such center s . For any $x > a$, choose s sufficiently large and compose reflection about s with reflection about $s + \pi$. Both reflected points stay in I , and the two sign changes give $p(x + 2\pi) = p(x)$. The same periodicity holds for p' .

Now take $y = x + 2\pi$ in the original differentiated identity. It gives $p(x) \sin x - p'(x) \cos x = 0$, or $(p(x) \cos x)' = 0$. The constant vanishes at a zero of cosine, so $p = 0$ everywhere by continuity. The remaining equation is $(q(x) - q(y))(\sin x - \sin y) = 0$. It makes q constant; when two sine values coincide, use a third point with a different sine value. $\square$

The unitary change $x = 2\pi X$ turns the physical kernel into a constant multiple of the displayed one. A commuting realization whose identity includes compactly supported smooth test functions must satisfy this formal interior equation, by integration by parts. The proposition excludes that smooth scalar second-order class. It does not exclude higher-order, matrix, nonlocal, or interior-singular differential models.

Corollary 7.14 (Polynomial expressions on a finite interval). *Let $\tau > 0$, $D = d/dx$, and p, q, s be real polynomials. If $L = DpD + q + \frac{i}{2}(sD + Ds)$ satisfies*

$$(L^x - \bar{L}^y)K_\tau(x, y) = 0 \quad (-1 < x, y < 1), \quad (7.25)$$

then $p = s = 0$ and q is constant.

Proof. The real part of the identity analytically continues to all real x, y : the divided difference has removable diagonal and is entire. Rescale by $2\pi\tau$ and apply Proposition 7.13 on a half-line, with the sign of p reversed, to obtain $p = 0$ and constant q . The imaginary part on the diagonal is $(s(x) \cos(2\pi\tau x))' = 0$. Its constant vanishes at the zeros of cosine, so $s = 0$. $\square$

8 Algebraic interaction of the boundaries

8.1 The leading interaction of the two boundaries

We evaluate the full interaction, including the complementary-space correction, which has the same order as the direct coupling. The reduction is analogous to Feshbach–Schur [16, Theorem 1.2 and equations (1.15)–(1.16)]; the physical sine-Loewner modes determine its explicit coefficient.

Fix a real $\alpha \in (-g, 0) \cup (0, g)$, and write

$$a = \sqrt{1 - 2\alpha^2}, \quad \beta = \pi^{-1} \arctan a, \quad \nu = \frac{1}{2} + \beta, \quad \gamma = 2\beta, \quad c_\alpha = -\frac{\pi a(1 + a^2)}{\alpha}. \quad (8.1)$$

For a normalized left boundary eigenfunction ψ_L on (l, ∞) , define its physical tail amplitude by

$$\eta_L = \lim_{T \rightarrow \infty} T^\nu \int_T^{T+1} \psi_L(l+x) dx = \frac{\Gamma(\nu)}{(2\pi)^\nu} \left(e^{-i\pi\nu/2} v_1 + e^{i\pi\nu/2} v_4 \right), \quad (8.2)$$

where v is the endpoint vector in Theorem 7.9, scaled so that its physical eigenfunction has norm one on the half-line. The limit follows from (7.17): integration over one unit interval removes the two nonzero leading carrier frequencies. For a normalized right boundary eigenfunction ψ_R on $(-\infty, h)$, define η_R by the same limit with $\psi_R(h-x)$ in the integrand. The amplitudes may vanish; their phase follows the eigenfunction's phase. The same amplitude is determined in modulus by the normalized physical tail mass:

$$\lim_{T \rightarrow \infty} T^\gamma \int_T^\infty |\psi_L(l+x)|^2 dx = \frac{|\eta_L|^2}{\alpha^2 \gamma}.$$

Indeed the tail direction in the proof has squared norm α^{-2} ; the nonmatching carriers integrate to a lower order. The right analogue holds with $\psi_R(h-x)$.

Theorem 8.1 (Closed leading boundary interaction). *Fix l, h , and suppose that α is an isolated eigenvalue of one or both half-line operators in Theorem 7.2. Their individual eigenvalues are simple by Theorem 7.5. Set*

$$T_m = P_{(l, m+h)} \mathcal{G} P_{(l, m+h)}, \quad L = m + h - l, \quad m \rightarrow \infty \quad (m \in \mathbb{Z}).$$

If α belongs to both boundary spectra, let $\mu_1 \leq \mu_2$ be the eigenvalues of the Hermitian matrix

$$\mathbf{M}_\alpha = c_\alpha \begin{pmatrix} 0 & \overline{\eta_L} \eta_R \\ \overline{\eta_R} \eta_L & 0 \end{pmatrix}. \quad (8.3)$$

The two finite-interval eigenvalues converging to α , in increasing order within the cluster, satisfy

$$\lambda_j(T_m) = \alpha + L^{-\gamma} \mu_j + o(L^{-\gamma}), \quad j = 1, 2. \quad (8.4)$$

If α belongs to only one boundary spectrum, its single eigenvalue satisfies $\lambda(T_m) = \alpha + o(L^{-\gamma})$. All parameters are fixed; no uniformity as α approaches zero or a band edge is asserted.

The two-boundary case implies $l + h \in \mathbb{Z}$. Indeed reflection identifies the right operator at h with the left operator at $-h$, and Theorem 7.5 assigns a fixed nonzero gap level to a unique boundary phase modulo one. Conversely, this congruence makes the two half-line operators unitarily equivalent. Centered intervals satisfy the congruence automatically.

Corollary 8.2 (Centered parity splitting). *Let α be an isolated eigenvalue of $L_{-r}\mathcal{G}L_{-r}$, with normalized eigenfunction of amplitude η in (8.2). For the centered operator $\mathcal{K}_{n+r}$, put $L = 2(n+r)$. Its unique even and odd eigenvalues converging to α satisfy*

$$\begin{aligned}\lambda_{\text{even}} &= \alpha + c_\alpha |\eta|^2 L^{-\gamma} + o(L^{-\gamma}), \\ \lambda_{\text{odd}} &= \alpha - c_\alpha |\eta|^2 L^{-\gamma} + o(L^{-\gamma}).\end{aligned}\tag{8.5}$$

Their midpoint is $\alpha + o(L^{-\gamma})$. If $\eta \neq 0$, their splitting has the nonzero leading magnitude $2|c_\alpha||\eta|^2 L^{-\gamma}$, and eventually $|\lambda_{\text{even}}| < |\lambda_{\text{odd}}|$. If $\eta = 0$, both shifts are $o(L^{-\gamma})$; strict ordering is not inferred. In fact both shifts are $O(L^{-1+\delta})$ for every fixed $\delta > 0$ in this case.

Section 8.2 proves the full correction and evaluates its scalar integral. Theorem 9.1 certifies $\eta \neq 0$ for the phase-zero mode.

8.2 Proof of the leading boundary interaction

Throughout, α is real and lies in an open central gap, and β, ν, γ, a are as in (8.1). Put $\epsilon = L^{-\gamma}$. Inner products are conjugate linear in the first variable.

8.2.1 The physical tail directions

Put $t = \pi\beta$, $q = (1 - ia)/2$, $\bar{q} = (1 + ia)/2$, and $c = (2 \cos t)^{-1}$. These are the endpoint multiplier values at $\omega = -i\beta$, so that $q = ce^{-it}$, $\bar{q} = ce^{it}$, and $q^2 + \bar{q}^2 = \alpha^2$. Undoing D_θ in (7.12), the kernel vector with first component one is

$$(1, (\alpha/c)e^{it}, -i(\alpha/c)e^{2it}, -ie^{3it})^T.$$

The second and third rows give $v_2 = \alpha v_1/q$ and $v_3 = \alpha v_4/\bar{q}$; the first gives $v_4 = i(q^2 - \alpha^2)/(cq) = -i\bar{q}^2/(cq)$, and the fourth then holds. If the first component is zero all four are zero.

Use global physical carriers in the order $(-1, 0, 1)$, and put

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}, \quad v_+ = \begin{pmatrix} \bar{q}/\alpha \\ 1 \\ q/\alpha \end{pmatrix}, \quad v_- = \overline{v_+}.$$

Let $A^\pm$ be the tail vectors of the full continuation of the normalized left mode, and $B^\pm$ those of the right mode, with the superscript denoting the direction $X \rightarrow \pm\infty$. Thus, for example,

$$\tilde{\psi}_L(X) = |X - l|^{-\nu} \sum_{p=-1}^1 A_p^\pm e^{2\pi i p X} + O(|X - l|^{-\mu}), \quad \nu < \mu < 1.$$

Substitution in (7.17) gives

$$A^+ = \eta_L v_+, \quad A^- = -i(c/\alpha)SA^+, \quad B^- = \eta_R v_-, \quad B^+ = i(c/\alpha)SB^-. \tag{8.6}$$

Here the factors of D_θ cancel the factors from replacing $e^{2\pi i p(X-l)}$ by $e^{2\pi i p X}$. To check the ratios explicitly, put $\varphi = \pi/4 + t/2$. The positive-tail vector before its common Gamma factor is $(e^{i\varphi}v_3, e^{-i\varphi}v_1 + e^{i\varphi}v_4, e^{-i\varphi}v_2)$; its middle component is $2 \cos(2t)e^{-i\pi/4+3it/2}$. Division by it gives $(\bar{q}/\alpha, 1, q/\alpha)$. Reversing the Gamma phases gives the second formula in (8.6); reflection gives the two right formulas. Since $\cos(2t) > 0$, η_L vanishes precisely when the endpoint vector vanishes, and likewise on the right.

8.2.2 The three-channel limit and residual profiles

Let $H_{\mathbb{R}}$ have kernel $[\pi(s-t)]^{-1}$, set $P_{\pm} = (I \pm iH_{\mathbb{R}})/2$, and define

$$\mathbb{G} = \begin{pmatrix} 0 & P_+ & 0 \\ P_+ & 0 & P_- \\ 0 & P_- & 0 \end{pmatrix}, \quad (\mathcal{U}_L f)(X) = L^{-1/2} \sum_{p=-1}^1 e^{2\pi i p X} f_p((X-l)/L).$$

These maps have norm at most $\sqrt{3}$ and, for fixed $f, g \in L^2(\mathbb{R})^3$, satisfy

$$\langle \mathcal{U}_L f, \mathcal{U}_L g \rangle \rightarrow \langle f, g \rangle, \quad \|\mathcal{G}\mathcal{U}_L f - \mathcal{U}_L \mathbb{G}f\|_2 \rightarrow 0. \quad (8.7)$$

For the first limit, off-diagonal carrier products are Fourier transforms of L^1 functions at nonzero frequencies tending to infinity. For the second, write $M_p f(X) = e^{2\pi i p X} f(X)$ and $D_L f(X) = L^{-1/2} f((X-l)/L)$. The exact commutator identity is

$$\mathcal{G} = (2i)^{-1} (M_1 H_{\mathbb{R}} - H_{\mathbb{R}} M_1 - M_{-1} H_{\mathbb{R}} + H_{\mathbb{R}} M_{-1}).$$

The transform $H_{pL} = M_{-pL} H_{\mathbb{R}} M_{pL}$ on the scaled variable has multiplier $-i \operatorname{sgn}(\omega + 2\pi pL)$. For $p \neq 0$ it converges strongly to $-i \operatorname{sgn} p$. The output from the carrier p is exactly

$$\frac{1}{2i} M_{p+1} D_L (H_{pL} - H_{(p+1)L}) f - \frac{1}{2i} M_{p-1} D_L (H_{pL} - H_{(p-1)L}) f.$$

The ± 2 outputs tend to zero. The remaining entries are P_+ between $-1, 0$ and P_- between $0, 1$, proving the claim.

Write $p = \mathbf{1}_{(0,1)}$ on the envelope space. Interval multiplication intertwines exactly with p , so T_m has the strong envelope limit

$$C = p\mathbb{G}p = \begin{pmatrix} 0 & Q & 0 \\ Q & 0 & I - Q \\ 0 & I - Q & 0 \end{pmatrix}, \quad Q = \frac{1}{2}(I + iH_{(0,1)}). \quad (8.8)$$

The commuting operators $Q, I - Q$ satisfy $Q^2 + (I - Q)^2 \geq I/2$; hence $\sigma(C) \subset \{0\} \cup [-1, -g] \cup [g, 1]$.

Translate the right mode by the integer m , and let Φ_m contain the restrictions of the one or two chosen boundary modes to $(l, m+h)$. Integer translation preserves all global carrier phases. The omitted-tail envelope columns are

$$Z_L(s) = \mathbf{1}_{s>1} s^{-\nu} A^+, \quad Z_R(s) = \mathbf{1}_{s<0} (1-s)^{-\nu} B^-.$$

These exterior profiles belong to L^2 , unlike the singular interior powers. The scaled squared tail error is $O(L^{\gamma+1-2\mu}) = o(1)$. Since the exact residual is minus $P_{(l,m+h)} \mathcal{G}$ applied to the omitted tail, (8.7) gives

$$\epsilon^{-1/2} (T_m - \alpha) \Phi_m - \mathcal{U}_L F \longrightarrow 0, \quad F(s) = \left(\frac{i}{2} \ell_{\nu}(s) S A^+, -\frac{i}{2} \ell_{\nu}(1-s) S B^- \right), \quad (8.9)$$

omitting an absent boundary column, where

$$\ell_{\nu}(s) = \frac{1}{\pi} \int_1^{\infty} \frac{t^{-\nu}}{t-s} dt = \frac{1}{\pi} \int_0^1 \frac{x^{\nu-1}}{1-sx} dx.$$

Its endpoint singularity is logarithmic, so $F \in L^2(0,1)^3$. The left exterior Hilbert transform is $-\ell_{\nu}(s)$ and the right one is $\ell_{\nu}(1-s)$, which fixes both signs in (8.9).

8.2.3 The direct interaction

Let $P = P_{(l,m+h)}$, $P^\perp = I - P$, and let z_i, z_j denote the half-line modes, extended by zero on their complementary half-lines. Their full continuations satisfy $\mathcal{G}z_i = \alpha \tilde{z}_i$. Self-adjointness gives the exact identity

$$\langle Pz_i, (T_m - \alpha)Pz_j \rangle = -\alpha \langle \tilde{z}_i, P^\perp z_j \rangle + \langle P^\perp z_i, \mathcal{G}P^\perp z_j \rangle. \quad (8.10)$$

The tails imply $\Phi_m^* \Phi_m = I + O(\epsilon)$: omitted squared tails are $O(\epsilon)$, and the cross overlap is bounded by $O(L^{1-2\nu}) \int_0^1 s^{-\nu}(1-s)^{-\nu} ds$ plus $O(L^{-\nu})$ endpoint terms. For (8.10), matching carriers converge by dominated convergence and the others vanish by Riemann–Lebesgue. The cross profiles $s^{-\nu}(1-s)^{-\nu}$ on $(0, 1)$ and $|s|^{-\nu}(1-s)^{-\nu}$ on $(-\infty, 0)$ are integrable; the better tail exponent controls the remainders.

The diagonal limit of $\epsilon^{-1} \Phi_m^* (T_m - \alpha) \Phi_m$ is $(A^+)^* (A/2 - \alpha) A^+ / \gamma$, and similarly on the right. It is zero, because

$$(A/2 + (ia/2)S)v_+ = \alpha v_+, \quad v_+^* S v_+ = 0,$$

and reflection gives the same right identity. The first term follows by substitution; its imaginary quadratic part gives the second. For the cross entry, put $b_\nu = B(1 - \nu, 2\nu - 1)$ and

$$\kappa_\nu = \frac{1}{\pi} \int_1^\infty \int_1^\infty \frac{x^{-\nu} y^{-\nu}}{x + y - 1} dx dy.$$

The first term in (8.10) has limit $-\alpha b_\nu (A^-)^* B^-$; the second has limit $i\kappa_\nu (A^+)^* S B^- / 2$. Therefore their sum is

$$-\frac{i}{2} j_\nu (A^+)^* S B^- = -\frac{a}{\alpha} j_\nu \bar{\eta}_L \eta_R, \quad j_\nu = \frac{b_\nu}{\sin \pi \nu} - \kappa_\nu = \int_0^1 s^{-\nu} \ell_\nu(1-s) ds. \quad (8.11)$$

For the last identity, integrate $s^{-\nu}/(s+u)$ over all $s > 0$, obtaining $\pi u^{-\nu}/\sin \pi \nu$, then integrate against $(1+u)^{-\nu}/\pi$. The portion $s > 1$ is κ_ν and the portion $0 < s < 1$ is j_ν . All integrands are positive.

8.2.4 The full complementary correction

Let Π_m be the spectral projection of the cluster converging to α . Theorem 7.2 supplies its exact rank and radii $0 < \delta_1 < \delta_2$ such that the cluster lies in $|x - \alpha| < \delta_1$ and no spectrum lies in $\delta_1 \leq |x - \alpha| \leq \delta_2$, for all sufficiently large m . Choose $[\alpha - \delta_2, \alpha + \delta_2]$ inside the same open central gap. Let b be continuous and real on $[-1, 1]$, zero for $|x - \alpha| \leq \delta_1$, equal to $(x - \alpha)^{-1}$ for $|x - \alpha| \geq \delta_2$, and interpolated across the empty annulus. Consequently

$$b(T_m) = (I - \Pi_m)(T_m - \alpha)^{-1}, \quad b(C) = (C - \alpha)^{-1},$$

where the first inverse is restricted to the spectral complement. Telescoping the intertwining defect proves polynomial transfer. Uniform polynomial approximation of b and the common norm bound on $\mathcal{U}_L$ then give

$$\|b(T_m)\mathcal{U}_L f - \mathcal{U}_L b(C)f\| \rightarrow 0 \quad (f \in L^2((0, 1))^3).$$

Write $V_m = (T_m - \alpha)\Phi_m$. The spectral theorem and the fixed complementary gap give $\|(I - \Pi_m)\Phi_m\| = O(\epsilon^{1/2})$. In conjunction with (8.9), the preceding transfer proves

$$\epsilon^{-1} V_m^* b(T_m) V_m \rightarrow F^* (C - \alpha)^{-1} F. \quad (8.12)$$

The exact identity

$$\Phi_m^* \Pi_m (T_m - \alpha) \Pi_m \Phi_m = \Phi_m^* (T_m - \alpha) \Phi_m - V_m^* b(T_m) V_m \quad (8.13)$$

keeps the whole complementary contribution. The Gram matrix of $\Pi_m \Phi_m$ is $I + O(\epsilon)$, so these vectors span the actual cluster and orthonormalization does not change its leading matrix.

For its explicit calculation, the unitary logit transform $s = e^x/(1+e^x)$, $f(s) \mapsto \sqrt{s(1-s)}f(s)$ sends $H_{(0,1)}$ to the convolution kernel $[2\pi \sinh((x-y)/2)]^{-1}$. Its angular Fourier multiplier is $-i \tanh \pi\omega$. Put $x_\omega = \tanh \pi\omega$, $q_\omega = (1+x_\omega)/2$, $d_\omega = (1+x_\omega^2)/2$. The symbol of C and its inverse at α are

$$c_\omega = \begin{pmatrix} 0 & q_\omega & 0 \\ q_\omega & 0 & 1-q_\omega \\ 0 & 1-q_\omega & 0 \end{pmatrix}, \quad R_\alpha(\omega) = -I/\alpha + \frac{\alpha c_\omega + c_\omega^2}{\alpha(d_\omega - \alpha^2)}.$$

The inverse identity follows from $c_\omega^3 = d_\omega c_\omega$; its denominator is positive because $d_\omega \geq 1/2 > \alpha^2$. The transformed scalar residual is

$$L_\nu(\omega) = \frac{B(\nu, 1/2 + i\omega)}{\sqrt{2\pi} \cosh \pi\omega}. \quad (8.14)$$

To verify the normalization, write $A = 1/2 - i\omega$, $B = 1/2 + i\omega$. The defining double integral is

$$\frac{1}{\pi\sqrt{2\pi}} \int_0^1 x^{\nu-1} \int_0^1 \frac{s^{A-1}(1-s)^{B-1}}{1-xs} ds dx = \frac{B(A, B)B(\nu, B)}{\pi\sqrt{2\pi}}.$$

Here $A + B = 1$, so the inner hypergeometric factor is $(1-x)^{-A}$, and $B(A, B) = \pi/\cosh \pi\omega$. Absolute Fubini follows from the same positive integrals with $\operatorname{Re} A = \operatorname{Re} B = 1/2$. Reflection has transform $L_\nu(-\omega)$.

Set $u_+ = Sv_+ = (1, ia/\alpha, -1)^T$ and $u_- = \overline{u_+}$. Direct multiplication gives

$$\begin{aligned} u_+^* R_\alpha u_+ &= u_-^* R_\alpha u_- = 0, \\ u_+^* R_\alpha u_- &= -\frac{4a}{\alpha(a - ix_\omega)}. \end{aligned}$$

For example the first contraction is $-2/\alpha + (x_\omega^2 + a^2)/[\alpha(d_\omega - \alpha^2)]$, which is zero. In the second contraction the real numerator is $-4a^2$ and the imaginary numerator $-4iax_\omega$, over $\alpha(x_\omega^2 + a^2)$. Thus the complementary diagonal vanishes pointwise. Combining the cross contraction, the two signs in (8.9), and (8.11), the full cross coefficient is

$$-\frac{a}{\alpha} \overline{\eta_L} \eta_R [j_\nu + I_\nu], \quad I_\nu = \int_{\mathbb{R}} \frac{L_\nu(-\omega)^2}{a - i \tanh \pi\omega} d\omega. \quad (8.15)$$

Exponential decay of L_ν and $a > 0$ give absolute convergence.

8.2.5 Evaluation of the scalar and conclusion

On $(0, 1)$ let $H = H_{(0,1)}$, let $\mathcal{R}f(s) = f(1-s)$, and put

$$f(s) = s^{-\nu}, \quad h(s) = s^{-\nu}(1-s)^{\nu-1}.$$

Their difference is in L^2 : it is $O(s^{1-\nu})$ at zero and $O((1-s)^{\nu-1})$ at one. The elementary principal-value identity

$$\frac{1}{\pi} \operatorname{pv} \int_0^\infty \frac{t^{-\nu}}{s-t} dt = -\cot(\pi\nu) s^{-\nu} = a s^{-\nu}$$

gives $Hf = af + \ell_\nu$. The substitution $x = s/(1-s)$, $y = t/(1-t)$ gives $Hh = ah$ by the same identity. Subtracting the principal values, which agree almost everywhere with the bounded L^2 transform of their difference, gives $(H-a)(f-h) = \ell_\nu$. No assertion that f or h separately

belongs to L^2 is used. Since $H^* = -H$ and $a > 0$, the operators $H \pm a$ are invertible. Plancherel and (8.14) therefore give

$$I_\nu = \langle \ell_\nu, (a + H)^{-1} \mathcal{R} \ell_\nu \rangle = -\langle f - h, \mathcal{R} \ell_\nu \rangle.$$

The last sign uses $((a + H)^{-1})^* = -(H - a)^{-1}$. Both separate pairings with f, h are absolutely integrable, so

$$j_\nu + I_\nu = \int_0^1 s^{-\nu} (1-s)^{\nu-1} \ell_\nu(1-s) ds = \frac{B(\nu, 1-\nu)^2}{\pi} = \frac{\pi}{\sin^2 \pi \nu} = \pi(1+a^2).$$

For the beta evaluation, substitute the integral for ℓ_ν and use Tonelli. With $y = 1 - s$ the inner integral is $B(\nu, 1-\nu)(1-x)^{-\nu}$; the remaining integral is the same beta function. This proves (8.3).

Finally, divide (8.13) by ϵ and orthonormalize $\Pi_m \Phi_m$. The resulting finite Hermitian matrix tends to $\mathbf{M}_\alpha$, proving (8.4) including coincident or zero leading coefficients. With only one boundary column its limit is zero. For $l = -r$, $h = r$, choose the right mode by reflection. Reflection about $m/2$ exchanges the two columns exactly, and $\eta_R = \eta_L$. The symmetric column has even parity and the antisymmetric column odd parity, so the two leading shifts are $c_\alpha |\eta|^2$ and $-c_\alpha |\eta|^2$. The centered family has $m = 2n$ and physical length $L = 2(n+r)$. The sign of c_α is opposite that of α , proving the strict absolute-value ordering when $\eta \neq 0$. If $\eta = 0$, the endpoint vector is zero and Theorem 7.9 gives every tail exponent $\mu < 1$. Choose $1/2 < \mu < 1$ and set $e = L^{1-2\mu}$. The two elementary bounds

$$\int_0^L (1+x)^{-\mu} (1+L-x)^{-\mu} dx = O(e), \quad \int_0^\infty (1+x)^{-\mu} (1+L+x)^{-\mu} dx = O(e)$$

follow by splitting at $L/2$ and L . Applied to the two-sided continuations, they give $\Phi_m^* \Phi_m = I + O(e)$ and $\|V_m\| = O(\sqrt{e})$. The exact continuation identity (8.10) gives $\Phi_m^*(T_m - \alpha)\Phi_m = O(e)$; same-boundary terms are products of omitted tails, and the second integral controls opposite-boundary continuation overlaps. The uniformly bounded $b(T_m)$ and (8.13) therefore give a projected Rayleigh matrix $O(e)$ and Gram matrix $I + O(e)$. Orthonormalizing proves $O(L^{1-2\mu})$, hence $O(L^{-1+\delta})$ for every $\delta > 0$.

9 A certified mode with nonzero splitting

9.1 A certified isolated boundary mode

A rational trial function, a full-operator negative-window exclusion, and a bounded residue functional give the following joint certificate.

Theorem 9.1 (A unique boundary mode with nonzero tail). *The phase-zero physical operator B_0 has no spectrum in $[-953/2000, -713/2000] = [-0.4765, -0.3565]$ and exactly one eigenvalue α_0 in $[713/2000, 953/2000]$. It is simple and lies in*

$$(81/200, 107/250) = (0.405, 0.428).$$

For every normalized physical eigenfunction at α_0 , the amplitude in (8.2) satisfies $|\eta| > 1/100$. Consequently, along $\mathcal{K}_n$ with $n \in \mathbb{N}$ and $L = 2n \rightarrow \infty$, the even eigenvalue is eventually smaller than the odd one, with leading difference $2|c_{\alpha_0}| |\eta|^2 L^{-\gamma(\alpha_0)} + o(L^{-\gamma(\alpha_0)})$.

Sections 9.2 and 9.3 prove the three assertions in order: a full-space residual gives existence, a global parametrix bound and signed alternation give uniqueness, and physical spectral projection transfers a certified residue value to the eigenfunction. All rounding errors, omitted cells and spectral complements are included. The nonvanishing statement concerns this mode; it imposes no condition on other phases or boundary levels.

9.2 Proof of the certified root enclosure

9.2.1 A full-space residual and the existence witness

Let $u \in L^2(-1, 1)$, let $\tilde{u}$ be its extension by zero, and let $\mathcal{H}$ be the full-line Hilbert transform with kernel $[\pi(x - y)]^{-1}$. Put

$$q_u = \langle u, Q_0 u \rangle, \quad c_u = \operatorname{Re} \langle Q_0 u, J u \rangle.$$

Choose any finite orthonormal family e_j supported outside $(-1, 1)$ and set $E_u = \sum_j |\langle e_j, \mathcal{H} \tilde{u} \rangle|^2$. For real a , Hilbert-transform isometry and Bessel's inequality give

$$\|(Q_0 - aJ)u\|^2 \leq q_u - \frac{1}{4}E_u - 2ac_u + a^2\|u\|^2 =: U_u(a). \quad (9.1)$$

Indeed $Q_0 = (I + iH)/2$, $H^* = -H$, and $\|\mathcal{H} \tilde{u}\|^2 = \|u\|^2$, so

$$\|Q_0 u\|^2 = q_u - \frac{1}{4}\|\mathbf{1}_{\mathbb{R} \setminus (-1, 1)} \mathcal{H} \tilde{u}\|^2.$$

For $q_u > 0$ let $f = V_0 u$, where V_0 is the restriction of inverse Fourier transformation to $(0, \infty)$. Then $\|f\|^2 = q_u$ and

$$(B_0 - a)f = V_0 J(Q_0 - aJ)u, \quad \operatorname{dist}(a, \operatorname{spec} B_0) \leq \sqrt{U_u(a)/q_u}.$$

Here the norm of $V_0 J$ is at most one. Thus an interval containing $[a - \sqrt{U_u(a)/q_u}, a + \sqrt{U_u(a)/q_u}]$ and lying in an open central gap contains a discrete physical eigenvalue.

Take u constant on cells in a J -invariant partition and the e_j to be normalized indicators of disjoint exterior cells. The Hilbert cell formula (9.9) evaluates q_u, c_u, E_u exactly, including principal values. Omitting the other exterior modes preserves the upper-bound direction; no tail quadrature is required.

Existence in Theorem 9.1. The exact rational trial data in `anc/boundary_gap_witness.json` specify a partition of $(0, 1)$, its exact translate by -1 , complex coefficients on the resulting 512 normalized indicator functions, and disjoint exterior indicator functions. Write u_0 for this trial. Its common positive parameter is

$$a_0 = \frac{3750237663924847}{9007199254740992}. \quad (9.2)$$

The independent checker `anc/verify_boundary_gap_certificate.py` evaluates the displayed double-cell logarithmic integrals with outward-rounded Arb ball arithmetic [25]. At both 192 and 256 bits it verifies the strict rational bounds

$$0.41635 < a_0 < 0.41637, \quad q_{u_0} > 0.32783, \quad 0 < U_{u_0}(a_0) < 0.000038641,$$

and directly verifies $\sqrt{U_{u_0}(a_0)/q_{u_0}} < 0.01086$. All inputs are exact rationals and all rounding errors are enclosed; no eigenvalue solver or quadrature estimate enters this check. Thus the residual interval lies strictly inside $(0.405, 0.428)$, which is contained in $(0, g)$. Equation (9.1) and the spectral theorem force a spectral point there. The reference gap makes it a discrete eigenvalue, and Theorem 7.5 makes it simple. $\square$

9.2.2 The negative-window model

We prove the exclusion at a single frequency parameter and then use a resolvent bound. Put

$$a = -\frac{833}{2000}, \quad P_a = Q_0 - aJ, \quad c_0 = \frac{\pi}{2}, \quad \chi(t) = \begin{cases} \cos(c_0 t), & 0 \leq t \leq 1, \\ 0, & t > 1. \end{cases}$$

Identify $L^2(-1, 1)$ with $L^2((0, 1); \mathbb{C}^2)$ by $f_1(t) = u(t)$, $f_2(t) = u(t-1)$. On four copies of $L^2(0, \infty)$ retain the endpoint model $Q_e, J_e, M_a = Q_e - aJ_e$ of Section 7.3, at phase zero. Define

$$Ef(t) = \chi(t)(f_1(t), f_2(t), f_1(1-t), f_2(1-t))^T$$

with zero extension outside $(0, 1)$. Since $\chi(t)^2 + \chi(1-t)^2 = 1$, one has $E^*E = I$. In particular no interior model is needed for this partition.

9.2.3 The exact model and the residual criterion

The intertwining error is $\Delta = EQ_0 - Q_eE$. For later use set

$$D_0(t, s) = \frac{\chi(t) - \chi(s)}{t - s}, \quad D_1(t, s) = \frac{\chi(t) - \chi(s)}{t + s}, \quad D_2(t, s) = \begin{cases} \frac{\chi(t)}{1 - t + s}, & t < 1, \\ 0, & t \geq 1. \end{cases}$$

For $t > 0$, $0 < s < 1$, its complete kernel is

$$\Delta(t, s) = \frac{i}{2\pi} \begin{pmatrix} D_0(t, s) & D_1(t, 1-s) \\ -D_2(t, s) & D_0(t, s) \\ -D_0(t, 1-s) & D_2(t, 1-s) \\ -D_1(t, s) & -D_0(t, 1-s) \end{pmatrix}. \quad (9.3)$$

Divided differences take their continuous diagonal values. The unmatched corner quotients are bounded because $\chi(t) = \sin(c_0(1-t))$ near one. The entries are bounded on bounded rectangles and are $O(t^{-1})$ at infinity. Thus $\Delta \in \mathfrak{S}_2$. The Lipschitz cutoff suffices for this assertion and the intertwining identity; no differentiated endpoint expansion is used here.

At the single parameter a the model satisfies

$$\|M_a^{-1}\| \leq 5. \quad (9.4)$$

Here is an elementary verification of the constant. Write $p = a^2$, $d = qr \in [0, 1/4]$, and let m be the matrix in (7.12). For $w = z(z-1)$, direct expansion gives

$$\det(m - zI) = w^2 - 2pw + p^2 - p + d(w + 2p).$$

Since $4/25 < p < 19/100$, this determinant at $z = 1/5$ is at most $p^2 - 9p/50 - 9/625 \leq -71/10000$, and at $z = -1/5$ it is at most $p^2 - 49p/50 + 147/1250 \leq -31/10000$. At $q = 0$ the eigenvalues are $-|a|, |a|, 1 - |a|, 1 + |a|$. Exactly one is below each of $-1/5$ and $1/5$. Continuity as q ranges over $[0, 1]$ shows that no eigenvalue enters $[-1/5, 1/5]$, proving (9.4).

Let C_∞ denote the Carleman operator with kernel $[\pi(t+s)]^{-1}$. The $(1, 4)$ block of Q_e is an orthogonal projection: its multiplier has diagonal entries q, r , off-diagonal entries $ic, -ic$, and $q + r = 1$, $c^2 = qr$. The two singleton channels satisfy $q^2 = q - c^2$, $r^2 = r - c^2$. Since $2c$ is the multiplier of C_∞ , one obtains the exact identity

$$Q_e^2 = Q_e - \frac{1}{4} \text{diag}(0, C_\infty^2, C_\infty^2, 0). \quad (9.5)$$

Choose any finite step-function projection Π on $L^2(0, \infty)$, and any source step-function projection Π_s on $L^2(0, 1)$. Let $W, D : L^2((0, 1); \mathbb{C}^2) \rightarrow L^2((0, \infty); \mathbb{C}^4)$ vanish on $(\Pi_s \otimes I_2)^\perp$ and have their outputs in $\Pi \otimes \mathbb{C}^4$. Put $Q_\Pi = (\Pi \otimes I_4)Q_e(\Pi \otimes I_4)$ and $C_\Pi = \Pi C_\infty \Pi$. All inner products below are Hilbert–Schmidt inner products. Define

$$\begin{aligned} \mathcal{R}(W, D)^2 &= \langle W, Q_\Pi W \rangle - \frac{1}{4}(\|C_\Pi W_2\|_2^2 + \|C_\Pi W_3\|_2^2) - 2a \text{Re} \langle Q_\Pi W, J_e W \rangle + a^2 \|W\|_2^2 \\ &\quad - 2 \text{Re} \langle (Q_\Pi - aJ_e)W, D \rangle + \|D\|_2^2. \end{aligned} \quad (9.6)$$

Then

$$\|M_a W - D\|_2^2 \leq \mathcal{R}(W, D)^2. \quad (9.7)$$

Indeed (9.5) gives the first line of (9.6) with the full Carleman norms. Their finite orthogonal projections have smaller norm and enter with a minus sign. Every other term is exact after compression, since J_e preserves the step space and both W and D take values there. Expanding the squared residual proves (9.7).

The intertwining identity now yields

$$S_a P_a = I + K_a, \quad S_a = E^* M_a^{-1} E, \quad K_a = E^* M_a^{-1} \Delta, \quad \|S_a\| \leq 5.$$

For every such trial W, D , the full correction therefore obeys

$$\|K_a\| \leq \|E^* W\|_2 + 5(\mathcal{R}(W, D) + \|\Delta - D\|_2). \quad (9.8)$$

This follows by subtracting $E^* W$ and using $M_a^{-1} \Delta - W = M_a^{-1}(\Delta - D + D - M_a W)$. The source complement is included in $\Delta - D$.

9.2.4 Exact finite quantities

All finite matrices use normalized cell indicators. For cells (a_i, b_i) of length h_i , write $F(x) = x \log |x|$, $F(0) = 0$. The Hilbert and Carleman matrices are

$$\begin{aligned} H_{ij} &= \frac{F(b_i - a_j) - F(a_i - a_j) - F(b_i - b_j) + F(a_i - b_j)}{\pi \sqrt{h_i h_j}}, \\ C_{ij} &= \frac{F(b_i + b_j) - F(a_i + b_j) - F(b_i + a_j) + F(a_i + a_j)}{\pi \sqrt{h_i h_j}}. \end{aligned} \quad (9.9)$$

Since $F''(x) = 1/x$ off zero, twice integrating gives these formulas; the Hilbert diagonal is zero in principal value. The Hilbert formula also applies to the interior and exterior cells in the existence test. With the channel structure of Q_e and exchanges $(1, 2), (3, 4)$, these matrices evaluate (9.6) exactly.

The projected trial norm is equally explicit. Suppose the source partition $I_i = (s_i, s_{i+1})$ is reflection symmetric and is also the initial part of the model partition on $(0, 1)$. Let $v_{j,i}$ be the constant row vector of W_j on I_i , and let i^* index its reflected cell. These row vectors are obtained by dividing normalized model-cell coefficients by $\sqrt{|I_i|}$. Then

$$\|E^* W\|_2^2 = \sum_i \sum_{j=1}^2 \left[A_i \|v_{j,i}\|^2 + B_i \|v_{j+2,i^*}\|^2 + 2C_i \operatorname{Re} \langle v_{j,i}, v_{j+2,i^*} \rangle \right], \quad (9.10)$$

where, with $h_i = s_{i+1} - s_i$,

$$A_i = \frac{h_i}{2} + \frac{\sin(\pi s_{i+1}) - \sin(\pi s_i)}{2\pi}, \quad B_i = h_i - A_i, \quad C_i = \frac{\cos(\pi s_i) - \cos(\pi s_{i+1})}{2\pi}.$$

They are the integrals of χ^2 , $\chi(1-t)^2$ and $\chi(t)\chi(1-t)$ on the cell, respectively.

9.2.5 A global midpoint error bound

Take D to be the exact midpoint step kernel of (9.3) on $(0, T) \times (0, 1)$, zero for $t > T$. The output partition includes one as an endpoint; the source partition is exactly reflection symmetric. For a scalar kernel with derivative bounds A, B on a rectangle of widths h, k , its squared midpoint error is at most

$$\mathcal{E}(h, k; A, B) = hk \left(\frac{A^2 h^2 + B^2 k^2}{12} + \frac{ABhk}{8} \right). \quad (9.11)$$

To verify this, bound the pointwise error by $A|t - t_0| + B|s - s_0|$ and integrate its square over the rectangle.

The following bounds specify every cell. For $t < 1$, both derivatives of D_0 are bounded by $c_0^2/2$, by the divided-difference integral and $|\chi''| \leq c_0^2$. Both derivatives of D_1 are bounded by $3c_0^2/2$: use $|\chi'(t)| \leq c_0^2 t$ and $|\chi(t) - \chi(s)| \leq c_0^2 |t^2 - s^2|/2$ in the differentiated quotient. For an output cell above one with lower endpoint t_- , and source lower endpoint s_- , put $d = t_- + s_-$. Since $D_1 = -\chi(s)/(t + s)$ there, its derivative bounds are

$$|\partial_t D_1| \leq d^{-2}, \quad |\partial_s D_1| \leq c_0 d^{-1} + d^{-2}.$$

The remaining nonzero kernels reduce, up to a sign, to $f(\alpha, \beta) = \sin(c_0 \alpha)/(\alpha + \beta)$, $0 \leq \alpha \leq 1$, $\beta \geq 0$. For D_2 below one take $(\alpha, \beta) = (1 - t, s)$; for D_0 above one take $(\alpha, \beta) = (1 - s, t - 1)$. On $[\alpha_-, \alpha_+] \times [\beta_-, \beta_+]$ with $d = \alpha_- + \beta_- > 0$, valid bounds are

$$A_\alpha = \frac{c_0 \beta_+ + c_0^3 \alpha_+^3 / 3}{d^2}, \quad A_\beta = \frac{c_0 \alpha_+}{d^2}. \quad (9.12)$$

Indeed $|x \cos x - \sin x| \leq x^3/3$, by integration of $-x \sin x$. This proves the first bound after differentiating f ; the second uses $|\sin x| \leq x$. Assign these bounds to t, s in the displayed coordinate order. When $d = 0$, use instead the squared error bound c_0^2 times the cell area, since both f and its midpoint value lie in $[0, c_0]$. This handles the corner cells without asserting a global square-integrable first derivative. Above one, D_2 vanishes.

Let E_j be the sum of these squared cell bounds for D_j . The multiplicities in (9.3) give the finite error $(4E_0 + 2E_1 + 2E_2)/(4\pi^2)$. The complete omitted output tail satisfies

$$\begin{aligned} \|\mathbf{1}_{(T, \infty)} \Delta\|_2^2 &= \frac{1}{4\pi^2} \int_0^1 \chi(s)^2 \left(\frac{4}{T-s} + \frac{2}{T+s} \right) ds \\ &\leq \frac{1}{8\pi^2} \left(\frac{4}{T-1} + \frac{2}{T} \right), \end{aligned} \quad (9.13)$$

using $\int_0^1 \chi(s)^2 ds = 1/2$. Finite and tail squared errors add on disjoint output supports. These inequalities provide a global bound for $\|\Delta - D\|_2$.

9.2.6 Exact witness and rational verification gates

The source grid has 48 cells and is specified by

$$s_j = \frac{j^4}{2 \cdot 24^4} \quad (0 \leq j \leq 24), \quad s_{48-j} = 1 - s_j.$$

The model grid is the sorted union of

$$\{s_j : 0 \leq j \leq 48\}, \quad \{1 + s_j : 1 \leq j \leq 48\}, \quad \{2^k(1 + \ell/6) : 1 \leq k \leq 39, 1 \leq \ell \leq 6\}.$$

It has 330 cells and ends at $T = 2^{40}$. The file `step_parametrix_witness.json` specifies a 1320×96 matrix W , whose real and imaginary coefficients are integers divided by 2^{48} in the normalized cell bases. Its SHA-256 digest begins `2b5d1dca6fbcdf3a`; the full digest is recorded with the checker output. The choice of trial is immaterial to the proof once these exact coefficients are fixed.

The accompanying negative-window checker and its global-error helper evaluate the preceding expressions with outward-rounded Arb arithmetic [25]; their filenames and execution instructions are listed in `anc/README.md`. For (9.11)–(9.13) it uses the rational upper bounds $c_0 < 11/7$, $1/(4\pi^2) < 1/36$ and $1/(8\pi^2) < 1/72$. Midpoint defect values are enclosed directly, so D has no separate saved-coefficient rounding error. All primitive arguments in (9.9) are formed as exact rationals before conversion; a zero argument uses $F(0) = 0$.

At both 128 and 192 bits, the checker verifies the strict rational gates

$$\|E^*W\|_2 < \frac{27}{100}, \quad \mathcal{R}(W, D) < \frac{9}{500}, \quad \|\Delta - D\|_2 < \frac{37}{1000}. \quad (9.14)$$

No eigenvalue solve, Fourier truncation, or numerical quadrature enters this verification. The gates and (9.8) give

$$\|K_a\| < \frac{27}{100} + 5 \left(\frac{9}{500} + \frac{37}{1000} \right) = \frac{109}{200} < \frac{3}{5}.$$

Exclusion and uniqueness in Theorem 9.1. The Neumann series makes $I + K_a$ invertible. Since $S_a P_a = I + K_a$ and $\|S_a\| \leq 5$, the self-adjoint pencil P_a is bounded below by a constant strictly greater than $2/25$. Its range is closed and its orthogonal complement is its zero kernel, so it is onto and

$$\|P_a^{-1}\| < \frac{25}{2}.$$

For real λ with $|\lambda - a| \leq 3/50$, $\|P_\lambda - P_a\| = |\lambda - a|$ and the inverse Neumann product is strictly below $3/4$. Thus every P_λ on $[-953/2000, -713/2000]$ is invertible. The exact frequency correspondence of Proposition 7.7 excludes physical spectrum there.

The existence argument above supplies a positive eigenvalue in $(81/200, 107/250)$. If another positive boundary level lay in $[713/2000, 953/2000]$, the signed alternation of Corollary 10.6 would put a negative gap magnitude strictly between them, contradicting the exclusion. The eigenvalue is simple by Theorem 7.5. $\square$

9.3 A global residue functional and its interval certificate

We work at phase zero, with $P_\alpha = Q_0 - \alpha J$. For real $0 < \alpha < g$ put

$$a = \sqrt{1 - 2\alpha^2}, \quad t = \arctan a, \quad \nu = \frac{1}{2} + t/\pi, \quad c = (2 \cos t)^{-1}, \quad q = ce^{-it}.$$

For $p > 0$ and real $z < 1$ define

$$\begin{aligned} \ell_p(z) &= \frac{1}{\pi} \int_0^1 \frac{x^{p-1}}{1 - zx} dx, \\ A_\nu(s) &= - \int_0^1 \frac{x^{-\nu}}{1 + s - x} dx = - \frac{\pi}{1 + s} \ell_{1-\nu} \left(\frac{1}{1 + s} \right), \quad B_\nu(s) = -A_\nu(1 - s). \end{aligned} \quad (9.15)$$

Let

$$k = (1, (\alpha/c)e^{it}, i(\alpha/c)e^{2it}, ie^{3it})$$

be a row, and define

$$\begin{aligned} j_1(s) &= k_2 A_\nu(s) + \pi[-\ell_\nu(s) + k_3 \ell_\nu(1 - s) + k_4 \ell_\nu(-s)], \\ j_2(s) &= k_3 B_\nu(s) + \pi[-\ell_\nu(s - 1) - k_2 \ell_\nu(s) + k_4 \ell_\nu(1 - s)], \\ p_\alpha &= \frac{-i\Gamma(\nu)e^{-i(\pi/4+3t/2)}}{8\pi^2 ac(2\pi)^\nu}, \quad F_\alpha(u) = p_\alpha \int_0^1 [j_1(s)u(s) + j_2(s)u(s - 1)] ds. \end{aligned} \quad (9.16)$$

The scalar p_α here is a prefactor, distinct from the pencil P_α . The two row kernels have only logarithmic endpoint singularities, so F_α is a bounded complex-linear functional on $L^2(-1, 1)$.

Lemma 9.2 (The global residue row). *If $P_\alpha u = 0$, then $F_\alpha(u)$ is the inward physical tail amplitude of $V_0 u$, with its induced physical normalization. Moreover, in the operator norm of functionals,*

$$\|F_\alpha\| < \frac{13}{100}, \quad \|\partial_\alpha F_\alpha\| < 1, \quad \frac{2}{5} \leq \alpha \leq \frac{11}{25}. \quad (9.17)$$

9.3.1 Derivation and the auxiliary cutoff limit

The right null column of the endpoint multiplier at $\omega = -i\beta$, with first component one, is

$$v = (1, (\alpha/c)e^{it}, -i(\alpha/c)e^{2it}, -ie^{3it})^T.$$

Direct multiplication in (7.12) shows that k is its left null row, without conjugation. Its adjugate has $(1, 1)$ entry $-q^3$ and

$$\partial_\omega \det m_{\alpha,0}(-i\beta) = i4\pi\alpha^2 ac^2.$$

For a smooth endpoint cutoff, the residue formula in the proof of Theorem 7.9 therefore gives the endpoint vector

$$V = \frac{q^3}{4\pi\alpha^2 ac^2} vN, \quad N = \int_0^\infty x^{-\nu} k(-\Delta_e u)(x) dx. \quad (9.18)$$

Indeed the factor $-i\sqrt{2\pi}$ in that residue formula cancels the Mellin normalization and the displayed determinant derivative. The positive zero-carrier Gamma factor for v is $2\cos(2t)e^{-i\pi/4+3it/2}$, and $\alpha^2 = 2c^2\cos(2t)$. Thus its physical amplitude is

$$\frac{\Gamma(\nu)e^{-i(\pi/4+3t/2)}}{4\pi ac(2\pi)^\nu} N.$$

This calculation uses only $EP_\alpha = M_\alpha E + \Delta_e$ and $\chi = 1$ near zero; the auxiliary cutoffs need not form an isometric partition. For two such cutoffs the difference localizes away from zero. On a kernel vector its Mellin transform is regular at $-i\beta$ and $km_{\alpha,0}(-i\beta) = 0$, so the source pairing is unchanged. No cutoff independence on arbitrary u is asserted.

Take smooth cutoffs equal to one up to $1 - 2\varepsilon$ and zero after $1 - \varepsilon$, and let $\varepsilon \downarrow 0$. Their source-pairing rows converge in L^2 to the row for $\chi = \mathbf{1}_{(0,1)}$. Here are the required operator bounds. In EQ_0 , move Q_0 onto the cutoff difference multiplied by $x^{-\nu}k$; this weight is bounded near one and Q_0 is bounded on L^2 . In $Q_e E$, move Q_e onto $x^{-\nu}k$ and restrict the result to a neighborhood of one. That result is bounded: the input near zero is integrable since $\nu < 1$, its tail is integrable against the Cauchy denominator, and its local principal value is the transform of a smooth power. Both error rows tend to zero with the L^2 norm of the cutoff difference. These statements may first be proved for smooth inputs and then extended by density. They require no uniform derivative bound on the steep cutoffs.

For this limiting cutoff, the kernel in (7.20), before its factor $i/(2\pi)$, has on $0 < x < 1$ only the entries $(2, 1) = (x - 1 - s)^{-1}$ and $(3, 2) = (2 - x - s)^{-1}$. On $x > 1$ it is

$$\begin{pmatrix} -(x-s)^{-1} & -(x+1-s)^{-1} \\ 0 & -(x-s)^{-1} \\ (x+s-1)^{-1} & 0 \\ (x+s)^{-1} & (x+s-1)^{-1} \end{pmatrix}.$$

Integration against $x^{-\nu}k$ gives exactly the two rows in (9.16), with $N = -i(2\pi)^{-1} \int_0^1 (j_1 u_1 + j_2 u_2)$. This proves the first assertion of Lemma 9.2, including its normalization.

9.3.2 Uniform row and derivative bounds

All decimals in the following estimates are finite rational numbers. For $2/5 \leq \alpha \leq 11/25$, elementary differentiation gives

$$a > 0.78, \quad c > 0.63, \quad 0.7 < \nu < 0.75, \quad \kappa := \alpha/c < 0.7, \\ |t'| < 0.71, \quad |\nu'| < 0.23, \quad |c'| < 0.35, \quad |\kappa'| < 2, \quad |k'_2| < 2.5, \quad |k'_3| < 3, \quad |k'_4| < 2.14.$$

For example, $t' = -2\alpha/[a(1+a^2)]$, $c' = -\alpha/(2c)$, and $\kappa' = 1/c + \alpha^2/(2c^3)$. In bounding ν' below we also retain the sharper bound $|\nu'| < 0.71/\pi$.

The power series $\pi\ell_p(z) = \sum_{n \geq 0} z^n/(n+p)$ for $0 \leq z < 1$, and the positive defining integrals for $z \leq 0$, imply

$$\begin{aligned} \pi\ell_\nu(s) &\leq \nu^{-1} - \log(1-s), & \pi\ell_\nu(-s), \pi\ell_\nu(s-1) &\leq \nu^{-1}, \\ |A_\nu(s)| &\leq (1-\nu)^{-1} + \log 2 - \log s, & B_\nu(s) &\leq (1-\nu)^{-1} + \log 2 - \log(1-s). \end{aligned}$$

For the last bounds apply the series at $z = (1+s)^{-1}$ and its reflection. Differentiation in ν gives the absolute bounds

$$\nu^{-2} + \pi^2/6 < 15/4, \quad \nu^{-2} < 21/10, \quad (1-\nu)^{-2} + \pi^2/6 < 18,$$

respectively for a positive-argument $\pi\ell$, a nonpositive-argument $\pi\ell$, and A, B . The series bounds use $\sum_{n \geq 1} (n+p)^{-2} \leq \pi^2/6$.

With $L_0 = -\log s$ and $L_1 = -\log(1-s)$, these estimates yield

$$\begin{aligned} |j_1| &\leq 7.2 + 1.4L_0 + L_1, & |j_2| &\leq 7.2 + L_0 + 1.4L_1, \\ |j'_1| &\leq 24 + 5.5L_0, & |j'_2| &\leq 26 + 2.2L_0 + 5.5L_1. \end{aligned} \tag{9.19}$$

For example the first undifferentiated constant is at most $0.7(4.7) + 2.7(10/7) < 7.2$. The first differentiated constant is at most

$$2.5(4.7) + (3 + 2.14)\frac{10}{7} + 0.23 \left(0.7(18) + 1.7\frac{15}{4} + \frac{21}{10} \right) < 24.$$

Replacing $2.5(4.7) + 3(10/7)$ by $3(4.7) + 2.5(10/7)$ bounds the second by 26. The logarithmic coefficients in (9.19) are respectively the sums of the corresponding $|k_j|$ or $|k'_j|$.

The scalar prefactor satisfies

$$|p_\alpha| < 0.0093, \quad |p'_\alpha| < 4|p_\alpha|. \tag{9.20}$$

For the first inequality use

$$|p_\alpha| \leq \frac{\Gamma(0.7)}{(2\pi)^{0.7} 8\pi^2(0.78)(0.63)} < 0.0093.$$

The digamma $\psi_0 = \Gamma'/\Gamma$ is increasing, since $\psi'_0(x) = \sum_{n \geq 0} (n+x)^{-2} > 0$. The scalar bounds $\psi_0(0.7) > -1.3$ and $\psi_0(0.75) < 0$ show that Γ decreases and $|\psi_0| < 1.3$ on $[0.7, 0.75]$. Consequently

$$|p'_\alpha/p_\alpha| \leq \frac{0.71}{\pi}(1.3 + \log(2\pi)) + \frac{0.88}{0.78^2} + \frac{0.35}{0.63} + \frac{3}{2}(0.71) < 4.$$

All scalar inequalities here are also checked with outward-rounded arithmetic by the accompanying verifier.

Finally, $\int L_j = 1$, $\int L_j^2 = 2$, and $\int L_0 L_1 = 2 - \pi^2/6 < 0.36$. Squaring (9.19) and integrating gives bounds 186.656 for $\|(j_1, j_2)\|_2^2$ and 2055.792 for $\|(j'_1, j'_2)\|_2^2$. Therefore

$$\|F_\alpha\| < 0.0093\sqrt{186.656} < 0.13, \quad \|\partial_\alpha F_\alpha\| < 4(0.13) + 0.0093\sqrt{2055.792} < 1.$$

The same logarithmic envelopes justify differentiation in operator norm. This finishes Lemma 9.2.

9.3.3 Exact integration of the rational witness

Define primitives

$$\begin{aligned} E_\nu(z) &= \frac{z\ell_\nu(z) + \pi^{-1}\log(1-z)}{1-\nu}, & E'_\nu &= \ell_\nu, \\ D_\nu(s) &= \frac{(1+s)A_\nu(s) - \log s}{1-\nu}, & D'_\nu &= A_\nu. \end{aligned} \quad (9.21)$$

Their continuous endpoint values are

$$E_\nu(0) = 0, \quad E_\nu(1) = -\frac{\psi_0(\nu) + \gamma_E}{\pi(1-\nu)}, \quad D_\nu(0) = \frac{\psi_0(1-\nu) + \gamma_E}{1-\nu},$$

where γ_E is Euler's constant. The derivative identities follow by integration by parts in the defining integrals; the endpoint values follow from $\pi\ell_p(z) + \log(1-z) \rightarrow -\psi_0(p) - \gamma_E$ as $z \uparrow 1$. Primitives of j_1, j_2 are consequently

$$\begin{aligned} J_1(s) &= k_2 D_\nu(s) + \pi[-E_\nu(s) - k_3 E_\nu(1-s) - k_4 E_\nu(-s)], \\ J_2(s) &= k_3 D_\nu(1-s) + \pi[-E_\nu(s-1) - k_2 E_\nu(s) - k_4 E_\nu(1-s)]. \end{aligned} \quad (9.22)$$

Use the same exact witness u_0 and parameter a_0 as in (9.2). Integrate (9.22) over each cell and multiply by the witness's constant value there. Its stored coefficients multiply normalized indicators, so this value is the coefficient divided by the square root of the exact cell width. The first 256 coefficients belong to the negative half and the last 256 to the positive half; they pair with j_2 and j_1 , respectively.

The accompanying interval verifier evaluates these endpoint differences using $\ell_p(z) = {}_2F_1(1, p; p+1; z)/(\pi p)$, Gamma and digamma functions with outward-rounded Arb arithmetic [25]. The exact parameter relations $b - c = -1$ and $a + b - c = 0$ are supplied when evaluating the zero-balanced hypergeometric function. At both 192 and 256 bits it verifies

$$|F_{a_0}(u_0)| > 0.069, \quad \|u_0\| < 1.001, \quad 0.41635 < a_0 < 0.41637, \quad (9.23)$$

as well as the scalar bounds used above. No quadrature, eigensolver, or saved floating-point result enters these checks. The existing residual certificate gives

$$\|P_{a_0}u_0\| < 0.00622 : \quad U_{u_0}(a_0) < 0.000038641 < (0.00622)^2. \quad (9.24)$$

9.3.4 The complete physical projection estimate

Nonvanishing in Theorem 9.1. By the proved isolation clause, $\alpha_0 \in (0.405, 0.428)$ is the only spectral point in $[0.3565, 0.4765]$. Let Π project onto it. Equation (9.23) puts every other spectral point at distance at least 0.049 from a_0 .

Put $f_0 = V_0 u_0$. The identities $B_0 = V_0 J V_0^*$, $V_0^* V_0 = Q_0$, and $\|V_0\| \leq 1$ give

$$(B_0 - a_0)f_0 = V_0 J P_{a_0} u_0, \quad \|(I - \Pi)f_0\| < \frac{0.00622}{0.049}.$$

For $\alpha \in (0.405, 0.428)$ define the bounded physical functional

$$\mathcal{L}_\alpha f = \alpha^{-1} F_\alpha(J V_0^* f), \quad \|\mathcal{L}_\alpha\| < \frac{0.13}{0.405}.$$

On the α eigenspace it equals the physical tail amplitude: $u = \alpha^{-1} J V_0^* f$ satisfies $P_\alpha u = 0$ and $V_0 u = f$. For the trial vector, the exact identity $J Q_0 = \alpha I + J P_\alpha$ instead gives

$$\mathcal{L}_\alpha f_0 = F_\alpha(u_0) + \alpha^{-1} F_\alpha(J P_\alpha u_0).$$

Thus no trial value is identified with an eigenvector value.

Now $|\alpha_0 - a_0| < 0.012$, and (9.17) yields

$$|F_{\alpha_0}(u_0) - F_{a_0}(u_0)| < 0.012(1.001), \quad \|P_{\alpha_0}u_0\| < 0.00622 + 0.012(1.001).$$

Combining these bounds with the full complementary projection error,

$$|\mathcal{L}_{\alpha_0}\Pi f_0| > 0.069 - 0.012(1.001) - \frac{0.13}{0.405} \left(0.00622 + 0.012(1.001) + \frac{0.00622}{0.049} \right) > 0.0103.$$

The final inequality is rational and is checked independently by the same verifier. Since $\|\Pi f_0\| \leq \|f_0\| \leq \|u_0\| < 1.001$, normalizing the nonzero eigenfunction Πf_0 gives $|\eta| > 0.01$. Simplicity makes this magnitude independent of the phase of the normalized eigenfunction. The asserted parity consequence now follows from Corollary 8.2. $\square$

10 Parity traces and relative counts

10.1 Spatial parity of traces

Reflection sectors have classical determinant analogues in scalar Toeplitz-plus-Hankel theory [7, Sections 2–3]. The following argument treats the operator-valued symbol and its seam directly, retaining the physical within-cell reflection.

Lemma 10.1 (A reflected Toeplitz limit). *Let r be a self-adjoint involution on a separable Hilbert space $\mathcal{H}$, and let $a \in C^3([0, 1]; \mathfrak{S}_1)$ be self-adjoint with $ra(1 - \theta)r = a(\theta)$. Put*

$$A = \frac{1}{2}(a(0) + a(1)), \quad B = \frac{1}{2}(a(0) - a(1)), \quad R_n = F_n \otimes r,$$

where F_n reverses the n cells and $M = \sup_{\theta} \|a(\theta)\|$. For every $f(0) = 0$ holomorphic near $[-M, M]$, along even n ,

$$\lim_{n \rightarrow \infty} \text{tr}(R_n f(T_n(a))) = \frac{1}{2} \text{tr}_{\mathcal{H}}(r[f(A) - f(a(1/2))]). \quad (10.1)$$

Proof. All finite sections are trace class: their trace norms are at most $n \int_0^1 \|a(\theta)\|_1 d\theta$, by the rank-one cell-vector representation of the Fourier coefficients. Set $h(\theta) = 1 - 2\theta$, $b = A + hB$, $c = a - b$, and $a_t = b + tc$. Then $c(0) = c(1) = 0$, R_n commutes with $T_n(a_t)$, and $\|a_t(\theta)\| \leq M$: both b and a_t are convex combinations of symbols with norm at most M .

Put $H_n = T_n(h)$. Since $F_n H_n F_n = -H_n$ and $T_n(b) = I_n \otimes A + H_n \otimes B$, reflection exchanges the nonzero H_n eigenspaces in opposite pairs, contributing zero to $\text{tr}(R_n f(T_n(b)))$. The kernel contributes $\text{tr}(F_n|_{\ker H_n}) \text{tr}(r f(A)) = 0$: for even n , $\text{tr} F_n = 0$ and its nonzero pairs have zero trace. Hence

$$\text{tr}(R_n f(T_n(b))) = 0. \quad (10.2)$$

If $d \in C^3([0, 1]; \mathfrak{S}_1)$ and $d(0) = d(1)$, Lemma 3.22 makes both one-sided Hankel blocks $\Pi_+ \mathcal{L}(d) \Pi_-$ and $\Pi_- \mathcal{L}(d) \Pi_+$ trace class: the principal matrix is $((j + l + 1)^{-2})_{j, l \geq 0}$ and the remainder has summable weighted Fourier trace norms. For a fixed bounded self-adjoint symbol a_0 and $k \geq 0$, this gives

$$\text{tr}(R_n [T_n(d)T_n(a_0) - T_n(da_0)]T_n(a_0)^k) \rightarrow 0. \quad (10.3)$$

Indeed split $-P_n \mathcal{L}(d)(I - P_n) \mathcal{L}(a_0) P_n$ into $j < 0$ and $j \geq n$ boundary terms. At the left, the Hankel factor converges in trace norm and all remaining factors and adjoints converge strongly. Its trace-class limit pairs to zero with the weakly null zero-extended R_n (check finite

cell supports). Conjugating by F_n gives the same argument at the right, since $F_n R_n F_n = R_n$. Finite-rank approximation makes these bounds uniform on compact C^3 symbol families.

Choose a fixed contour Γ in the holomorphy domain enclosing $[-M, M]$ at positive distance. For $\zeta \in \Gamma$, put $d_{t,\zeta} = c(\zeta - a_t)^{-1}$. These symbols are C^3 in trace norm, vanish at both endpoints, and have uniform C^3 bounds on $[0, 1] \times \Gamma$. With $Q_n = I - P_n$, exactly

$$T_n(c)(\zeta - T_n(a_t))^{-1} - T_n(d_{t,\zeta}) = -P_n \mathcal{L}(d_{t,\zeta}) Q_n \mathcal{L}(a_t) P_n (\zeta - T_n(a_t))^{-1}.$$

Put $\delta = \text{dist}(\Gamma, [-M, M])$. The resolvents have norm at most δ^{-1} ; strong convergence of the self-adjoint finite sections and the resolvent identity give strong convergence of resolvents and adjoints at both ends. Finite nets make this uniform on fixed vectors for $(t, \zeta) \in [0, 1] \times \Gamma$. The symbols $d_{t,\zeta}$ form a compact $C^3(\mathfrak{S}_1)$ family, so their two Hankel trace norms have a common bound C_H . The displayed error thus has trace norm at most MC_H/δ . Its boundary limits are trace-norm uniform by the preceding localization; finite-rank approximation and weakly null reflection give uniform vanishing of its reflected trace. Cauchy calculus therefore gives

$$\text{tr}(R_n T_n(c) f'(T_n(a_t))) - \text{tr}(R_n T_n(c f'(a_t))) \longrightarrow 0.$$

Differentiating the Cauchy integral and cycling the trace gives

$$\frac{d}{dt} \text{tr}(R_n f(T_n(a_t))) = \text{tr}(R_n T_n(c) f'(T_n(a_t))).$$

For $e_t = c f'(a_t)$ the zero-endpoint Fourier sums are uniformly absolutely summable in trace norm, say by C_e . Thus the derivative integrand is dominated, independently of n, t , by

$$C_e + \frac{\text{length}(\Gamma)}{2\pi\delta} \sup_{\Gamma} |f'| MC_H.$$

This justifies the contour limit and dominated convergence in t . For even n , absolute trace-norm Fourier convergence gives

$$\text{tr}(R_n T_n(e_t)) = \sum_{|k| < n, k \text{ odd}} \text{tr}(r \hat{e}_t(k)) \longrightarrow \frac{1}{2} \text{tr}(r[e_t(0) - e_t(1/2)]).$$

Reflection symmetry gives $[r, a_t(1/2)] = 0$, so cyclicity identifies $\text{tr}(rc(1/2)f'(a_t(1/2)))$ with $d \text{tr}(rf(a_t(1/2)))/dt$. Since $e_t(0) = 0$, integration in t gives $\frac{1}{2} \text{tr}(r[f(A) - f(a(1/2))])$. This proves (10.1) for this f . The exact linear reference cancellation works unchanged by its spectral decomposition. □

Theorem 10.2 (Spatial-parity limit). *Let $Rf(x) = f(-x)$ on $L^2[-1, 1]$ and $g = 1/\sqrt{2}$. For every $f(0) = 0$ holomorphic near $[-1, 1]$,*

$$\text{tr}(Rf(\mathcal{K}_\tau)) = \frac{1}{2}[f(g) + f(-g) + f(1) - f(-1)] + o_f(1) \quad (\tau \rightarrow \infty). \quad (10.4)$$

In particular, for every fixed integer $m \geq 1$,

$$\text{tr}(R\mathcal{K}_\tau^{2m}) \longrightarrow 2^{-m}, \quad \text{tr}(R\mathcal{K}_\tau^{2m-1}) \longrightarrow 1.$$

Proof. At integer $\tau = q$, dilation and integer translation identify the operator with the $2q$ -cell section of its exact symbol $\sigma(\theta) = W_\theta^* X_{1,0} W_\theta$. Spatial reflection becomes $F_{2q} \otimes r$, where $(rf)(t) = f(1-t)$. This symbol is smooth in trace norm and satisfies $r\sigma(1-\theta)r = \sigma(\theta)$. Its endpoint average $A = (X_{1,0} + X_{0,-1})/2$ has nonzero eigenvalues $\pm g$, both in the r -even space $\text{span}\{e_0, (e_1 + e_{-1})/\sqrt{2}\}$. At $\theta = 1/2$, the $+1$ eigenvector of σ is proportional to $\cos(\pi t)$ and is

r -odd, while the -1 eigenvector is proportional to $\sin(\pi t)$ and is r -even. Hence Lemma 10.1 gives precisely (10.4) along integer τ .

For $\tau = n + r$, $r \in [0, 1]$, set $m = 2n$, $A_m = P_{(-r, m+r)} \mathcal{G} P_{(-r, m+r)}$ and $B_m = P_{(0, m)} \mathcal{G} P_{(0, m)}$; both commute with $\mathcal{R}_m u(X) = u(m - X)$. The already proved strip localization (6.15) gives

$$A_m - B_m = F_{-r}^L + S^m F_r^R S^{-m} + o_{\mathfrak{S}_1}(1).$$

For a contour Γ around $[-1, 1]$ in the holomorphy domain, insert this identity into

$$f(A_m) - f(B_m) = \frac{1}{2\pi i} \int_{\Gamma} f(\zeta) (\zeta - A_m)^{-1} (A_m - B_m) (\zeta - B_m)^{-1} d\zeta.$$

Each boundary product converges in trace norm after fixing that boundary. The reflections $\mathcal{R}_m = S^m R$ are weakly null at the left, and $S^{-m} \mathcal{R}_m S^m = S^{-m} R$ are weakly null at the right. Consequently $\text{tr}(\mathcal{R}_m[f(A_m) - f(B_m)]) \rightarrow 0$ for fixed r .

Cauchy calculus bounds $\|f(A) - f(B)\|_1$ by a fixed multiple of $\|A - B\|_1$, while $\|A_m(r) - A_m(s)\|_1 \leq 4b(|r - s|)$ independently of m . Thus finite nets give uniform convergence on $r \in [0, 1]$. Dilation and integer translation recover the reflected traces at $n + r$ and n , proving the full limit. $\square$

Remark 10.3 (Quantitative moment transfer). Writing $G_L = P_{(-L, L)} \mathcal{G} P_{(-L, L)}$, cyclic differentiation of the reflected multiple-integral trace gives $d \text{tr}(R G_L^m)/dL = 2m G_L^m(L, -L)$: all m boundary terms agree, and the two endpoints agree by reflection. With $w(s) = (1 + |s|)^{-1}$, $|G(X, Y)| \leq 4w(X - Y)$ and $\int_{-L}^L w(X - Z)w(Z - Y)dZ \leq 8 \log(1 + 2L)w(X - Y)$. Indeed one of the two distances is at least $|X - Y|/2$, while the other factor has integral at most $2 \log(1 + 2L)$. Induction gives

$$|G_L^m(L, -L)| \leq \frac{4^m [8 \log(1 + 2L)]^{m-1}}{1 + 2L}.$$

Thus replacing L by $\lfloor L \rfloor$ changes its reflected m th moment by $O_m(\log^{m-1}(2 + L)/(1 + L))$.

Corollary 10.4 (Parity trace balance and determinants). *For the spatial even and odd restrictions $\mathcal{K}_{\tau, \pm}$ and $f(0) = 0$ holomorphic near $[-1, 1]$,*

$$\text{tr } f(\mathcal{K}_{\tau, \pm}) = \frac{1}{2} \text{tr } f(\mathcal{K}_{\tau}) \pm \frac{1}{4} [f(g) + f(-g) + f(1) - f(-1)] + o_f(1).$$

Thus the volume and logarithmic coefficients divide equally. For each real $w > 0$,

$$\frac{\det(I + w \mathcal{K}_{\tau, +}^2)}{\det(I + w \mathcal{K}_{\tau, -}^2)} \longrightarrow 1 + \frac{w}{2}. \quad (10.5)$$

For $|z| < 1$, locally uniformly, the analytic logarithms normalized at zero satisfy

$$\log \frac{\det(I - z \mathcal{K}_{\tau, +})}{\det(I - z \mathcal{K}_{\tau, -})} \longrightarrow \frac{1}{2} \log(1 - z^2/2) + \frac{1}{2} \log \frac{1 - z}{1 + z}. \quad (10.6)$$

Proof. The commuting projections $(I \pm R)/2$ and Theorem 10.2 give the first identity. Apply it to $\log(1 + wx^2)$, holomorphic near $[-1, 1]$, and $\log(1 - zx)$ to obtain the determinant formulas. For $|z| \leq \rho < 1$ choose a common contour Γ around $[-1, 1]$ with $\rho \sup_{\Gamma} |\zeta| < 1$. The normalized logarithms and their derivatives form a compact family there; the proof's Hankel, Fourier and strip bounds and finite nets give local uniformity. $\square$

10.1.1 Sharp parity counts inside the central gaps

Theorem 10.5 (Exact eventual parity counts in the central gaps). *Fix $r \in \mathbb{R}$ and $0 < t < g$. If $t \notin \text{spec } B_{-r}$, then, for all sufficiently large integers n ,*

$$\#\{\lambda_j(\mathcal{K}_{n+r,+}) > t\} - \#\{\lambda_j(\mathcal{K}_{n+r,-}) > t\} = 1. \quad (10.7)$$

If instead $-t \notin \text{spec } B_{-r}$, then eventually

$$\#\{\lambda_j(\mathcal{K}_{n+r,+}) < -t\} - \#\{\lambda_j(\mathcal{K}_{n+r,-}) < -t\} = 0. \quad (10.8)$$

Here the subscripts $+$ and $-$ denote even and odd spatial parity. Each assertion requires regularity only at its own signed threshold.

Proof. Put $m = 2n$ and use physical coordinates $(-r, m+r)$. Write $\rho f(t) = f(1-t)$ on the fiber and $\mathcal{R}_m = F_m \otimes \rho$ on the full cell space, where the full cell reflection sends j to $m-1-j$. Retain the gauge of Lemma 3.21 $\mathcal{U} = \mathcal{L}(V)$ and set

$$\tilde{\mathcal{R}}_m = \mathcal{U}^* \mathcal{R}_m \mathcal{U}, \quad Z(\theta) = \rho V(1-\theta)^* \rho V(\theta), \quad d = Z - I.$$

Then

$$\tilde{\mathcal{R}}_m = \mathcal{R}_m \mathcal{L}(Z), \quad \rho Z(1/2) = V(1/2)^* \rho V(1/2), \quad (10.9)$$

and d is smooth in trace norm with $d(0) = d(1) = 0$.

Let $P_m^{A,\sigma}$ and $P_m^{C,\sigma}$ be the sharp spectral projections of $\mathcal{A}_m(-r, r)$ and $\mathcal{C}_m$ above t for $\sigma = +$, or below $-t$ for $\sigma = -$. Choose a rectangle enclosing the appropriate spectral band, crossing the real axis at the regular threshold and beyond 1 or -1 . The gap theorem gives uniform contour resolvent bounds for large m and both half-line limits. Their resolvents and adjoints converge strongly at each end, including the real crossing by the resolvent identity and common bound. Inserting (7.1) therefore localizes $P_m^{A,\sigma} - P_m^{C,\sigma}$ in trace norm at both boundaries. The gauge commutes with shifts, so both $\tilde{\mathcal{R}}_m$ and $S^{-m} \tilde{\mathcal{R}}_m S^m$ are weakly null. Consequently

$$\text{tr}(\tilde{\mathcal{R}}_m(P_m^{A,\sigma} - P_m^{C,\sigma})) \rightarrow 0. \quad (10.10)$$

The star itself has exactly zero reflected count in each sign for even m . To see this, write it as $\begin{pmatrix} 0 & B_m \\ B_m^* & 0 \end{pmatrix}$, where $B_m B_m^* \geq I/2$, and put $W_m = (B_m B_m^*)^{-1/2} B_m$. Its signed spectral subspaces are the graphs $x \mapsto 2^{-1/2}(x, \pm W_m^* x)$. The star reflection symmetry intertwines these graphs with F_m on the central component. Since $\text{tr } F_m = 0$ and all nonzero star eigenvalues have magnitude at least g , $\text{tr}(\mathcal{R}_m P_m^{C,\sigma}) = 0$ exactly. Thus (10.9) reduces the remaining trace to

$$\text{tr}(\tilde{\mathcal{R}}_m P_m^{C,\sigma}) = \text{tr}(\mathcal{R}_m T_m(d) P_m^{C,\sigma}).$$

For z on the contour put $d_z = d(z-a)^{-1}$. An exact identity is

$$T_m(d)(z - \mathcal{C}_m)^{-1} - T_m(d_z) = -P_m \mathcal{L}(d_z)(I - P_m) \mathcal{L}(a) P_m (z - \mathcal{C}_m)^{-1}.$$

The symbols d_z have zero endpoints and trace-class one-sided Hankel blocks, uniformly on the contour. The localization argument of Lemma 10.1, now using the reference resolvent, makes the reflected trace of the right side tend to zero uniformly. Contour integration and the odd Fourier coefficient sum therefore give

$$\text{tr}(\mathcal{R}_m T_m(d) P_m^{C,\sigma}) \rightarrow -\frac{1}{2} \text{tr}_{\mathcal{H}}(\rho d(1/2) p_\sigma(a(1/2))), \quad p_\sigma(a) = \frac{1}{2}(a^2 + \sigma a).$$

Both signed star eigenvectors at $1/2$ are ρ -even. The physical positive eigenvector is proportional to $\cos(\pi t)$ and is ρ -odd; the negative one is proportional to $\sin(\pi t)$ and is ρ -even. Equation (10.9) makes the two fiber traces respectively -2 and 0 . Together with (10.10), the physical reflected counts tend to 1 and 0 . They are integers because physical reflection commutes with the physical sharp spectral projections, so both equalities hold eventually. $\square$

In particular the gauge-invariant two-boundary positive count offset is odd, and its negative counterpart is even. These congruences do not fix an individual reference index or evaluate a boundary eigenvalue.

Corollary 10.6 (Alternation of signed boundary magnitudes). *There is a continuous strictly decreasing lift $\mathfrak{b} : (0, g) \rightarrow \mathbb{R}$ such that, for every real b ,*

$$t \in \text{spec } B_b \iff \mathfrak{b}(t) \in b + \mathbb{Z}, \quad -t \in \text{spec } B_b \iff \mathfrak{b}(t) \in b - \frac{1}{2} + \mathbb{Z}.$$

Thus the positive gap levels and the magnitudes of the negative gap levels alternate at each fixed boundary phase. If $0 < a < c < g$ and $\pm a, \pm c \notin \text{spec } B_b$, their counts in the magnitude interval (a, c) differ by at most one. For the centered operators $\mathcal{K}_{n+r}$, these two signed counts are eventually even and differ by at most two; in each parity sector they differ by at most one.

Proof. The simple strictly decreasing branches of Theorem 7.5 have continuous strictly decreasing local inverses. Unique phases make them agree modulo one; the interval $(0, g)$ admits a continuous real lift. Subdivision of compact subintervals makes it globally strictly decreasing. Half-period sign reversal gives the second equivalence; differentiability of the inverse is unnecessary.

The two lattices in the assertion alternate. More explicitly, at the stated regular endpoints the two counts are

$$\lfloor \mathfrak{b}(a) - b \rfloor - \lfloor \mathfrak{b}(c) - b \rfloor, \quad \lfloor \mathfrak{b}(a) - b + \frac{1}{2} \rfloor - \lfloor \mathfrak{b}(c) - b + \frac{1}{2} \rfloor.$$

Their difference has magnitude at most one. For $b = -r$, the centered finite statement follows from Theorem 7.2 and Corollary 7.6: each boundary level contributes exactly one eigenvalue of each parity, and there are eventually no additional eigenvalues in either fixed compact interval. $\square$

Proposition 10.7 (Order of a stationary boundary crossing). *Let $\lambda_0 \in (-g, 0) \cup (0, g)$ be an eigenvalue of B_{b_0} , and let h_0 be its normalized physical eigenfunction, continued to its entire bandlimited representative. Let $d \geq 0$ be its finite vanishing order at b_0 , with $d = 0$ when $h_0(b_0) \neq 0$. The local eigenbranch satisfies*

$$\lambda(b_0 + s) = \lambda_0 - \frac{\lambda_0 |h_0^{(d)}(b_0)|^2}{(2d+1)(d!)^2} s^{2d+1} + O(s^{2d+2}).$$

In particular, any stationary crossing has odd finite order.

Proof. Use the unshifted frequency pencil $L(b) = Q_b - \lambda(b)J$ from (7.10). Choose an analytic normalized kernel vector $u(b)$ with $\langle u, u' \rangle = 0$, and set

$$z(b) = \langle v_b, u(b) \rangle, \quad q(b) = \langle u(b), Q_b u(b) \rangle > 0, \quad j(b) = \langle u(b), J u(b) \rangle = \frac{q(b)}{\lambda(b)} \neq 0.$$

Differentiating $L(b)u(b) = 0$ gives

$$\lambda' = -\frac{|z|^2}{j}, \quad L(b)u' = v_b z + \lambda' J u.$$

The inverse of $L(b)$ on the orthogonal complement of its simple kernel is locally uniformly bounded. Hence $\|u'\| \leq C(|z| + |z|^2)$. The analytic function z cannot vanish on an interval, since then λ would be constant. If its zero order at b_0 is d_0 , integration gives $u(b_0 + s) - u(b_0) = O(s^{d_0+1})$. Consequently $z(b_0 + s)$ and $\langle v_{b_0+s}, u(b_0) \rangle$ have the same first nonzero Taylor coefficient. The latter is $\sqrt{q(b_0)} h_0(b_0 + s)$, so $d_0 = d$ and

$$z(b_0 + s) = \sqrt{q(b_0)} \frac{h_0^{(d)}(b_0)}{d!} s^d + O(s^{d+1}).$$

Substitute this in the formula for λ' and integrate once. $\square$

Theorem 10.8 (Relative signed counts at a regular gap threshold). *Fix $r \in \mathbb{R}$ and $0 < t < g$ with $\pm t \notin \text{spec } B_{-r}$. Write*

$$N_+(t; n, r) = \#\{\lambda_j(\mathcal{K}_{n+r}) > t\}, \quad N_-(t; n, r) = \#\{\lambda_j(\mathcal{K}_{n+r}) < -t\}.$$

For all sufficiently large integers n , putting $b = -r$, one has

$$N_+(t; n, r) - N_-(t; n, r) = 2[I_+(t; b) - I_+(t; b + \tfrac{1}{2})] - 1 \in \{-1, 1\}.$$

The value is 1 precisely when the unique positive crossing phase lies in $(b, b + 1/2)$ modulo one. The two physical count offsets relative to $2n$ are therefore adjacent integers; their absolute values are not determined here.

Proof. Write $i_L(t; l) = \iota_+(t; l)$ and $i_R(t; h) = \iota_-(t; h)$ for the positive left and right indices in (7.5). The references cancel in their differences. Since physical reflection commutes with $\mathcal{G}$ and exchanges the right boundary at h with the left boundary at $-h$, unitary invariance of the trace-class projection difference gives

$$i_R(t; h_2) - i_R(t; h_1) = i_L(t; -h_2) - i_L(t; -h_1)$$

at regular endpoints. This does not require a reflection-equivariant gauge or an equality between individual reference indices.

Put $m = 2n$. Half-translation changes the kernel's sign, so the negative count on $(b, m - b)$ equals the positive count on $(b + 1/2, m - b + 1/2)$. Apply Corollary 7.3 to these two fixed endpoint pairs, using the same m . Their reference counts cancel, and reflection then gives

$$\begin{aligned} N_+ - N_- &= i_L(t; b) - i_L(t; b + \tfrac{1}{2}) + i_R(t; -b) - i_R(t; -b + \tfrac{1}{2}) \\ &= 2i_L(t; b) - i_L(t; b + \tfrac{1}{2}) - i_L(t; b - \tfrac{1}{2}) \\ &= 2[i_L(t; b) - i_L(t; b + \tfrac{1}{2})] - 1. \end{aligned}$$

The period law and monotonicity in Theorem 7.5 make the bracket either zero or one; its value is one exactly when the unique crossing occurs in that half-period. Regularity at its endpoints follows from the two signed regularity assumptions in the statement. $\square$

Corollary 10.9 (Relative ranks of a centered boundary pair). *Let α be a nonzero central-gap eigenvalue of B_{-r} . Choose $0 < a < |\alpha| < c < g$ so that α is the only boundary eigenvalue of either sign with magnitude in $[a, c]$. For all sufficiently large n , let k be the number of opposite-sign eigenvalues of $\mathcal{K}_{n+r}$ with magnitude greater than c . The two eigenvalues converging to α occupy same-sign ranks $k, k + 1$ and combined absolute-value ranks $2k, 2k + 1$, counting from largest magnitude and counting multiplicity.*

If $\alpha > 0$, then $k = 2q$ and their even and odd positive-sector ranks are respectively $q + 1$ and q . If $\alpha < 0$, then $k = 2q + 1$ and their respective negative-sector ranks are both $q + 1$.

Proof. Local gap convergence adds exactly two own-sign eigenvalues and no opposite-sign eigenvalue between the thresholds. The latter count stays k ; the former increases by two. The signed-count theorem forces its value above c to be $k - 1$, giving both rank pairs.

For a positive cluster, Theorem 10.5 makes $k = 2q$ and divides the $k - 1$ positive levels above the cluster into q even and $q - 1$ odd levels. For a negative cluster it makes $k = 2q + 1$ and divides the $k - 1$ negative levels into q of each parity. Corollary 7.6 then adds one member in each sector. The internal ordering of the two parities is not asserted. $\square$

11 Further questions

The transition counting error obtained here is $O((\log \tau)^{2/3})$ on fixed compact subsets of the outer bands. Determining the sharper error for this signed operator requires more than the smooth comparison used in Theorem 3.5. Uniform laws for thresholds approaching zero or a band edge remain open. The finite-frequency results determine the bulk graph and low-order logarithmic terms; a full second-order law and transition classification for an arbitrary commensurate family are not asserted.

11.1 Boundary questions

The isolated phase-zero mode has a certified nonzero amplitude; the corresponding statement at arbitrary phase or for every gap eigenvalue is not proved. Determining the absolute boundary indices would identify specific fixed-offset ranks near the central transition. Uniform laws as the spectral parameter approaches zero or an outer band require additional analysis. Other questions concern an elementary evaluation and convergence rate for the phase-dependent Fredholm product, and extension of the reflected limit to the full band-local Lipschitz class. The kernel also has the exact integrable representation

$$K_\tau(x, y) = \frac{\mathbf{u}_\tau(x)^\top \mathbf{v}_\tau(y)}{x - y}, \quad \mathbf{u}_\tau(x) = \begin{pmatrix} \sin(2\pi\tau x) \\ 1 \end{pmatrix}, \quad \mathbf{v}_\tau(x) = \frac{1}{\pi} \begin{pmatrix} 1 \\ -\sin(2\pi\tau x) \end{pmatrix}.$$

Its diagonal numerator vanishes, as required for a rank-two integrable kernel. Evaluating the global connection in the boundary equation of Theorem 7.8 remains open. The scalar differential obstruction in Section 7.5 leaves matrix, higher-order and nonlocal reductions available. Generalized sine-kernel Riemann–Hilbert methods [27] provide a methodological comparison, not an evaluated connection formula for this boundary problem.

Reproducibility and use of AI tools

The analytic theorems are accompanied by coefficient, finite-covariance, edge, counting, endpoint, interaction and finite-frequency diagnostics. These floating-point checks detect errors; they do not replace proofs. The three boundary interval verifiers use exact rational witnesses, exact cell primitives and outward-rounded Arb arithmetic [25], with the separate existence, exclusion and residue dependencies and analytic error estimates proved in Section 9. All programs and witness data are in `anc/`; the source READMEs and pinned requirements accompany them. No external mathematical supplement or private input is needed. The complete source archive is deposited with this manuscript on Zenodo, doi:10.5281/zenodo.22662221. A browsable copy is available in the author’s public research repository.

Generative AI tools, including Anthropic Claude and OpenAI Codex, assisted drafting, literature discovery, code, proof exploration and adversarial checking under the author’s direction. Their outputs were treated as untrusted and checked before inclusion. The author takes full responsibility for the mathematics, computations, citations and text.

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